

Extended lexical frames in academic journal writing

by

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DEDICATION

To my family

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GLOSSARY

Frame Description	Formatting	Explanation
Lexical frame with single-word variable slot (traditional)	THE * OF THE	- fixed words capitalized - single-word variable slot denoted by “*”
Lexical frame with multi-word variable slot	THE + OF THE	- fixed words capitalized - multi-word variable slot denoted by “+”
Lexical frame variant with filler	THE first draft OF THE	- fixed words capitalized - filler words bold
Lexical frame variant with filler and context	This text appears at THE beginning OF THE proposal	- fixed words capitalized - filler words bold - context words plain text

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ABSTRACT

Many previous studies have explored the phraseological characteristics of academic journal writing. One frequently studied feature is the lexical frame, a recurrent discontinuous sequence of words containing a variable slot filled by a single word. However, occurrences of this frame containing noun fillers with modifiers, such as THE true beginning OF THE and THE frequent use OF THE are not captured using traditional single-word filler approaches. These examples suggest that the true frequency of lexical frames in academic writing has been underestimated in previous research, giving an incomplete picture of the register's phraseological characteristics, as well as the extent of creative language use within formulaic structures.

To address this gap, this study operationally defines extended lexical frames as a group of fixed words surrounding an internal variable slot that can include any number of words, though limited to six here for practical purposes. For example, the sequences THE * OF THE, THE * * OF THE, THE * * * OF THE, etc. are combined as THE + OF THE, with the plus sign (+) indicating one or more words. Because frames in academic writing have been found to most often consist of function words surrounding a noun filler, and due to the importance of noun phrase complexity in academic writing, this study focuses only on frames with noun phrases in the variable slot. A series of custom Python scripts were written and tested to explore the possibilities of using existing natural language processing (NLP) tools to efficiently identify these extended frames in the Academic Journal Register Corpus (AJRC) (Gray, 2015), a corpus of academic journal writing representing a variety of disciplines.

Results of this analysis show that lexical frame variants with multi-word fillers make up a large proportion (~45%) of all noun phrase-based lexical frames identified in the corpus, indicating a potentially fruitful area for future research. Additionally, the script developed for

this dissertation achieved a high recall level (99%), though precision was lower (~75%) due to the need for manual analysis of certain syntactic structures that challenge existing parsing tools. Findings also describe structural patterns of multi-word fillers across frequently occurring frames, illustrating the presence of filler “families” consisting of a root noun and its elaborations and highlighting a previously underexplored dimension of linguistic creativity and productivity within formulaic sequences.

By highlighting extended lexical frames and their multi-word noun phrase fillers, this study provides additional insight into the characteristics of phraseology and nominal complexity in academic writing, addressing the still-relevant call for more research on longer formulaic sequences. Additionally, the extended frame approach may have pedagogical value as a framework for the use of both formulaic patterns and extended noun phrase structures.

CHAPTER 1. INTRODUCTION

Words do not make meaning in isolation but are highly context dependent, and their combinations follow structured, recurrent patterns. These ideas are central to phraseology, or the study of multi-word units and recurring word sequences. Research has shown that language is much more formulaic than might be expected (Wray, 2002), with some estimating that as much as 59% of language consists of multi-word formulas (see discussion in Nelson, 2018). The dual nature of language use as both formulaic and creative is described well by Sinclair (1991), who contrasts the Idiom Principle, seeing language as consisting of prefabricated recurrent sequences, with the Open Choice principle, which focuses on language as a sequence of words freely chosen based on grammatical rules.

Some categories of phraseological sequences are highly fixed and serve as examples of the Idiom Principle in action. For example, the phrase “under the weather” is a fixed idiom, with no possibility of variation; it would be highly unlikely to hear a person describing themselves as “over the weather”. These types of forms are not frequently used, but are highly salient, and were the subject of early phraseology research. Other formulaic sequences are identified based on frequency rather than saliency and have become a more common subject of research as computational tools have facilitated their identification. For example, *lexical bundles* are defined as “recurrent expressions, regardless of their idiomaticity, and regardless of their structural status” (Biber et al., 1999, p. 990). Lexical bundles commonly found in academic writing include sequences like “on the other hand” and “in the present study”, which occur frequently and unlike idioms, can be understood as a combination of individual word meanings. Other phraseology studies have focused on discontinuous sequences that allow some variation (e.g. “in the **present/current** study”). This type of variation, illustrating both the Open Choice and Idiom

principles, is manifested in a discontinuous phraseological form known as *lexical frames* or *phrase frames*, which are defined as recurrent fixed words surrounding a variable slot.

Lexical frames traditionally consist of frequently recurring patterns of two or more fixed words with one variable word between them. For example, the pattern THE * OF THE appears often in academic English, with fillers such as THE **end** OF THE, THE **beginning** OF THE, and THE **nature** OF THE. It is also possible for frames to include multiple words in the variable slot. For example, the fixed words THE * OF THE might also enclose 2-word (THE **true nature** OF THE) 3-word (THE **constantly evolving nature** OF THE) or even 4-word sequences (THE **complex and multifaceted nature** OF THE) while maintaining the same structure of *the* + *Noun Phrase* + *of the*. These multi-word fillers illustrate how frames may be more productive than previously thought in analyses limited to only single word fillers (e.g. Forsyth & Grabowski, 2015), which may have underestimated the degree of productivity and creativity in the use of formulaic language. Lexical frames can serve as valuable linguistic resource that is formulaic while also offering choice and creativity (Casal & Yoon, 2023), and frames with multi-word fillers demonstrate an additional degree of flexibility and functionality that may be an important phraseological characteristic of academic writing. While a few studies have included lexical frames with multiple words in the internal slot to a limited degree (Eeg-Olofsson & Altenberg, 1994; Geluso, 2022; Hsu, 2020; Römer, 2010), there remains a need for a more thorough look at these phraseological forms and the structural characteristics of their fillers. The first goal of this study is to address that gap by extending the scope of lexical frames to accommodate longer variable internal slots, providing a fuller description of the phraseological characteristics of academic journal writing in English and illustrating how formulaic sequences and creative language use are integrated in this form.

One reason for the limited amount of research on extended lexical frames is the challenge of finding them in a corpus, and no tools are currently available for that purpose. Thus, in order to overcome this problem, the first goal of this study is to develop a method of identifying extended frames in a corpus with a minimum of manual analysis needed. Addressing this methodological goal involves first creating an operational definition of what constitutes a valid frame variant, specifically focusing on frames with noun-based fillers as they are the most frequently occurring type in academic writing (Biber, 2009). Based on this definition, a custom Python script is developed and tested to explore the possibilities of using existing natural language processing (NLP) tools to efficiently identify these extended frames in a corpus of academic journal writing representing a variety of disciplines. With the ability to efficiently identify extended lexical frames in a corpus, it is possible to explore a second goal: to investigate the frequency and structural characteristics of these multi-word frames.

In pursuing these two goals, one concerned with methodology and the other with language description, this study contributes a new methodological approach in phraseological research and illustrates how it can provide insight into some under-explored characteristics of academic journal writing, as well as illustrating additional ways the open choice and idiom principles interact in language use (see Sinclair, 1991).

1.1. Background

Formulaic language has been shown to play an important role in academic writing, serving as a fundamental component in constructing knowledge and creating coherence in texts (Hyland, 2008). One type of recurrent multi-word sequence that has received considerable attention is the lexical bundle, defined by Biber et al. (1999) as a continuous sequence of three or more words (e.g. *in the case of, the end of the, on the other hand*) that meet a set frequency of

recurrence (e.g. 10 occurrences per million words) and appear in multiple texts. These bundles have systematic functions, making them important components of building academic discourse (Biber, 2009). Their importance is also reflected in their overall frequency, as 21% of words in academic prose occur in a recurrent bundle (10 times per million words) (Biber et al., 1999, p. 995). Structurally, lexical bundles most often are incomplete units. In conversation, bundles most often consist of a clausal fragment, such as a pronominal subject + verb phrase + complement clause (e.g. *I don't know what*). In contrast, bundles in academic writing most often contain a part of a noun phrase or prepositional phrase (e.g., *in the case of the*), specifically function words such as prepositions or determiners (*in the, of the*) surrounding a content word (*case*), usually a noun (Biber et al., 1999).

This common bundle structure in academic writing reflects Renouf and Sinclair's (1991) observation that grammatical, or function, words (pronouns, articles, and conjunctions) co-occur much more commonly than lexical words (nouns, verbs, and adjectives). These grammatical words often combine in 'frameworks' that enclose characteristic groups of words such as A * OF, which the authors found attracted nouns, specifically nominalizations, as filler words. Renouf and Sinclair (1991) began with a set of pre-selected 2-word frameworks, but later studies have used a corpus-driven approach to identify recurrent 'lexical frames' based on frequency. In a study of variation in speech and writing, Biber (2009) analyzed variability of the components in frequent continuous bundles to identify recurrent frames; for example, the author found that the frequent bundle *it is necessary to* is most variable in the third position, suggesting the frame IT IS * TO. The results of the study showed that lexical frame form and variability are distinguishing characteristics of academic writing, which is phrasal and more variable, compared to conversation, which is clausal and more fixed.

While Biber (2009) extracted frames from a corpus by first identifying frequent continuous bundles and then looking for sequences with variable slots, Gray and Biber (2015) took a 'frames-first' approach to identify frames not associated with frequent continuous bundles. Their study found several highly variable frames that were not associated with any one continuous bundle that met minimum frequency for identification in previous studies. The present study argues that this frames-first approach might be extended further to identify additional frame instances by allowing the variable slot in a lexical frame to include more than one word. For example, Gray and Biber's (2015) frames-first approach uncovered the previously unidentified frame OF THE * OF, with the most frequent filler being "*use*". However, occurrences of this frame containing noun fillers with premodifiers, such as OF THE **frequent use** OF and OF THE **use pattern** OF are not captured using this approach, nor are coordinated nouns (OF THE **production and use** OF) or nouns with prepositional phrase postmodifiers (OF THE **point of view** OF). Limiting the variable slot to a single word thus prevents exploration of how lexical frame usage relates to noun phrase complexity, an important distinguishing feature of academic writing (e.g. Biber & Gray, 2016).

Lexical frames differ both in frequency and structure across registers, or text types defined by situational characteristics (Biber & Conrad, 2009). In the register of written academic English, lexical frames are typically filled by content words, most of which are nouns (Biber, 2009). However, it is important to note that nouns can take both pre- and post-modifiers, and complex noun phrases are an important characteristic of academic writing (Biber & Gray, 2011), with almost 60% of noun phrases in academic prose containing a modifier (Biber et al., 1999, p. 578). Given the frequency of function word based frames in academic writing, and the importance of complex noun phrases in the register, it is worthwhile to explore how noun

phrases might occupy longer variable slots in lexical frames. For example, previous studies have identified the most frequent frame in academic writing as THE * OF THE (Gray & Biber, 2015). In the academic subcorpus of the Corpus of Contemporary American English (COCA) (Davies, 2008), one common realization of this frame is THE **purpose** OF THE, which appears 1962 times. Allowing for an additional word in the variable slot (THE * **purpose** OF THE) we get 499 additional entries or 25% more, the most frequent of which is "THE **primary purpose** OF THE " (98 occurrences). Extending the window further (THE * * **purpose** OF THE) gives 117 additional occurrences (a 6% increase), mostly coordinated noun phrases such as THE **nature and purpose** OF THE.

Description of the phraseological characteristics of academic journal writing has been the subject of numerous studies targeting both the register as a whole and discipline-specific features. However, the examples above suggest that the true frequency of lexical frames in academic writing has been underestimated with the traditional single-word filler approach. A more detailed look at the internal variation of frequent frames could provide a fuller picture of the phraseological patterning of academic journal writing in English. To address this gap, this study defines a lexical frame based on the fixed words, allowing for multiple words in the internal filler. The sequences THE * OF THE, THE * * OF THE, THE * * * OF THE, etc. would be combined as THE + OF THE, with the plus (+) sign indicating one or more words. Extending the variable slot to capture examples like these raises questions about a useful operational definition for frames and their variable slots, as well as methodology for identification, which will be discussed below.

Frames with multi-word fillers other than noun phrases may also be a valuable path to explore. For example, Biber (2009) identified the frame IT IS * THAT as one of the most

frequent in academic writing, with the most common filler being the adjective **possible**. A study allowing multi-word adjective phrases as fillers may uncover instances such as IT IS **also possible** THAT, IT IS **quite possible** THAT, and IT IS **entirely possible** THAT, all of which appear frequently in the academic subcorpus of COCA (Davies, 2008). However, this study is limited to noun phrase-based frames to keep the scope manageable, particularly given the new methodology being tested, and because of the important role of noun phrase elaboration in academic writing. By highlighting multi-word noun phrase fillers, this method will enable additional insights into the nature of noun phrase complexity in academic writing, addressing the still-relevant call for more research on longer formulaic sequences (Biber, 2009).

1.2. Lexical frames with multi-word fillers

Though relatively limited in number, some studies have given attention to lexical frames with multi-word fillers or similar patterns. This includes research focused on frames containing two variable words, such as Eeg-Olofsson and Altenberg (1994), which hints at the possibilities of a more detailed look at extended fillers but provides only a very limited analysis of the two-word fillers they identify. Following that study's example, Geluso (2020) focuses on noun fillers that can include an optional second word as a premodifier. This allows the study to capture more frame instances, though the analysis focuses only on the noun. While useful as a starting point, these studies suggest a need for further development of analytical methods for multi-word fillers.

Other approaches have also been used to identify sequences of fixed and variable words whose form sometimes overlaps with extended lexical frames. For example, *concgrams* are defined as "all of the permutations of constituency variation and positional variation generated by the association of two or more words" (Cheng et al., 2006, p. 414), with no limit on the number of intervening (variable) words or the order of the fixed words in the sequence. While allowing positional variation of fixed words gives a useful insight into extended collocations, the

concgram method provides a different picture of word association from the recurrent word sequences targeted by lexical frames studies. In order to make an argument for cohesiveness of a lexical frame with multi-word fillers, frequency of co-occurrence is not sufficient; it is necessary to identify syntactic relationships between fixed elements. However, concgram searches identify word co-occurrences with no need for a syntactic relationship, making connections across clause and even speaker turn boundaries (though the degree to which that happens may depend on setting chosen for word span). Römer (2010) combined the concgram approach with a lexical frames analysis in order to identify more meaningful extended lexical frames (e.g. IT IS NOT * * CLEAR). However, the study's method of identifying concgrams that constitute meaningful units was based on "semantic integrity" (p. 101), and the criteria for meaningfulness are difficult to clearly identify. The study raises valuable points about the importance of getting a fuller picture of the phraseological profile of texts but also highlights some of the methodological challenges that remain.

1.3. The current study

In order to address the need for a fuller and more nuanced description of lexical frame use in academic writing, this dissertation examines the frequency and structure of lexical frames with multi-word fillers in a corpus of academic journal writing (i.e., published research articles). Specifically, this study focuses on one type of frame: fixed function word frames with a noun phrase in the variable slot (e.g., THE **initial draft** OF THE). This structure is the focus of this study due to the prevalence of function-word frames in academic writing (Biber, 2009; Gray & Biber 2013), as well as the importance of complex noun phrases (the most likely realization of multi-word noun phrase fillers) in this register (Biber & Gray 2010, 2016). Because the target domain for this study is general rather than discipline-specific, the analysis is based on a corpus that includes articles representing a variety of fields of study. Also, language use varies along

many different situational variables in addition to discipline, so research paradigms (e.g. qualitative, quantitative), methodological approaches, and article types common to the included fields are represented (see Section 3.2 for more detail). Using this collection of texts, lexical frames that are frequent across the entire corpus are identified¹ and analyzed for frequency of occurrence and structural characteristics. The structural analysis aims to describe the syntactic characteristics of frame fillers to illustrate noun phrase complexity features within lexical frames. That is, the study considers the distribution of simple vs. complex noun phrases as frame fillers (e.g., 50% of a frame’s instances may be found to have a single noun as a filler, 30% an adjective + noun combination, and 20% a noun with a prepositional phrase as postmodifier).

The primary issue in identifying lexical frames with extended slots is that many sequences of words may not represent coherent lexical frames due to the structure or content of the filler. For example, a search for the frame THE + IN THE with 1+ words in the variable slot may return variants such as

Another similarity of THE **two groups lies** IN THE fact that

where the filler words “*two groups*” and “*lies*” do not belong to the same sentence constituent. Broadly, “*two groups*” belongs to the subject noun phrase constituent of the sentence, while “*lies*” heads the verb phrase in the sentence. As most previous studies have not allowed for extended variable slots, what constitutes a valid frame variant has not been defined. Therefore, for the purposes of this study, the words in a multi-word filler must belong to the same constituent in order for the frame variant to be considered valid. When extending frames whose

¹ Comparison of similarities and differences between disciplines are beyond the scope of the current study and will be pursued in future work.

1-word fillers are nouns, all of the filler words must belong to the same noun phrase, which can include modifiers such as adjectives and postmodifying prepositional phrases, among others. Additionally, the fixed words in a frame must belong to the same constituent to ensure that the frame elements do not cross a constituent boundary. A more detailed discussion of noun phrase structure as it relates to filler validity is presented in Section 3.3.

Even with a clear operational definition of frame validity, judging which examples are acceptable requires careful attention to the surrounding context of the frame, and when relevant, this has been previously done using human judgment (e.g. Römer 2010; Tan & Römer, 2022). However, this process is very time consuming and requires a coder experienced with grammatical analysis. To address these challenges and facilitate analysis of larger databases, this study examines the possibilities of using natural language processing tools including dependency parsers to make acceptability judgments automatically by identifying sentence constituents and their relationship with lexical frames and their fillers. This approach provides a new perspective on phraseological patterns in academic writing, opening the door to additional questions about patterns of use in these previously unexplored multi-word fillers. For example, because no study has focused specifically on extended fillers in this way, it is not yet known how frequent longer fillers may be as compared to single-word fillers. Additionally, it would be valuable to gain insight into the structure of these extended fillers for particular frames, whether they tend to be filled by premodified or postmodified noun phrases, for example. This new window into lexical frame variability could provide a valuable addition to measures used in previous studies, such as predictability and type-token ratio (TTR) (Gray & Biber, 2015; Römer, 2010).

This type of analysis may have pedagogical value as well. Previous research has provided lists of lexical frames for use in guiding novice writers (Casal & Yoon, 2023; Cunningham, 2017;

Golparvar & Barabadi, 2020; Liu & Chen, 2022; Lu et al., 2018), so similar value might be drawn from information about extended frames, which offer more flexibility and generalizability in their application and illustrate uses of function words, which can be more difficult for L2 learners to acquire (Geluso, 2019). It is also possible that increased use of longer frame fillers may be a marker of learner proficiency, similar to Garner's (2016) finding of increased variability and decreasing predictability of lexical frames in English learner writing as proficiency increased, and Parkinson and Musgraves' (2014) study of increased noun phrase complexity as a marker of learner development.

The next chapter explores previous phraseology literature, with a focus on discontinuous sequences and details of existing lexical frames research, as well as information about noun phrase complexity in academic writing. Also included is a discussion of NLP tools used in previous frames studies, and others that may offer additional possibilities. Chapter 3 provides an overview of the two Phases of this study and includes a description of the corpus used for analysis. Chapter 4 describes Phase 1 of the study, which is focused on methodological research goals, including operationalizing what constitutes a valid frame + filler combination, and development and testing of the Python scripts used to identify extended lexical frames. Chapter 5 describes the methods used in Phase 2, which is the part of this study focused on language description, and provides descriptions of the frequency and structural characteristics of the extended lexical frames identified by the script developed in Phase 1. Finally, Chapter 5 discusses these findings in relation to previous research while exploring limitations of the study and potential implications for future research and pedagogy.

CHAPTER 2. LITERATURE REVIEW

In order to situate this study in relation to previous research, this literature review covers several topics. Relevant information about previous phraseology research and the theoretical background supporting this work will be provided to illustrate how discontinuous formulaic language has been conceptualized and identified in previous research. Also, the goals of this dissertation require particular focus on the syntactic structure of lexical frames and their fillers, particularly noun phrases due to their frequency as frame fillers. Therefore, this chapter will also review previous approaches to categorizing lexical frames based on structure, as well as research on noun phrase structure and complexity and their relationships with the variables of register, discipline, and proficiency level. The chapter will conclude with a discussion of the natural language processing (NLP) tools used in previous frames research.

2.1. Phraseology

Central to the topic of this dissertation is the 'phraseological tendency' of language, or the way meaning is created by combinations of words rather than individual words (Sinclair, 2004). The importance of recurring word combinations is well-established in corpus linguistics literature (Biber, Conrad & Cortes, 2004; Biber & Barbieri, 2007; Hyland, 2008; Stubbs, 2007), as well as from other methodological and theoretical perspectives such as Construction Grammar (Goldberg, 2006), Pattern Grammar (Hunston & Francis, 2000), and Lexical Priming theory (Hoey, 2005). However, what constitutes an item of phraseological interest varies among researchers (Table 2-1), and the characteristics of phraseological items have been described by Gries (2008) along six parameters:

1. the **nature** of the elements involved (e.g. words, lemmas, grammatical patterns)
2. the **number** of elements involved

3. the required **number** of occurrences
4. the **distance** between elements
5. the lexical and syntactic flexibility of the elements
6. the role of semantic unity and semantic non-compositionality / non-predictability

Table 2.1

Selected phraseological terms from previous research

Phraseological term	Definition	Characteristics	Example	Example study
idiom	A (mostly) fixed word sequence whose meaning is not predictable from its components	Infrequent but salient, non-compositional in meaning	<i>cut to the chase</i>	Simpson & Mendis, 2012
collocation	co-occurrence of two or more words more often than expected by chance	Statistically defined, semantically transparent	<i>play + role</i>	Hoey 2005
colligation	A word and the grammatical patterns associated with it	Lexically flexible: word + abstract pattern	<i>think + that clause</i>	Hunston & Francis, 2000
construction	pairing of form/pattern and meaning at varying levels of abstraction, such as morphemes, single words, or grammatical patterns	Lexically flexible, can focus on any unit of meaning	<i>verb + direct object + indirect object</i>	Goldberg, 1996
collostruction	Method of identifying words that are attracted to a “slot” in a given construction	Can bridge abstract constructions with lexically specified words	<i>Cause + direct object + indirect object</i> e.g. <i>cause harm to others</i>	Stefanowitsch & Gries, 2003
lexical bundle	3+ word continuous sequences meeting a set frequency of recurrence and range of texts	Semantically transparent, structurally incomplete	<i>the end of the</i>	Biber et al., 1999
lexical frame/ p-frame/ collocational framework	Recurrent discontinuous sequence of words containing a variable slot	Semi-flexible, semantically transparent, structurally incomplete	THE * OF THE	Gray & Biber, 2013 Fletcher, 2006 Renouf & Sinclair, 1991

Some phraseological studies have focused on **idioms**, which are perceptually salient sequences of words whose meaning cannot be derived from the meanings of the constituent words (e.g. *kick the bucket, in a nutshell*). Among the parameters above, the sixth is most important for this type of phraseological item, because the number of words that make up an idiom is dependent entirely on the semantic unity of the sequence. The number of occurrences is not a consideration with idioms, as they are often uncommon, despite often drawing attention due to their non-compositionality and perceptual salience. While some idioms allow a degree of lexical or syntactic flexibility, such as allowing the past tense (*kickeded the bucket*), they are generally characterized by low flexibility.

Another type of phraseological form is the **collocation**, or the co-occurrence of two or more lexical words more often than would be expected by chance, based on the initial concept in Frith (1957). Unlike idioms, collocations are defined statistically rather than based on saliency, and the combinations are generally semantically transparent, with the individual words retaining their meaning. Collocation patterns can also be used to illustrate differences in semantically similar words. For example, the words *little* and *small*, though similar in meaning, collocate with different words (*little kid, little bit, little while* vs. *small amount, small piece, small size*) (Biber et al., 1999). The distance between elements can vary, as collocations may consist of words that are adjacent (*little baby*) or separated by intervening words (*little newborn baby*) depending on the preferences of the researcher. Also, researchers may conduct collocate searches with some lexical flexibility by using lemmas, or base forms of words (e.g. combining *read, reads, reading*, etc. under 'read'). The idea of collocation is a key component of the theory of Lexical Priming (Hoey 2005), which emphasizes the psychological association of words with

other words (collocates), pragmatic functions, and grammatical structures as the source of phraseological patterning. **Colligations** represent a further level of abstraction, describing a lexical item (word) and the grammatical patterns with which it is associated, also known as lexico-grammatical associations (Biber et al., 1999). For example, the word *think* frequently occurs with *that*-complement clauses (e.g. *he thinks that the weather is nice*). This type of phraseological construction is central to the theory of Pattern Grammar as well, which focuses on “the interaction between the particular lexical items in it and the grammatical patterns that they form a part of” (Hunston & Francis, 2000, p. 1), and emphasizes the inseparability of lexis and grammar.

Construction grammar (Goldberg, 1996) allows for a further level of abstraction, with **constructions** being defined as a pairing of form/pattern and meaning at varying levels of abstraction, such as morphemes, single words, fixed expression, or grammatical patterns. Constructions are not required to contain any lexical items and can include abstract argument structures such as *Verb + Direct Object + Indirect Object* with no specified words. One approach to exploring phraseological characteristics based in Construction Grammar is the **collostructions** method (Stefanowitsch & Gries, 2003). With the intent of addressing the limitations of traditional collocation-based methods, which rely primarily on linear proximity of collocates, this approach considers grammatical structure, allowing identification of related items at a distance. This method begins with a construction of interest, which can be defined at varying levels of abstraction, from single words to argument structures (e.g. intransitive, ditransitive). The analysis involves finding which words are attracted to (or repelled by) a slot in the construction using the Fischer exact test as a measure of collocational strength. In this model, a collexeme is the lexeme (word) attracted to construction, a collostruct is a construction

associated with a particular lexeme, and a **collostruction** is the combination of collexeme and collostruct. For example, when examining the verb “cause” and the words that co-occur with it as object nouns, the authors find that different words are more strongly associated with its transitive use (*caused harm to others*) and its ditransitive use (*caused you some inconvenience*). While this approach provides much more detail about collocation at multiple levels, it requires extensive manual inspection and coding. In addition, collostruction analysis must begin with a construction in mind, which means function word based constructions are not often a target. One solution to these limitations is using frequency of occurrence to identify phraseological items of interest, a more corpus-driven approach that allows findings to emerge more directly from the data with fewer theoretical assumptions.

2.2. Frequent continuous sequences

Lexical bundles are 3+ word continuous sequences meeting a set frequency of recurrence and range of texts, such as *in the case of*, *the end of the*, and *on the other hand* (Biber et al., 1999; Biber, Conrad, & Cortes, 2004). Lexical bundles are identified based on frequency rather than being pre-selected based on the interest of the researcher. They are most often non-idiomatic and semantically transparent, though semantic unity is not generally required. Structurally, they typically represent incomplete units, crossing clause boundaries in speech (e.g. *I don't know why*) or phrase boundaries in academic writing (e.g. *as a result of*) (Biber et al., 1999). Bundles occur frequently in academic writing, with Biber et al. (1999) estimating that 4-word bundles occur over 5,000 times per million words in the register. Additionally, nearly 21% of the words in their academic writing corpus were part of the most frequent bundles. Lexical bundles function in academic writing to construct knowledge, organize texts, and reflect register differences (Biber, 2009; Hyland, 2008). Use of register-appropriate bundles is also an important aspect of communication within a discourse community (Chen & Baker, 2010).

2.3. Discontinuous sequences

In addition to continuous word sequences, some researchers have explored sequences with one or more variable elements, most often a single slot in an internal position (e.g. A * B C). These have been conceptualized in different ways in various studies. One early example is Renouf and Sinclair (1991), who defined *collocational frameworks* as "discontinuous sequence[s] of two words, positioned at one word remove from each other"(p. 128), such as A * OF and TOO * TO. In academic writing, these fixed words are most often function words such as prepositions, articles, or conjunctions. Others have referred to similar sequences as "frames" (Eeg-Oloffson & Altenberg, 1994; Mintz, 2003), "phrase frames" or "p-frames" (Fletcher, 2006) and "lexical frames" (Gray & Biber, 2013).

Initial studies of this type focused on previously selected frames of interest for study, but a more corpus-driven approach was developed by Biber (2009), who began by identifying frequent continuous lexical bundles, then measured the variability of each word position to find instances representing frame structures. For example, analysis of *at the end of*, *at the time of*, and *at the beginning of* shows highest variability in the third position, leading to the frame AT THE * OF. A more thoroughly corpus-driven approach was developed by Gray and Biber (2013), who showed that the 'bundles-first' approach (Biber, 2009) missed some highly variable frames where no individual filler occurred often enough to meet the minimum frequency threshold. For example, the frame A * RANGE OF was not identified in Biber (2009) because its most common filler, *a **wide** range of*, does not meet the minimum frequency to be identified in the first step as a bundle, though the combination of all of its fillers (e.g. *a **broad** range of*, *a **whole** range of*) met the minimum threshold and marked it as an important frame for consideration.

In addition to frequency, frames studies have used a variety of other criteria for specifying which frames to focus on or exclude. For example, most studies do not include

frames with variable slots in either initial (* AS WELL AS) or final position (IN ORDER TO *) (e.g. Römer, 2010; Gray & Biber, 2013; Garner, 2016; Tan & Römer, 2022). Frames may also be removed from consideration based on crossing punctuation or turn boundaries (Gray & Biber, 2013), structural incompleteness (Lu et al., 2018), or whether a frame is filled primarily by numbers or proper nouns (Tan & Römer, 2022). Other approaches to identifying which frames are of interest rely on judgment of meaningfulness (e.g. Römer, 2010). However, the criteria for meaningfulness are often not described in detail and may rely on some amount of subjective judgment, which suggests the need for more fully specified approaches to defining frame validity in order to improve replicability.

2.3.1. Variability measures in discontinuous sequences

Another important consideration in frames research is the way measures such as variability are addressed. Type-token ratio (TTR) describes the variability of the slot in a lexical frame, with *type* meaning the number of unique words occurring in the variable slot and *token* meaning the total number of occurrences of the frame (Renouf & Sinclair 1991; Eeg-Olofsson & Altenberg 1994; Römer, 2010; Gray & Biber, 2013, etc.). For example, a frame occurring 100 times in a corpus with 20 different fillers would have a TTR value of 0.20. However, it is important to note that TTR is sensitive to corpus size (see discussion in Eeg-Olofsson & Altenberg, 1994; Hsu, 2020), because as corpora grow larger token numbers will continue to increase while the number of filler types grows more slowly, if at all, resulting in progressively lower TTR values.

Another property of lexical frames is predictability (Biber, 2009; Gray & Biber, 2013), which is a measure of how easily a frame's filler can be predicted. This is measured by dividing the frequency of the most common filler by the total number of occurrences of the frame and is reported as a percentage. For example, if the frame THE * OF THE occurs 100 times in a

corpus, with the filler "beginning" filling the variable slot most often at 40 times, the predictability of the frame is 40%. This value suggests a moderate level of variability, while a frame with many equally likely fillers would have a value closer to zero, and a frame dominated by a single filler would have a value closer to 100% indicating high predictability/low variability. While useful, this measure does not provide a full picture of the variation in fillers, as frames with one filler in 60% of instances and a second filler in the other 40% has the same predictability value as a frame with one filler 60% of the time and four others at 10% each. For this reason, there remains a need for another measure that gives an overall picture of filler type dispersion.

The value of these measures has been shown by the number of studies using them, and their limitations have been explored as well. One alternative to type-token ratio suggested by Eeg-Olofsson and Altenberg (1994) is entropy, a statistical measure often interpreted as randomness or uncertainty. The value of this measure is that it is not sensitive to corpus size and takes into consideration the number of different fillers and their statistical distribution. In the equation below, $p(x)$ represents the probability of the value of the variable (x) being measured. In this case, our variable is the filler appearing in a frame, so the values it can take on are all of the words found to fill that frame. The probability of a value is the frequency of that filler's occurrence divided by the total frequency of occurrence of the frame. For example, if the frame IN THE * OF occurs 100 times, and the word "beginning" is the filler 30 times, $p(\text{beginning}) = 30/100$, or 0.30. This value is then multiplied by the logarithm of the value. This is done for all fillers, and all of them are summed with the total multiplied by -1 to give the entropy value (H).

$$H = - \sum p(x) \log p(x)$$

If a frame only has a single filler, its entropy value is 0, indicating no uncertainty in predicting the word filling the variable slot. If the frame has multiple fillers, entropy increases as their frequencies become more similar; if they occur at the same frequency, predicting which filler will appear in the frame is essentially random, reflected by a high entropy value. For example, if a frame has two fillers that both occur 10 times in the corpus, the entropy value is 1, which can be thought of as maximum entropy. It should also be mentioned that the entropy value can exceed 1 when there are more categories (i.e. fillers), but essentially any number at or above 1 can be interpreted the same way, as a high level of unpredictability.

Entropy has previously been used as a measure of the variability of a frame's filler. However, it also has potential value for the purpose of this study, which is to identify frames with multi-word fillers and examine the characteristics of those fillers. When multiple words can appear, we can think of filler length (e.g. A * * B C, A * * * B C) as the variable being measured by entropy. The values taken by the variable would then range from 1-6, for example, representing the different possible filler lengths. The probability of each filler length would be the number of times any filler of that length appears in the frame, divided by the total frame occurrences. This measure could provide a useful picture of the new dimension of frame variability being examined in this study.

2.4. Frame studies

2.4.1. Register and lexical frames

Lexical frames have been studied by some researchers in relation to registers including conversational speech, academic writing, dissertations, and hotel website writing. In a study comparing conversation and academic writing, Biber (2009) found that academic writing uses variable sequences, usually with an internal variable slot (A * B C ; A B * C), more often than

conversation, which included more fixed sequences and sequences with the first or last item variable (* A B C ; A B C *). Though variable sequences are used in both registers, their composition was found to differ as well. It was found that frames in academic writing most often contain function words (e.g. prepositions, determiners) in fixed slots, with content words (e.g. nouns, verbs) in variable slots. For example, the fixed words in the frame IN THE * OF include only function words, with the variable slot most often containing content words like absence, context, and process. In conversation, however, the fixed slots including comparatively more content words, and variable slots included more function words. Gray and Biber (2013) built on these findings by using a frames-first approach to identify a more complete set of lexical frames in conversation and academic writing. The findings confirmed the patterns found in Biber (2009), showing that conversation is characterized by more fixed, continuous constructions, while academic writing makes use of more variable construction built around grammatical patterns in the form of function word-based frames.

2.4.2. Disciplinary associations in lexical frames

The relationship between lexical frame use and discipline has also been explored by researchers in fields including as mathematics (Cunningham, 2017), engineering (Nekrasova-Beker, 2019; Nekrasova-Becker & Becker, 2020), and environmental studies (Barabadi et al., 2020). Others have explored frame use in more specific areas within academic writing, such as conference abstracts in applied linguistics (Yoon et al., 2020) research article discussion sections in higher education studies (Golparvar & Barabadi, 2020), figure legends within scientific research articles (Liu et al., 2024), and commentaries on visuals (Wu et al., 2024). Contexts outside of academic writing have also been the subject of research, including hotel website language (Fuster-Marquez, 2014), pharmaceutical texts (Grabowski, 2015), translation (Grabowski, 2020), and promotional writing (Casal & Kessler, 2020).

Researchers have also compared differences in lexical frame use across disciplines, such as Reppen and Olson (2020), who compared course readings from nine design-related disciplines, though the study was primarily focused on continuous bundles. Yilmaz and Römer (2020) compared the use of fixed and variable sequences by users of English as a Lingua Franca (ELF) in hard and soft science texts such as blog posts, discussions, reports, and unedited research articles. Disciplinary differences in frame use were also found by J Ren (2021), who compared applied linguistics and pharmaceutical sciences and found relationships between variability and discipline as well as discourse functions. In a study of social science research article introductions Lu et al. (2021) also considered discourse function in a study of the association of lexical frames with rhetorical move-steps, finding some frames to be rhetorically specialized while others were more general in use. Additionally, Cao and Wu (2022) examined the use of frames for appraisal and evaluative functions in dissertation conclusions sections in biomedicine and engineering.

While these studies are only a sample of the work being done with lexical frames, they illustrate the importance of this feature in studies of phraseological characteristics of texts across a variety of contextual variables. Given the findings that lexical frames in academic writing tend to consist of fixed function word frames enclosing nouns as fillers, it begs the question of whether more complex noun phrases might occupy that same variable slot, as they represent the same syntactic constituent. Additionally, while lexical frame fixedness is shown to vary across disciplines when a single filler word is allowed, analyzing frames with multi-word noun phrases as fillers could provide a fuller picture of frame use and its variation. In both cases, the lack of attention to the possibility of extended frames may mean that a significant part of these researchers' areas of interest is being overlooked.

2.4.3. Lexical frames and pedagogy

Lexical frames have also been a topic of interest for pedagogical research, both from first language (L1) and second language (L2) perspectives. In a study of the writing section of the TOEFL iBT exam, Staples et al. compared frequency and fixedness of lexical bundles used by writers of three proficiency levels (2013). Following a bundles-first approach based on Biber (2009), fixedness in this study was defined as when all four slots in a bundle contained the same words in at least 50% of occurrences. The results found that for all learner levels, the majority of variable slots were filled by function words. This is a characteristic of speech, as compared to academic writing, which shows a preference for content words as fillers (Biber, 2009). Additionally, the study found an overall preference (>75%) for continuous rather than discontinuous bundles, which again is a characteristic found in speech rather than academic writing.

Frames have also been investigated in relation to L2 writer development, and many recent studies have examined L2 writers' use of lexical bundles across proficiency levels or in comparison with L1 and expert writers. For example, while primarily focused on continuous bundles, Chen and Baker (2010) compared both L1 and L2 writers to published academic article writing, finding that both student groups had a lower overall frequency and a smaller range of fillers when using the frames THE * OF THE and IN THE * OF. In a study using the Michigan Corpus of Upper-level Student Papers (MICUSP), Walcott (2021) found that compared to L1 writers, the L2 texts show lower levels of predictability, variability, and range of function of the phrase frames they used. In a study of L2 learners enrolled in an online English class, Garner (2016) looked at the use of lexical frames in the texts written by students during tests administered in the class. Differences were found across proficiency levels, with higher level learners employing more variable frames with a larger variety of functions. Similar results were

found by Tan and Römer (2022) in a study of L1 Chinese learners of English at five proficiency levels. Results showed that variability in 3-4-word lexical frames increased with proficiency, and predictability decreased. In addition, it was found that discourse functions of frames vary between proficiency levels. Zhou and Li (2024) examined the use of prepositional phrase frames in bachelor's theses, master's theses, and doctoral dissertations of Chinese L1 writers, comparing these to research articles from the field of applied linguistics. The study's findings showed a developmental trend of increasing diversity and discipline specific language and functions as writers advance to higher levels. Similar results were found by Yan et al. (2024) in a study of Chinese L1 English as a Foreign Language (EFL) learner essays across four proficiency levels (pre-B1, B1, B2, C1, and C2). They found that advanced learners used more varied phrase frames, and at the C1 and C2 level their frames usage is comparable in variability and structure to that of L1 English speaking students. In another recent study, Appel et al. (2024) looked at phrase frames produced in L2 English essays by Japanese university students, with results showing that higher level writers use more 4- and 5-word frames, though with lower variability, which was unlike previous findings.

While the studies mentioned above illustrate the interest among researchers regarding lexical frames use as a developmental marker in student writers, both L1 and L2, others have also argued for the value of explicit teaching of lexical frames and creation of pedagogical lists. Building on previous work that developed pedagogically useful lists of vocabulary and phraseology such as the Academic Formulas List (Simpson-Vlach & Ellis, 2010) and the Lexicogrammar of Academic Vocabulary lists (Green, 2019), Lu et al. (2018) created a list of phrase frames from social science RA introduction sections for English for Academic Purposes (EAP) contexts. Another list was created by Liu and Chen (2022), who developed a list of cross-

disciplinary frames taken from university lectures. Others have identified lists of pedagogically useful frames for specific disciplines such as mathematics (Cunningham, 2017) and higher education (Golparvar & Barabadi, 2020), both using keyness as a measure to identify frames specific to their disciplinary areas. Casal and Yoon (2023), pointing to the lack of research on direct instruction on phrase-frame use, identified phrase frames from conference abstracts from the field applied linguistics. The phrase frames were then used in a pedagogical intervention, a graduate academic writing course for L2 English students from multiple disciplines. Results showed that the learners had a positive reaction to the intervention, describing a heightened awareness of formulaic patterns in writing, particularly when connected with functional aspects of the patterns.

The studies described in the above sections demonstrate the value of lexical frames in describing phraseological characteristics of a target domain and as a marker of learner writing proficiency development, as well as their value for creating pedagogical lists. All have focused on single word fillers in a frame's variable slot but based on other research emphasizing the importance of noun phrase elaboration in academic writing (see Section 2.7.2), there may be value in exploring how extended fillers could contribute a more detailed phraseological description of academic writing and of learner writing development. However, to carry out such analyses, a methodological approach to identifying and analyzing such frames would first be necessary. The following section will review methods used by previous researchers in order to inform the present study's goals and identify potential research gaps.

2.5. Extended variable slots in lexical frames

The majority of research on lexical frames or similarly defined structures have only examined sequences with a single word allowed in the variable slot. However, some studies have taken a limited look at multi-word fillers. In an early example, Eeg-Olofsson and Altenberg

(1994) allowed two-word fillers in the recurrent discontinuous sequences they identified, which included a single fixed word on either side of the variable slot (e.g. THE * OF, THE * THAT). The most frequent frames identified consisted most often of function (grammatical) words in fixed positions and content (lexical) words in the variable slot. Their analysis focused primarily on semantic characteristics of fillers but also found that the grammatical function of fillers varied considerably, though that variation was not quantified or examined in detail. Additionally, the results showed that many 2-word fillers were extensions of frequent single word fillers. Both results suggest the need for a more detailed examination of multi-word fillers to identify possible structural and elaboration patterns that may be more visible with a larger variable frame window. Vincent (2013) also focused on preselected frames, based on pedagogical usefulness. In the study, searches were conducted for fillers with 1-3 words that were expansions of frequent single word filler examples; for example, the frequency of IN **many** WAYS led to a search that identified IN **any number of** WAYS, which was also common. While this analysis of extended fillers was limited and required manual analysis, it illustrated the value of identifying multi-word fillers to uncover a broader picture of frame use.

A less linear approach was taken by Chen et al. (2006), whose concgram methodology offers the flexibility to identify co-occurrences of two or more words with both positional variation (AB, BA) and constituent variation (A * * B, B * * A). For example, the study illustrates the concgram based on the co-occurring words **low** and **high**:

[...] *buy when it's **low** and sell when it's **high*** [...]

[...] *taking advantage of the **low** cost and **high** quality* [...]

[...] *is not preferred for its **high** peak current especially when **low*** [...]

These examples show the value of the concgram method in identifying co-occurrence patterns for words that have a meaningful association at a distance or appear in different orders.

However, whether an instance shows a meaningful relationship requires subjective and time-consuming analysis. The authors explore statistical methods (t-test, mutual information) to highlight more meaningful associations and reduce but find both to have drawbacks.

Results of a concgram search include sequences similar to lexical frames with multi-word fillers, but extensive manual inspection is necessary to remove results irrelevant to that type of study. This is due partly to positional variation being allowed, because when working from a lexical frames perspective, "part of the formulaic nature of the sequence is the fixed word order itself" (Biber, 2009, p. 288). Manual filtering may also be needed because co-occurring words may be part of different constituents, as in the example "buy when it's low and sell when it's high", where the collocates are part of separate verb phrases. Also, because concgram searches generate such a large number of results, the user must begin with associated words already chosen in order to efficiently identify patterns of interest. Though useful for giving a broad picture of word associations, the method is not intended for a frequency-based approach. In particular, the methodology is unlikely to be useful for the function word-based frames that most often occur in lexical frames in academic writing.

Römer (2010) addressed that aspect with a hybrid approach by first identifying frequent lexical frames (referred to as phrase frames or p-frames in the study) with single-word slots using kfNgram (Fletcher 2002-2007). It is important to note that this study took a bundles-first approach, basing the frame search on a previous bundles search. Continuous bundles were also identified using Collocate (Barlow, 2004). Results from both kfNgram (frames) and Collocate (bundles) were then used as starting points for concgram searches to find instances including

positional or constituency variation. For example, the frame IT * BE INTERESTING identified by kFNgram was used in a concgram search, giving examples including:

IT **would** BE **very** INTERESTING

IT **should also** BE INTERESTING

IT **seems to me that it would** BE INTERESTING

Römer's method is valuable for its ability to uncover examples not only with multi-word fillers ("it should also be interesting") but also examples with additional variable slots ("it would be very interesting"). However, this also requires extensive manual contextual analysis to determine which examples are desirable or meaningful, and the definition of what is meaningful is not fully defined. The extended filler approach in the present study is not able to capture all of the items identified by Römer's method because it does not allow for positional variation of the frame components. However, it would considerably reduce the amount of manual checking and the subjectivity of judging meaningfulness of potential extended frames.

While adopting a different method, Warren and Leung (2016) also chose not to limit the number of filler words in the collocational frameworks they explored in their study of "local grammar". This term, first used by Gross (1993), refers to behavior of collocations and the words in their immediate environment which cannot be fully described by conventional grammar rules. The authors began with a concgram analysis of two Hong Kong English corpora, the Hong Kong Engineering Corpus (HKEC, 9.2 million words) and the Hong Kong Financial Services Corpus (HKFSC, 7.3 million words), and a general English corpus, the BNC. The concgram list showed that the pairs any/may and any/shall co-occurred more frequently in Hong Kong English than the BNC, and these pairs were interpreted as four frameworks, ANY + MAY, MAY + ANY, ANY + SHALL, and SHALL + ANY. Their concordance lines were then

examined "to determine whether the co-occurring words were meaningfully associated or simply co-occurring by chance." (Warren & Leung, 2016, p. 4). Meaningful instances were then analyzed both semantically and structurally. Structural analysis was focused on broader categories, for example, the pattern "any + noun phrase + shall + verb phrase", and focus was on meaning and rather than noun phrase structure, for example. Like Römer (2010) this method provides a flexible way of identifying words' co-occurrence relationships but also requires extensive manual checking in order to identify valid or meaningful instances.

Multi-word fillers were investigated to a limited degree in Geluso (2020), a study of the semantic characteristics of the fillers in the colligational framework "THE (**Noun1**) OF THE (**Noun2**)" among L1 Japanese and English writers. Based on the approach in Eeg-Olofsson and Altenberg (1994), an optional one-word modifier is allowed before each of the nouns in the framework, taking the form THE (*) (**Noun1**) OF THE (*) (**Noun2**). This flexibility allows more instances to be captured, but the subsequent semantic analysis targets only the main noun. Along with semantic similarities in the two L1 groups' usage patterns, the results of the study showed that L1 Japanese writers used a larger number of fillers than L1 English speakers, which the author suggests may be attributable to native speakers having a greater awareness of collocational patterns and restrictions.

Hsu (2020) took a different approach by using the Google Books ngram viewer tool to identify lexical frames in Chinese. The study targets 3–5-word sequences, allowing multiple internal variable slots, either consecutive or non-consecutive (e.g. A * B, A ** B, A * B * C). The study adapts the structural classification system of Gray and Biber (2013) to class the 100 most frequent frames as verb-based, content word-based, and function word-based frames, and the syntactic role of the frames is analyzed as well. The results found that in contrast to English

language frames studies, function word-based frames are lower in frequency than other structural types. The authors suggest this may be due to language typology differences, as Chinese lacks equivalents to the English *a/an* and *the*, which are frequent components of frames in academic English. The study also explored how type-token ratio (TTR) loses its usefulness when working with very large corpora. Most relevant to the present study, findings showed that allowing longer fillers increases the number of frame types identified, which despite working with a different language, points to the value of the present study in giving a fuller phraseological picture. However, Hsu's study, like most others in the lexical frames literature, does not provide details about whether all of the multi-word fillers identified are considered to be valid, or if a specific procedure was followed to determine validity. It is hoped that the present study can provide a more fully defined approach to making such determinations.

While limited in number and scope, these studies exploring multi-word fillers in lexical bundles or similar sequences have shed some light on the possibilities of this approach. They have also highlighted some methodological challenges related to identifying and analyzing multi-word fillers, such as a need for extensive manual analysis, and a reliance on sometimes subjective notions of meaningfulness or coherence. The method of addressing these challenges that this study proposes will require a closer look at the syntactic structure of frames and their fillers, and the following section will describe approaches used by previous studies.

2.6. Structural analysis of lexical frames

Gray and Biber (2015) found that previous structural categories used in (continuous) lexical bundles studies were not always appropriate for discontinuous lexical frames due to the fact that the syntactic role of the fixed words can change based on the intervening variable word. For example, the frame HAS BEEN * TO could represent a verb phrase if the filler is a verb (e.g. HAS BEEN going TO), while it could be an adjective followed by a dependent clause (HAS

BEEN **able** TO). To address this ambiguity, the authors created simplified categories describing the role of the dominant fixed word in the frame. Verb-base frames contain one or more modal, auxiliary or main verb (e.g., MUST BE * TO, WAS * IN THE, I * GOING TO, WHAT DID * DO), content-word frames contain one or more nouns, adjectives, or adverbs but no verbs (e.g., ON THE * HAND, IT * NECESSARY TO, I * NO IDEA), and function word frames contain only function words such as prepositions, determiners, conjunctions, pronouns, complementizers, etc. (e.g., THE * OF THIS, IN THE * THAT, AS * AS YOU, A * IN THE) (Gray & Biber, 2015). Their analysis using these categories showed that function word-based frames were most frequent in academic writing (e.g. THE * OF THESE), while verb-based frames were more common in spoken conversation (e.g. I WAS * TO).

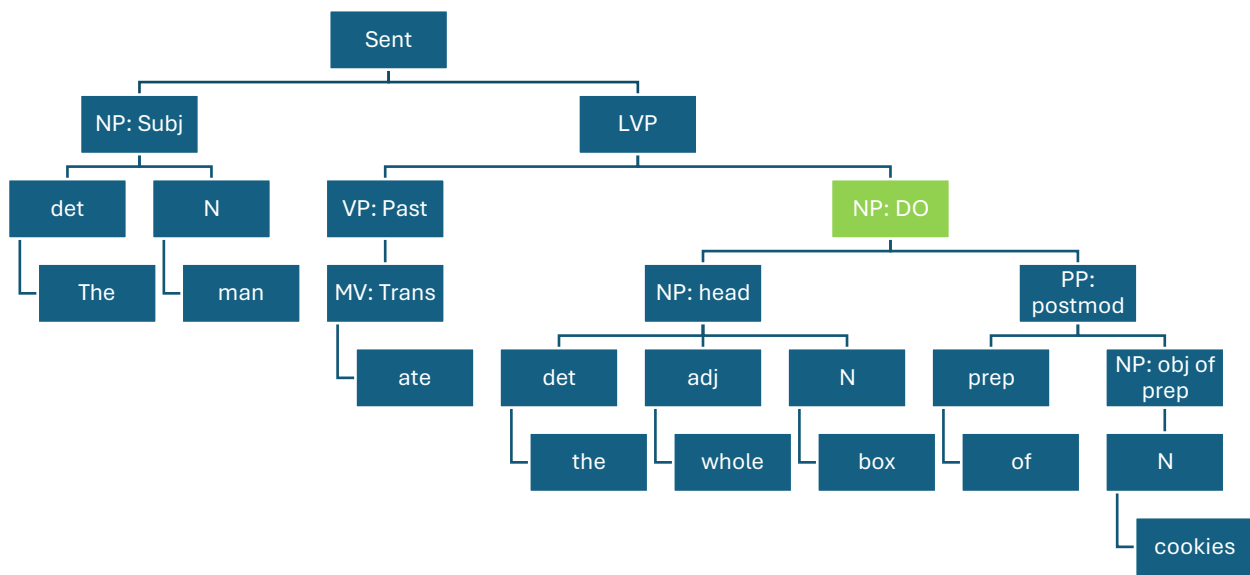
This system of categorization is based on the assumption that only a single word can fill the variable slot in a lexical frame. Therefore, in order to analyze and categorize frames with multi-word fillers it will be necessary to develop a method that considers the structural characteristics of the fillers as well.

2.7. Noun phrases

Previous lexical frames studies have described the structure of fixed components in lexical frames, but because their work focused on single-word fillers, there was no need to analyze the structure of the variable components of the frame. As frame fillers have been found to most often be nouns (Biber, 2009), analyzing multi-word fillers requires consideration of the ways that noun phrases can be structurally elaborated. Additionally, noun phrase complexity has been identified as a marker of proficiency (Parkinson & Musgrave, 2014) and an important marker of register (Biber et al., 2011). In particular, complex noun phrases play an important role in academic writing as a way of enabling a high level of information density (Biber & Gray, 2016).

2.7.1. Noun phrase structure

The approach to sentence structure analysis used in this study is based in constituency or phrase structure grammar (e.g. Chomsky, 1957), in which a single word can function alone, or be expanded into a longer constituent or phrase. From another perspective, a string of words is a constituent if the words can be traced back to a single point in sentence diagram representing a syntactic role, as in the example below where "the whole box of cookies" can be traced back to the direct object noun phrase node.



This means groups of words may behave as a single unit or phrase, which in the case of a noun phrase is sometimes tested using pronoun substitution, such as in the examples:

*The man ate **the whole box of Thin Mints***

*The man ate **it***

This is an important concept when analyzing lexical frames with multi-word fillers because this study takes the position that a frame + filler combination is valid when the filler words represent a single constituent.

Grammatical units such as noun phrases can be described in terms of their internal structure and their syntactic role (Biber et al., 1999). In the case of noun phrases, possible syntactic roles include subject, direct object, prepositional object, and many others. The structural elements of noun phrases include the head noun and optionally a determiner and pre- or post-modifiers. As one of the goals of this study is to understand the structural characteristics of multi-word noun phrase fillers for frequent lexical frames, it is important to consider the various ways in which noun phrases may be modified. Premodifiers include adjectives, participial modifiers, and nouns. Postmodifiers include clauses (relative clauses, *ing*-clauses, *to*-clauses, *ed*-clauses, *to*-infinitive clauses) and prepositional phrases, as well as appositive noun phrases. Another distinct type of modification following head nouns is complement clauses, which include *that*- and *to*-clauses (e.g. the concern that it was a bad idea; the last chance to turn back). Noun phrases can include multiple modifiers, which means a single sentence constituent (e.g. a direct object noun phrase) can potentially include many words, which is one of the primary motivations of this study's interest in multi-word lexical frame fillers. This is particularly relevant for our register of interest because in academic writing 25% of noun phrases have a premodifier, 20% have a postmodifier, and 12% have both (Biber et al., 1999 p. 578)

2.7.2. The noun phrase in academic writing

Noun phrase complexity in academic writing has been the subject of considerable research. Halliday (1989) showed that, in contrast to the clausal complexity of spoken English, academic writing uses elaborated noun phrases much more often, functioning as a means of increasing information density. Biber (1988) provided empirical evidence of these differences, using multidimensional analysis to show that clausal elaboration is more frequent in spoken communication, while noun phrase complexity is a common feature of informational writing.

Subsequent research showed that this type of complexity became increasingly frequent over the course of the 20th century (Biber & Gray, 2016).

Extensive noun phrase modification in academic writing allows for a 'compressed' style of discourse with less explicitness of meaning. This enables efficient communication between expert readers but can cause challenges for less experienced readers (Biber & Gray, 2010). For this reason, noun phrase complexity has been examined as a means of measuring learner proficiency in academic writing. For example, in Biber et al. (2011) the authors show that learners begin with a more clausally elaborated style characteristic of conversation and increase phrasal complexity as they become more experienced academic writers. The stages of complexity development in that study were supported by Parkinson & Musgrave (2014), who showed that more advanced L2 writers used nouns as premodifiers more often than less experienced writers, who used attributive adjectives more frequently. A similar trend was found in L1 writers from first-year undergraduate to graduate level, with increased use of phrasal complexity and decrease in clausal complexity features as L1 writers advance in their post-secondary studies (Staples et al., 2016)

As illustrated by the above studies, the importance of noun phrase complexity in academic writing, as well as the challenges it can pose to novice writers, point to the value of better understanding noun phrase structure as an element in lexical frames with multi-word fillers.

2.7.3. Why the RA register?

The research article (RA) register is a key area of knowledge construction in academia and understanding its writing conventions is essential for participating in the discourse of the scholarly community. This has made RAs an important area of study from a variety of research perspectives. From a phraseological perspective, this register is of particular interest to this study

of extended lexical frames with noun-based fillers. First of all, previous frames research has shown that lexical frames in academic writing most often consist of noun or prepositional phrase fragments (Biber 2009). Additionally, and especially relevant for the concept of extended frames, the RA register is characterized by phrasal complexity, though the degree of that complexity varies by discipline, with hard sciences showing the greatest degree of phrasal complexity and humanities the least (Gray, 2015). This prevalence of complex noun phrases makes academic research articles a fertile area to explore how extended lexical frames may provide a fuller picture of a register's phraseological characteristics.

2.8. Use of natural language processing tools in frames research

Previous studies of lexical frames or phrase-frames have made use of various natural language processing (NLP) tools to identify frames in corpora. Some early related studies used a combination of pre-selected frames, concordance lines, and manual analysis to analyze frame occurrences, though methodological details of the scripts or tools they used were not always provided (Renouf & Sinclair, 1991; Eeg-Olofsson & Altenberg, 1994). Building on the approach of Eeg-Olofsson and Altenberg (1994), Geluso (2020) began with a pre-selected frame (*the N1 of the N2*) and used a custom script to find instances with one or two filler words.

Other studies have relied on custom scripts built by the researcher. Biber (2009) took a more corpus-driven approach, identifying lexical frames based on analyzing the variability of each word position in frequently occurring continuous bundles. Details of how much of this analysis was automated using computational tools was not provided, however. Gray and Biber's (2013) frames-first approach required integration of a MySQL database into the study's custom Perl script in order to handle the amount of data needed for a fully inductive, corpus-driven

approach. More recently, Geluso (2022) also used a custom tool for frame extraction, though it is not clear how the Python script in the study may differ from Gray and Biber (2013).

Other studies have used pre-existing tools for frame identification and analysis. While less customizable, this option offers both convenience and more methodological transparency, as the details of the tool's functionality are more widely available. For example, Fletcher's *kfNgram* (2002-2007) is capable of identifying continuous ngrams or discontinuous phrase frames (equivalent to lexical frames) and has been widely used for frames studies (e.g. Barabadi et al., 2020; Cunningham, 2017; Fuster-Márquez, 2014; Garner, 2016; Golparvar & Barabadi, 2020; Grabowski, 2015; Lu et al., 2018; Römer & Arbor, 2009; Yoon & Casal, 2020). It has also been used by Römer (2010) in combination with other tools, including *Collocate* (Barlow, 2004) and *ConcGram* (Greaves, 2009) to find variants of p-frames with positional or filler length. While Römer's use of *kfNgram* in combination with other tools allowed identification of frames with multi-word fillers (among other structures of interest in that study), it required extensive manual analysis and subjective judgment of acceptability. In addition, the approach begins by identifying continuous forms of interest before exploring discontinuous forms. Similar to the discussion of the bundles vs frames-first approaches in Gray and Biber (2013), this method risks missing some extended frames that may not have frequent continuous forms.

The more recently developed phrase frame capability of *Antconc* (Anthony, 2024), while less widely used than *kfNgram* in frames research (Tan & Römer, 2022; Zhou & Li, 2024), is capable of identifying frames with multi-word fillers. However, frames of a particular filler length must be searched for separately because the program does not conceptualize frames with different numbers of variable slots as belonging to the same overall extended frame based on fixed words. For example, using this tool one might first search for 4-word frames with a single

variable slot, then search for 5-word frames with two variable slots, etc. If the same fixed word pattern is found in each search (e.g. THE * OF THE, THE * * OF THE), by performing multiple searches and manually combining the results from frames with the same fixed words, it would be possible to identify an extended lexical frame as defined in this study. However, the tool is not practical for the ‘extended frames first’ approach this study pursues. Additionally, search results must be inspected manually to determine whether they are meaningful or of interest to a given study, which becomes increasingly difficult and time consuming as filler length and corpus size increase.

While less often mentioned in corpus linguistics studies and more particularly in lexical frames studies, many other natural language processing tools are available as Python libraries. Though they require more background knowledge to implement, they can handle important corpus analysis tasks such as tokenization and PoS tagging, as well as more advanced features such as dependency parsing and model training. For example, the Natural Language Toolkit (Bird et al., 2009) library offers powerful functions for corpus linguistics analysis and advantages over more typically used retrieval tools (Wang & Hu, 2021; Spring & Johnson, 2022) but has been used by few applied linguists (Asención-Delaney & Collentine, 2011; J Ren, 2021; Kostromitina, 2022). Another Python library widely used in natural language processing but not often encountered in corpus linguistics research is spaCy (Honnibal et al., 2017), which offers features comparable to nltk but with some advantages in processing speed. Though it has not been extensively explored in the applied linguistics context, it was found to provide measures of lexical richness in EFL writing that correlated well with human raters and TOEFL ITP scores (Spring & Johnson, 2022). SpaCy is notable for its parsing speed when compared to other tools (Schmitt et al., 2019), which is important for the needs of this study, specifically when checking

structural features of large numbers of extended lexical frame fillers across a full corpus. Its accuracy in part of speech tagging is reported as 97%, similar to the state of the art in tagging, though those numbers can be misleading when taking a closer look, for example at whole sentence tagging accuracy (Manning, 2011) or across less studied domains (Al Omran & Treudi, 2017). It should be noted that spaCy's tagging and parsing features have high reported accuracy, but the model used was trained on the OntoNotes 5.0 corpus, which contains text from news, conversational telephone speech, weblogs, usenet newsgroups, broadcast, and talk shows (Weischedel et al., 2013). Therefore, further research may be needed on its specific accuracy for specialized domains such as academic writing. Additionally, more detailed reporting of accuracy that considers variation across words and tag types is necessary, as the overall accuracy measure is inflated by high-frequency words that are easily tagged (e.g. 'the' is always tagged as an article). With accuracy measures for specific features, tools using the spaCy library could take steps to avoid or remediate limitations of the parser and tagger. While these issues must be considered when using spaCy, limitations such as these are present in all current PoS tagging and parsing tools, and their strengths and weaknesses must be considered in any corpus analysis (Kyle & Eguchi, 2024). In any case, there is a need for greater transparency in choice of NLP library or other tools by researchers, as well as recognition of the ways accuracy of these tools can vary across domains (Al Omran & Treudi, 2017).

2.9. Rationale

The literature review presented here illustrates two important points motivating this study. First, there is a need for more research on longer formulaic sequences, and on formulaic language that also includes variability, allowing for fixed structures to function not only as formula, but also as a foundation for extended language production. Lexical frames with multi-word fillers in particular have received very limited attention but provide an opportunity for just

this sort of broader analysis of formulaic language. Secondly, there are no existing tools or methods to my knowledge that can efficiently identify these sequences based on clearly defined validity criteria.

To address these needs, this study has two overarching goals: (1) to explore to what extent frames with multi-word noun phrase fillers can be identified automatically, and (2) to describe the frequency, distributional, and structural characteristics of frames with multi-word noun phrase fillers in academic journal writing. The following two research questions address the first goal:

RQ1. What level of precision and recall can be achieved with automatic detection of lexical frames with multi-word noun phrase fillers?

RQ2. What syntactic features can be used to distinguish valid and invalid frame variants using existing NLP tools?

The ability to identify lexical frames with multi-word fillers raises additional language description questions that could not be explored using previous methods. Thus, after establishing methods for identifying extended lexical frames, the second goal of the dissertation is addressed through two additional research questions that investigate the frequency distribution and structural characteristics of extended frames in academic writing, and investigate to what extent allowing for multi-word fillers enables a fuller picture of frame use in discourse:

RQ3a. What is the frequency of noun-based lexical frames in academic writing when multi-word fillers are considered?

RQ3b. What is the frequency distribution of filler lengths in lexical frames with multi-word noun phrase fillers?

RQ4. What structures (e.g. *adj + n*, *n + pp*, etc.) are found in frame fillers of different lengths (and in what proportions)?

CHAPTER 3. METHODS OVERVIEW

3.1. Introduction

Previous lexical frames research has focused almost entirely on frames defined as a sequence of frequently recurring fixed words surrounding a 1-word variable slot (e.g. A **proportion** OF THE, A **topic** OF THE). The initial spark for this study was the question of why frame variants with multi-word fillers (e.g. A **large proportion** OF THE, A **relatively small proportion** OF THE) were not included in those analyses, despite these multi-word fillers playing the same syntactic role as the single word fillers.

Exploring that question involved two goals, as laid out in the previous chapter: one methodological and the other descriptive. The methodological goal, discovering to what extent frames with multi-word noun phrase fillers could be identified automatically, began with establishing a definition of what constitutes an acceptable frame variant, as not all possible variants are coherent as a frame (e.g. A **proportion is the ratio** OF THE). The tools normally used in lexical frames studies are not able to identify extended fillers as defined in this study, so the next steps were to develop and test a tool capable of implementing this definition and identifying these extended lexical frames. Together, these steps constitute Phase 1 of this study, and the details of their methods and results will be described in Chapter 4.

The methodological steps of Phase 1 enabled pursuit of the goals of Phase 2: describing the frequency, distributional, and structural characteristics of extended lexical frames in academic journal writing. The steps in this Phase included frequency analysis to discover how common multi-word lexical frame fillers are in relation to single word fillers, as well as structural description of these extended lexical frames. The details of these analyses, as well as the subsequent findings, are described in Chapter 5.

Before moving on to the details of the study's two phases in the following chapters, the remainder of this chapter describes the corpus chosen to represent the target register of academic journal writing, explaining its design, collection, and suitability for this study.

3.2. Corpus

A corpus can generally be defined as "a large and principled sample of texts designed to represent a target domain of language use (e.g., a language, dialect, or register)" (Egbert, Biber & Gray, 2022, p. 7). A key factor in the design of a corpus is representativeness, or "the extent to which a corpus permits accurate generalizations about the quantitative linguistic patterns that are typical in a target language or discourse domain" (Egbert et al., 2022, p. 11). One aspect of representativeness is *linguistic representativeness* (Biber, 1993) (also see *distribution considerations* in Egbert et al., 2022), which refers to the degree to which a corpus represents the range of linguistic features in a domain. In order to achieve linguistic representativeness, the second aspect of representativeness, must first be addressed. *Situational representativeness* (Biber, 1993), or *domain considerations* (Egbert et al., 2022), describes how well a corpus accounts for the range of texts in the target domain. Achieving this involves analyzing nonlinguistic or qualitative characteristics of the domain, identifying boundaries and operationally defining the domain, and sampling from within that domain.

To represent the target domain of published academic journal writing, this study is well served by the use of an existing corpus, the Academic Journal Register Corpus (AJRC) (Gray, 2015). This corpus provides a detailed representation of research articles not simply as the published work of disciplinary experts communicating research results but also accounts for internal variation in the register based on situational characteristics such as research methodology (e.g. qualitative vs quantitative) and data type. To understand and represent this

variation, Gray surveyed and analyzed journals from a variety of disciplines, resulting in a taxonomy of published journal writing with three article types: empirical research reports (both quantitative and qualitative), theoretical articles, and evaluative documents (e.g. commentaries, book reviews). However, given the comparative infrequency of evaluative papers, only empirical and theoretical were included in the corpus.

The selection of disciplines to include was influenced by several factors beyond simply including disciplines from a wide range of academic areas. Other considerations included representing each of the sub-registers (quantitative, qualitative, and theoretical) with at least two disciplines each, and including more than one register from each discipline when possible (no disciplines included all registers and most included a narrow range). Articles were chosen to represent methodological conventions as well, thereby allowing insight into the linguistic variation that may accompany different situational characteristics. To that end, the applied linguistics sub-corpus includes 30 qualitative and 30 quantitative articles, biology includes 30 quantitative articles, history includes 30 qualitative articles, philosophy includes 30 theoretical articles, physics includes 30 quantitative and 30 theoretical articles, and political science includes 30 qualitative and 30 quantitative articles. In addition to qualitative and quantitative research articles, theoretical (non-empirical) research is also included from the disciplines of philosophy and physics. High quality journals were identified based on detailed research of the represented disciplines, including consultation with disciplinary experts, and both generalist and specialist journals are included. Articles were selected randomly from three journal issues (2006, 2007, and 2008) and analyzed to ensure they fit the characteristics of the research domains and methodologies identified in the situational analysis.

The finalized corpus of 270 research articles includes 1,952,568 words from three registers (theoretical, quantitative, and qualitative research articles) and six disciplines, including: applied linguistics (60 articles), biology (30 articles), history (30 articles), philosophy (30 articles), physics (60 articles), and political science (60 articles). This set of texts presents a picture of academic journal writing across a variety of disciplines and research approaches, including hard sciences (biology, physics) as well as social sciences (applied linguistics, political science) and humanities (history, philosophy). Given the careful approach to achieving situational representativeness, enabling linguistic representativeness and thus generalizability, this corpus is well suited for the present study's goals, and serves as valuable data set for analysis of the broader phraseological characteristics of academic journal writing.

This corpus was used both in Phase 1 and 2 of this study. In Phase 1, a 10-text subset of the corpus was used for development and testing of a Python script for identifying extended lexical frames. In Phase 2, the script was used to analyze the full corpus, identifying the full set of noun-based extended frames occurring above a set frequency level. Details of both follow in the next two chapters.

CHAPTER 4. PHASE 1 METHODS AND RESULTS

Phase 1 of this study was focused on the methodological questions involved in identifying extended lexical frames in a corpus. This involved several steps, including establishing an operational definition for extended lexical frames, developing a Python script to identify the frames in a corpus, and testing the script's accuracy. This chapter details these methodological steps, as well as reporting the results of accuracy testing, addressing research questions 1 and 2.

4.1. Operationalizing the target feature

4.1.1. Defining extended lexical frames

Most previous frames researchers have defined lexical frames as a set of fixed words with a single variable slot, usually internal. Many of these studies have focused on frames with three fixed words and a single variable slot, such as the frame THE * OF THE, which encompasses variants such as THE **beginning** OF THE, THE **end** OF THE, and THE **start** OF THE (e.g. Römer, 2010; Fuster-Márquez, 2014; Gray & Biber, 2015, etc.). However, this study takes a new approach, specifying only the number of 'fixed' words in the frame and allowing the variable slot to include more than one word. In this study's definition, the previously mentioned frame could therefore include variants such as THE **true beginning** OF THE and THE **beginning and end** OF THE as well. This new structure will be represented as the sequence A + B C, with the letters A, B, and C representing fixed words and the 'plus' sign indicating 1 or more filler words, in contrast to the asterisk (*), which represents a single word filler. This means, for example, that the pattern A + B C encompasses A * * B C, A * * * B C, etc. when they contain the same fixed words (A, B, C) in the same positions.

However, not all sequences of words beginning with THE and ending in OF THE are included as variants of the frame THE + OF THE, as they may not represent a coherent unit. For example, in the sequence THE **car was parked in the front** OF THE, the filler words include a subject noun, verbs, and part of an adverbial phrase. Because this study is focused only on frames with noun phrase fillers, this sequence would not meet the criteria to be included. Frame variants are distinguished by their filler words, and for this study the structural characteristics of those filler words determine whether a frame variant is included in the analysis. A filler not meeting the structural criteria described in this study is considered to be **invalid**. The following sections will go into more detail about the criteria for **filler validity** used in this study.

4.1.2. Defining filler validity

Lexical frames can be filled by a variety of word types, though the types of fillers most often found vary by register. For example, in spoken academic lectures, frames are most commonly made up of content words (nouns, verbs) in fixed positions, often with verbs occupying the variable slot, for example I AM GOING TO [**show/tell/ask**] YOU. In academic writing, however, nouns are the most frequent type of filler word, with function words in fixed positions (Biber, 2009), such as in the common form THE [**case/context/form**] OF THE. In addition to their frequency as frame fillers in academic writing, nouns are particularly well-suited to this study's focus on longer fillers, as they can be expanded with a variety of pre- and postmodifiers. This type of noun phrase complexity has shown increasing prominence in academic writing (Biber & Gray, 2016), which suggests the value in exploring the use of complex noun phrases as lexical frame fillers. Based on these points, this study was limited to identifying only frames with noun-based fillers.

With the decision to focus on multi-word noun-based lexical frame fillers, it was necessary to operationalize the concept of a valid filler based on structural criteria, both to enable computational analysis and facilitate replicability of the methods for future frames studies. Validity criteria were developed through a process of diagramming example sentences and identifying constituent boundaries (see following sections for details), with a focus on both more typical structures and "edge cases" that might challenge the criteria. In particularly challenging cases, a disciplinary expert was consulted. Validity criteria were further revised and refined after manually coding lexical frame examples from small test corpus (~75,000 words) to create a 'gold standard' for later testing of the Python script (see Section 3.4).

For the purposes of this study, the filler in an extended lexical frame must occupy the same syntactic role as the filler in a **single slot** frame, so the filler words must belong to the same sentence constituent. In the case of the frame THE + OF THE, only nouns are found to occupy the slot, so a longer filler in this frame should contain a single, though possibly complex, noun phrase in order to be valid. For example, 'the **nature** of the', 'the **arbitrary nature** of the', and 'the **politically charged nature** of the' would all be considered variants of the frame THE + OF THE because the filler words belong to the same constituent, in this case a noun phrase with or without premodifiers. On the other hand, examples like 'THE **experimenter kept records** OF THE' would be excluded. In this case, there are three separate constituents in the filler: '**experimenter**' is the subject noun, '**kept**' is the main verb, and '**records**' is a direct object noun. The fixed elements THE and OF THE do not belong to the same constituent; THE is the determiner for the subject noun '**experimenter**', while OF THE function as part of a prepositional phrase postmodifying the object noun '**records**' rather than the subject noun "**experimenter**".

The following sections will detail characteristics of valid fillers in the context of more specific and complex structures. These examples will later be used in implementing validity checks in the Python script (see Section 3.4.3).

4.1.2.1. *Presence of a noun*

At a minimum, a valid filler must contain a noun or pronoun. However, some other forms may serve as nouns at times as well. For example, cardinal numbers, though most often used as determiners, can also be used as nouns, as in the example "*in **three** of the four articles*". Ordinal numbers can take on a nominal role as well, as in "*the **first** of the month*". Semi-determiners can also serve noun roles, as in "*the importance of the **former** in relation to the latter*", as can adjectives, as in "*the impact of these events **ON THE young AND the elderly** in the city*". Clauses can also fill a nominal position in a sentence, as in complement clauses like "*they dislike **going to the store***". In this case ‘going to the store’ is a non-finite clause appearing where a nominal would normally be, as the object of the verb ‘dislike’. In all of these cases, a filler is considered valid only if the entire constituent is contained within the variable frame, between the fixed words. This may be a noun phrase, pre- and post-modifiers included, or an entire clause taking a nominal role in the sentence.

Given the presence of a noun or noun-like word in a filler along with possible premodifiers (nouns, adjectives) and postmodifiers, some special cases that may affect validity will be described in more detail in the following sections.

4.1.2.2. *Noun phrases containing conjunctions*

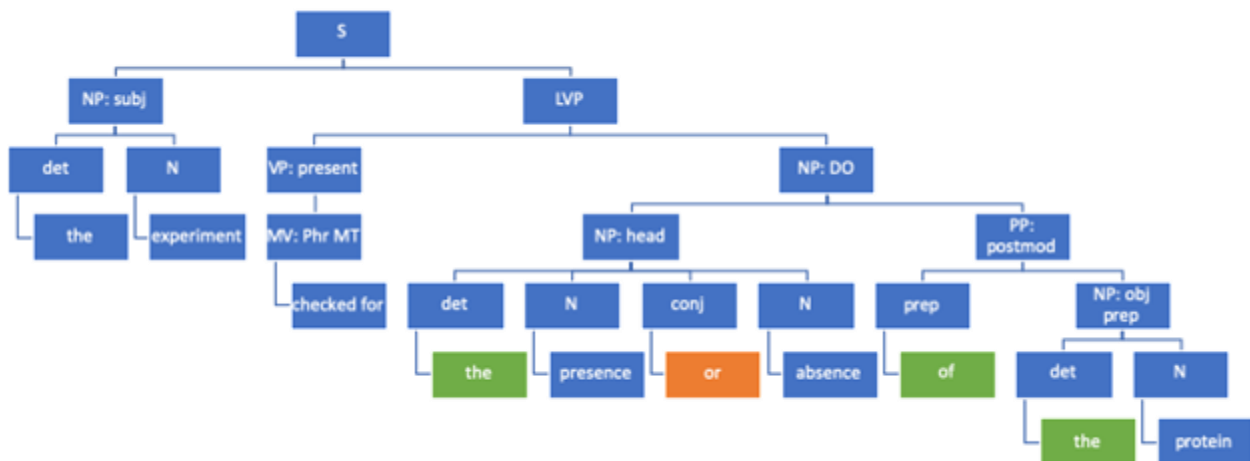
One case in which a noun phrase can contain a conjunction is with coordinated premodifying adjectives (e.g. *strict and systematic discipline*), a structure that occurs more commonly in academic writing than in other registers (Biber et al., 1999, p. 601). This type of

structure is valid as a frame filler as long as all premodifying words occur within the filler, though there were some minor complications when using automatic POS tagging (see Section 3.4.3.1).

In some situations, two (or more) nouns joined by a conjunction can occupy the same constituency level, or syntactic role in a sentence. For example, in "He liked **THE beginning and the end** OF THE movie, but not the middle", the words *beginning* and *end* are both functioning as direct objects of the verb *liked*. Also, both are postmodified by the prepositional phrase *of the movie*, so the fixed words 'OF THE' attach to the same constituent as the initial fixed 'THE'. Coordinated noun phrases of this form are considered to be valid fillers given that the coordinated words occupy the same constituency level, and in this example the first and last fixed elements are attached to the same constituent. However, determining the validity of forms like this often requires analysis of the wider context of words outside the frame. To illustrate, although both examples below contain two nouns separated by a conjunction within the filler, only Example 1 is valid.

1. The experiment checked for **THE presence or absence** OF THE protein (VALID)

In this example, both 'presence' and 'absence' function as direct object nouns for the phrasal verb 'checked for'. Additionally, 'OF THE' is part of a prepositional phrase that postmodifies both nouns. Based on this evidence, the filler represents a single constituent, and both fixed elements are part of this constituent.



2. There is no reason to publicize THE **issue and the name** OF THE sinner
(INVALID)

In Example 2, though the filler has the same structure, two nouns separated by a conjunction, it is invalid because the two nouns and the two fixed frame elements belong to different constituents. The first 'THE' belongs to the noun phrase headed by '**issue**', while 'OF THE' is part of a prepositional phrase postmodifying only the noun '**name**'. In other words, this is the 'name of the sinner', but not the 'issue of the sinner'. The conjunction '**and**' is not part of either noun phrase, but instead a part of the verb phrase, so its head is 'publicize', which lies outside of the frame.

4.1.2.3. *Adverbials*

Due to the positional flexibility of adverbs, special attention had to be paid to their effect on the validity of a frame. When an adverbial occurs within a filler, it is considered valid when it modifies a word within the filler. For example, adverbs may modify adjectives, so Example 1 below would be valid, as "very" modifies the participial adjective "filling". However, in the

second example the adverb "only" modifies the verb "acceptance", which is not a part of the filler.

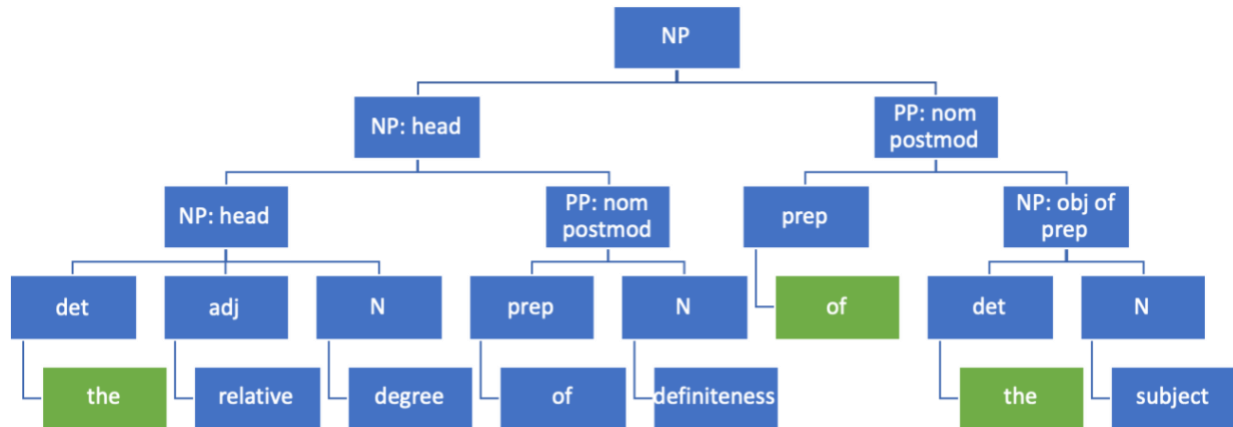
- 1) they ate A **very filling meal** IN THE restaurant (VALID)
- 2) acceptance OF **only the simpler features** OF THE usage (INVALID)
- 3) IT IS **hardly a fruitful line** OF investigation (INVALID)

Because some relatively frequent frames contain a verb in fixed position, it is possible that an adverb in the filler could modify one of the fixed words, as in Example 3, where the adverb "hardly" modifies the verb "is". It was decided that this type of form would be considered invalid for the purposes of this study as the focus is on noun phrase-based fillers, and in this case one of the filler words does not belong to the noun phrase.

4.1.2.4. *Postmodified nouns: prepositional phrases*

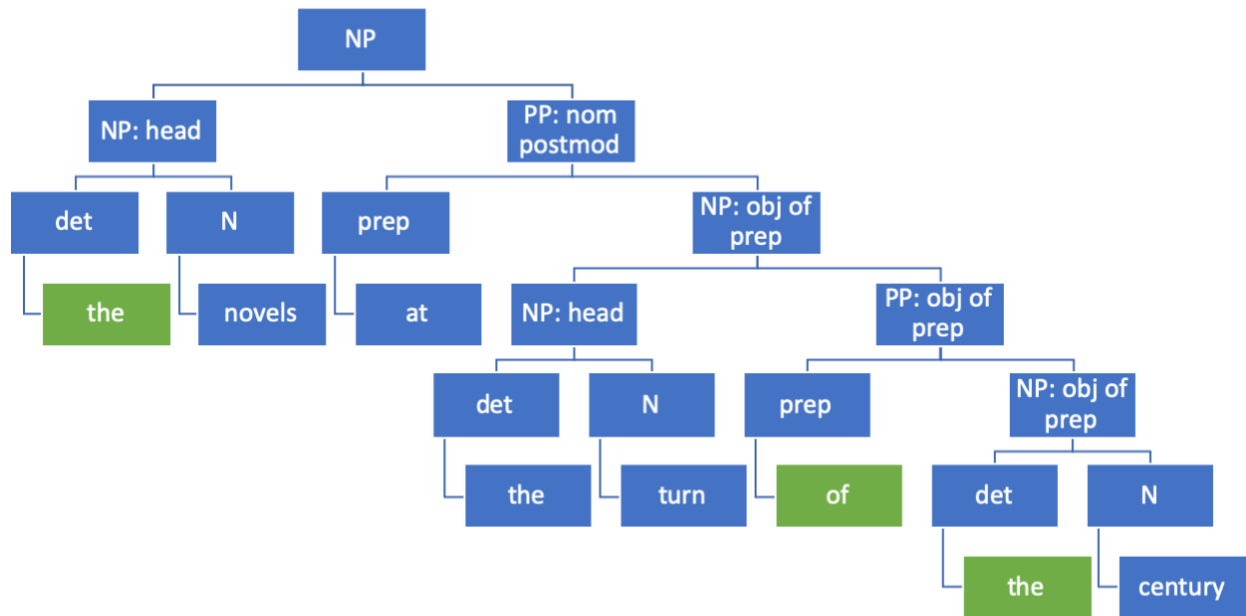
By far the most common type of noun postmodification in academic writing is the use of prepositional phrases (Biber et al., 1999, p. 606). For the purpose of this study, if a prepositional phrase occurs within a filler, it must be complete in order for the filler to be valid. However, determining this is not always straightforward, as it can sometimes be ambiguous where a prepositional phrase is attached. One test of validity is whether the initial and final fixed components appear to belong to the same constituent. In example 1 below, two groupings could be argued for: *relative degree of (definiteness of the subject)* or *(relative degree of definiteness) of the subject*. There is not a definite way to determine which is correct, though in the judgment of this author, the second option seems more accurate, as the word "degree" (in the sense used here) is less likely to stand alone than the word "subject". Given the challenge in making an

informed human judgment in cases like these, it is no surprise that it would offer challenges in automatic identification (see further discussion in Section 3.4.3.9).



In example 2, in contrast, the final “OF THE” modifies “turn”, which is not the head noun but part of a prepositional phrase postmodifying the head noun. Therefore, the final fixed words belong to a different constituent, making the example invalid.

2) THE **novels at the turn** OF THE century (INVALID)



It should be noted also that examining the full context of the sentence is important, because a prepositional phrase such as “at the turn of the century” could also function as an adverbial (e.g. *the book was written at the turn of the century*) and given the flexibility of placement of adverbials (Biber et al., 1999, p. 763), this introduces additional parsing challenges. Overall, these examples illustrate the importance of looking beyond the surface similarity of filler structures in order to identify relationships among the constituents and determine filler validity.

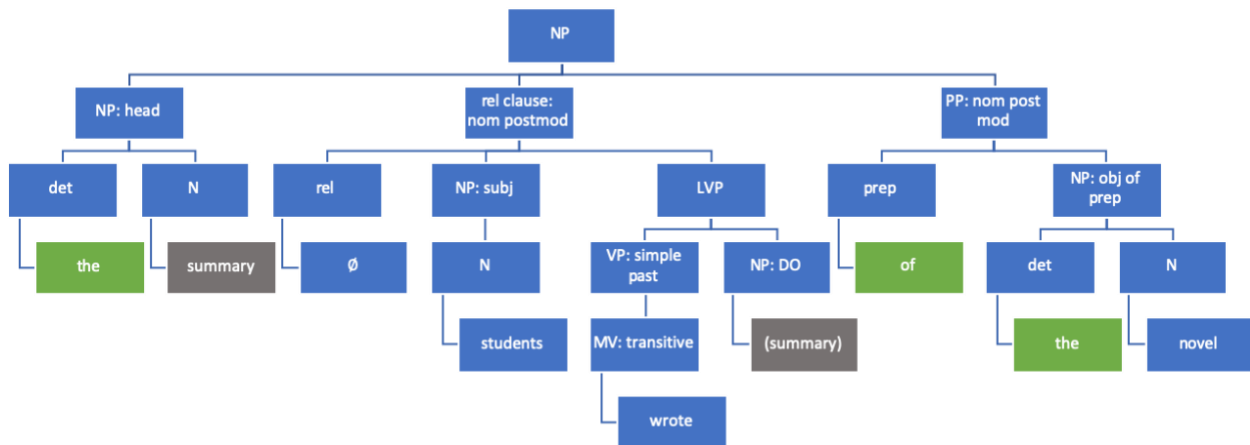
4.1.2.5. *Postmodified Nouns: Clausal postmodifiers*

Nouns can be postmodified by finite (relative) clauses and non-finite clauses (-ing, -ed, and to- clauses) (Biber et al., 1999, p. 575). Fillers with any of these types of modification will

be considered valid if the entire postmodifying clause is contained within the filler and the fixed components of the frame belong to the same constituents.

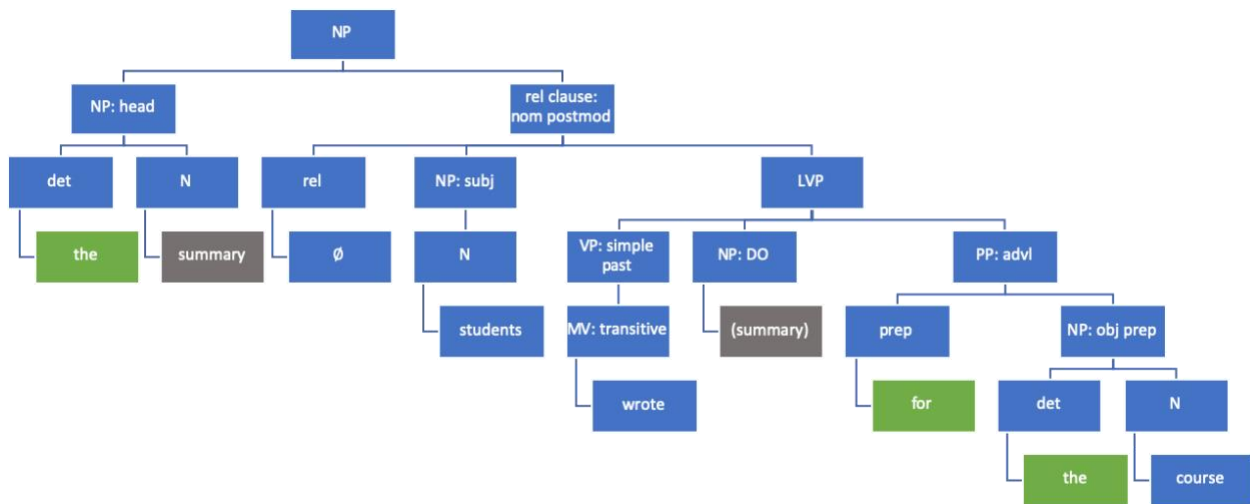
For example, the filler in the example below contains a noun (summary) with a relative clause as a postmodifier (the students wrote). The ending fixed components “OF THE” are the beginning of a prepositional phrase postmodifying summary, so they belong to the same constituent as the initial “THE”, meeting the validity criteria.

THE summary students wrote OF THE novel (VALID)



The next example below, however, is invalid because the ending of the frame (FOR THE) begins an adverbial that modifies the verb “wrote”, so it does not belong to the same constituent as the initial “THE”

THE summary students wrote FOR THE course (INVALID)



Determining whether a postmodifying relative clause is complete within the filler is not always straightforward, however. This is related to the issue of prepositional phrase attachment ambiguity, as in examples like:

We were constrained by THE information that was recorded in the patients' charts AT THE time of replantation

This example may or may not be valid depending on whether "at the time of replantation" is modifying the verb "constrained", or the verb "recorded". In the first case, the filler is valid because the postmodifying clause would include only the words " that was recorded in the patients' charts ", all of which are contained within the filler. If "at the time of replantation" modifies "recorded", however, the relative clause extends beyond the filler, making it invalid. This type of ambiguity prevents a fully automated approach to validity judgment, and even human decisions may be less than 100% certain.

4.1.2.6. *Postmodifiers: appositive noun phrases*

Another type of noun postmodification is appositive noun phrases, in which a second noun phrase, usually enclosed in parentheses in academic writing contexts, provides a gloss or paraphrase of the head noun. This may include explanations of the head noun, introduction of acronyms, or giving examples. Appositive noun phrases include job titles (*Joe Biden, the president of the United States*), acronyms (*The Corpus of Contemporary American English (COCA)*) and technical glosses (*the optical propagation direction (z-direction)*). Despite their presence in academic writing (~15% of noun postmodification in the register) (Biber et al., 1999, p. 638), appositive noun phrases as postmodifiers were not fully included in this study, as any text of 3+ words inside parentheses was not considered, and comma boundaries were not crossed. However, it might be argued that due to their equal syntactic status with the head noun, appositives do not represent the same type of syntactic complexity that this study has chosen to focus on.

4.2. Script development

The Python scripts developed for this study were designed to accomplish several steps, taking a corpus of individual text files and processing them in order to identify lexical frames with variable length fillers. During the development process, computational efficiency was a key consideration, as allowing multi-word fillers for lexical frames results in a much greater number of potentially valid instances compared with traditional frames studies. Each of the steps in the process, from text cleaning to script evaluation, will be described in detail in the following sections.

4.2.1. Text Cleaning

Before identifying lexical frames, the text files were cleaned to remove irrelevant text that might interfere with processing. Because this study uses an existing corpus, some aspects of cleaning and processing of texts was already done. In the development of the AJRC, Gray (2015) removed headers and footers from text files, though footnotes were preserved and moved to the end of the text to avoid interrupting body text. Reference lists, figures, tables, and (non in-line) formulas were also removed.

The first step of text cleaning for this project was removing metadata, with several additional steps that were tested iteratively to best allow identification of frames later (Table 3.1 below). Some cleaning steps were simple and follow common approaches from other studies. However, in some cases cleaning decisions were not straightforward. For example, phrases of the form "pre- and post-testing" were found in the sample corpus texts, and the dash on "pre-" caused a problem with SpaCY's tagger, which tagged "pre-" as an interjection. Removing the "-" caused the token "pre" to be tagged as an adjective instead. This is not correct by most standards, but because "pre-" is not a free morpheme, tagging it is difficult and the tagset used by spaCy (Penn Treebank) does not contain a tag for prefixes. However, for the purposes of this study, the adjective tag will be acceptable. More importantly, at least with the default settings for the tokenizer for the English model used (*en_core_web_lg*, see Section 3.4.3 for more detail), the word "post-testing" was split into three tokens: "post", "-", and "testing". For this reason, hyphens followed by a space ("pre- ") were removed, while hyphens surrounded by characters ("post-testing") were replaced with a space, creating two separate words. While some studies treat hyphenated words as a single word, counting such examples as separate words helps to more easily illustrate noun phrase complexity, which is a key topic of interest in this study. Additional cleaning steps are detailed in Table 3.1 below.

Table 3.1*Steps for cleaning texts before extracting ngrams*

Cleaning step	Reason
Delete text between angle brackets ('<>')	Remove unnecessary metadata
Replace double white space with single	Some writers use two spaces after period, for example, which interferes with tokenization in spaCy
Remove parentheses and their enclosed text, including nested parentheses, if either >3 words or a date are present	Remove text that is not part of the main body, as well as citations (operationalized as parentheses containing a 4-digit date). If < 3 words are present, it may be something like "(in)convenient" or "the (very) hungry caterpillar"
Replace double quotes (3+ words only)	Quoted material doesn't represent the author's own writing, so it was left out. When 1-2 words are enclosed in double quotes, it serves a different function (e.g. <i>they chose the "best" plan</i>)
Replace single quotes (checks that previous character is a space to avoid apostrophes)	same as previous, but single quotes are used in some situations such as nested quotes
replace dashes ('--') with newline ('\n') (e.g. "[...] native Chinese speakers commonly reject or deflect compliments or use circumlocution to avoid saying no -- speech acts in Chinese that require the speaker to take into account [...]")	Lexical frames cannot cross dash boundaries, so treat parts separately
Replace : ; , with newline ("\n")	Constituents cannot cross colon and semicolon boundaries; commas within constituents are rare (e.g. "the bulky, heavy box")
Split into sentences	Uses spaCy's sentence segmentation function, which makes use of the dependency parser to identify sentence boundaries
Split strings by newline character ("\n")	Separate headings from other text
Count tokens and increment word count total	Word count needed for frequency normalization later

After initial cleaning, the text was split along sentence boundaries using spaCy's sentencizer tool, a rule-based sentence boundary detector that uses a customizable list of sentence-final punctuation (Honnibal & Johnson, 2014). Accuracy measures of the sentencizer tool are not reported in the spaCy documentation, but because this study does not allow lexical frames that cross punctuation boundaries, it was acceptable for this purpose. SpaCy also provides a more accurate sentence boundary detector that makes use of the dependency parser, but it is much more computationally expensive and would be unnecessary for the needs of this study. While previous lexical bundles researchers have chosen to avoid crossing other punctuation boundaries such as commas (e.g. Gray & Biber, 2015) the initial pilot version of this tool for this project did not. This decision was made in order to aid in identification of complex coordinated noun phrases such as "*comfort, happiness, and accommodation*" and coordinate adjectives such as "*bulky, heavy box*" which may represent valid fillers. However, during piloting it was found that the number of valid lexical frames containing commas was very small compared to the number of false positives resulting from this choice. Therefore, in the final analysis comma boundaries were not crossed when identifying lexical frames, in line with previous approaches. In addition, sentences were split at occurrences of semicolons and colons, as they are not typically used within sentence constituents.

4.2.2. Frame database creation

After text cleaning, ngrams (a continuous sequence of n words) were extracted from each cleaned string using the Natural Language Toolkit's (Bird et al., 2009) ngrams method. In the piloting stage, ngrams of all lengths 4 words or more were extracted. For example, a sentence ten words in length would yield seven 4-grams, six 5-grams, five 6-grams, four 7-grams, three 8-grams, two 9-grams, and one 10-gram. However, results of piloting showed no valid fillers

longer than 9 words (and only a very small number more than 6 words), so in order to reduce memory load and computing time the final version on the script was limited to a maximum of 12-word sequences (3 fixed words + 9 filler words).

When each of these ngrams were identified, they were stored using SQLite, an open source, embedded relational database (Hipp, R.D., 2020). Each entry was stored as a 3-item tuple, with the ngram, the clause in which it originally appeared, and the name of the file in which it was found. Using the context of the full sequence/clause allowed the parser to work more accurately to identify the dependencies needed and judge whether the sequence represents a valid lexical frame + filler (see Section 3.4.3). The file name was important to include as well for calculating range (the number of texts in which a frame appears), and information encoded in the file name enabled further detail, such as disciplinary breakdown of results.

Early versions of the script saved output in Excel spreadsheets, and then JSON files, but storing the data using SQLite was later found to provide several advantages. First of all, SQL database files allow much more data than Excel files, which are limited to 1 million rows. In most corpus studies of this type, that limit will quickly be exceeded. For example, a single sample text with 5,318 words resulted in 57,626 4+word ngrams, so analyzing a typically sized corpus would not be possible using .csv files. Additionally, SQL databases allow for quick access, making later analysis more efficient, and they can be built in batches, which was found to be important given the long processing times needed for large corpora (see further discussion in Section 3.4.3).

Using the SQLite database of continuous ngrams, all frames containing three fixed words and one internal variable slot were identified by taking the first, second, and last word of each ngram (A B + D) and the first, second-to-last, and last (A + C D), creating two frames. For

example, given a database entry of the ngram "the cookies were delicious", we would get the frames THE COOKIES + DELICIOUS and THE + WERE DELICIOUS. This approach allows for longer fillers as well, so "the chocolate chip cookies were delicious" would also give the A + C D frame THE + WERE DELICIOUS in addition to the A B + D frame THE CHOCOLATE + DELICIOUS.

These frames were saved to a second SQLite database, this time with more detailed information for each entry. In addition to the *frame*, *context*, and *file*, frames database entries also included the *filler*, *validity*, *catch*, and *filler pattern*. *Filler* includes the words that fill the frames variable slot in the particular instance. *Validity*, which is initialized represents the decision of whether the frame + filler combination is valid according to the operational definition described in Section 3.3.2. This value was initialized as 'UNKNOWN' when the frames database was initially created, but was later updated by the filtering script. The *catch* value represents which of the filtering script's functions 'caught' the entry as invalid. This is initialized as 'N/A' and will only be changed if the frame is found to be invalid. Finally, the *filler pattern* value was created to store the string of POS tags representing the frame's filler, which will later be used to answer questions about structural patterns that appear in extended lexical frame fillers.

4.2.3. Identifying valid fillers

An important difference between this study and traditional frames studies is that when multi-word fillers are allowed, it is necessary to distinguish fillers that are valid from all other sequences that may be enclosed by recurring frames. Doing this manually would be highly impractical, as even a modestly sized corpus would have millions of potential frame + filler combinations to check. Thus, finding an automated solution to this challenge is a key part of this study.

In order to identify valid fillers, this study made use of the part-of-speech tagging and dependency parsing features of spaCy. More specifically, the study uses spaCy's largest English model, `en_core_web_lg`, a statistical model trained on web text. According to the documentation for the spaCy library, this model's POS tagging has an accuracy level of 0.97 and dependency tagging accuracy is 0.92 (Facts & Figures, n.d.). A more detailed breakdown of POS tag accuracy would be valuable, as even state-of-the-art taggers have issues with unfamiliar vocabulary or some linguistic forms that are sometimes hidden due to the higher frequency of more easily tagged common words (see Manning, 2011); however, this information is not provided for this tagger. Some mis-tagging issues were noticed during piloting and taken into account when developing the script, but limitations in POS tagging must be recognized as a limitation in this study.

Validity rules were implemented in the Python script as a series of individual functions designed to remove or “filter out” invalid frame + filler combinations (See Appendix A for details of the code)². For example, as the scope of this study is limited to noun phrase-based fillers, the most basic requirement for a valid filler is that it must contain a noun. Therefore, the first filter in the script marks as invalid any filler that contains no nouns. Any filler that passes this check is then checked against additional rules. Fillers that pass through all of the checks are assumed to be valid, subject to a final human check for criteria that could not be checked using the available NLP tools (e.g. nested prepositional phrases, clausal postmodifiers)

² During the piloting process and subsequent revisions a constituency parsing tool, Constituent TreeLib (Halvani, 2023) was found. In theory, such a tool could simplify the analysis process considerably, as the operational definition of a valid filler is related to constituent boundaries. However, in practice this tool was not suitable for the project. Accuracy, while reasonably good, was lower than the hand-coded filtering algorithm that had already been developed. More importantly, the tool was much slower (> 10x), to the degree that analyzing a corpus of this size would be impractical, or even impossible. For this reason, development continued with the original approach, taking advantage of POS tags and dependency tags to develop rule-based validity checking functions.

The following sections described the how the operational definition presented in section 3.3.2 was implemented in functions using spaCy (Python code can be found in Appendix B).

4.2.3.1. Filter function 1: No Nouns

The first, and most essential, requirement for a valid filler in this study (see Section 3.3.2.1) is the presence of a noun. Therefore, the first function in the validity filtering script marks as invalid those entries that do not contain any nouns. To do this, the function first test for any words with POS tags for noun, pronoun, or proper noun.

In some cases, words that are not strictly nouns serve a typically nominal role in a clause. For example, a word that is tagged by spaCy as a verb (running) can act as the head of a noun phrase (the **running** of the bulls), as can numbers ('in **four** of the experiments'). These types of forms are acceptable when judging validity in this study, and to check for them, so identifying them requires looking at both a word's POS tag and its dependency tag. Dependency tags describe the role played by a given word in relation to its head, and all words other than the root of a sentence (which is most commonly the main verb). For example, in the sentence "They made biscuits on Sunday", the root is the verb "made". The word "they" would have the dependency tag "nsubj" because it is the subject of the verb "made" (more specifically it is a nominal subject, as clauses can also take a subject role).

Therefore, the second check was for verbs and numbers with the dependencies described in Table 3.3 below.

Table 3.3

Dependency tags for roles typically filled by nouns

Dependency Tag	Meaning (adapted from Choi & Palmer, 2012)	Example (tagged word bolded)
dobj	direct object	<i>She likes napping</i>
pobj	object of a preposition	<i>They looked at the painting</i>
nsubj	subject noun	<i>30 was the target number</i>
csubj	clausal subject	<i>Making biscuits is easy</i>
attr	a noun phrase predicate of a copular verb	<i>He is a teacher</i>
nsubjpass	subject noun in a passive construction	<i>Much was learned</i>
pcomp	clausal complement of preposition	<i>She is good at writing</i> <i>He had a new approach to solving the problem</i>

4.2.3.2. Filter 2: Last filler word check

One relatively straightforward validity function implemented in the script involved checking the final word of a filler. Because a valid filler must contain a noun with optional pre- or post-modification, with the full constituent contained within the bounds of the filler, there are some word types that indicate incompleteness. For example, the dependency tag 'mark' in spaCy is applied to words that introduce a subordinate clause, such as "if", "because", and "that" (e.g. *they knew that the answer was wrong*). If a filler ends with a word from this category, it is necessarily incomplete, so any filler ending with one of these is invalid. Another dependency that indicates incompleteness is the tag 'expl', which is used for existential "there" (e.g. "there is a problem here") and dummy "it" ("it is going to rain").

In addition to dependency tags, some POS tags were used as a sign of a filler's incompleteness. These included coordinating and subordinating conjunctions, adpositions (prepositions and postpositions), and determiners. None of these can serve as the final word in a

noun phrase, with the exception of cases where a preposition is part of a phrasal verb (e.g. "slowing down") that is part of a nominal, as in the sentence "He saw THE **car that broke down** IN THE rearview mirror". In this case, the noun "car" has a clausal postmodifier containing a phrasal verb, so the filler ends with a preposition. To allow for this type of structure, when an adposition is found as the final word of a filler, the function checks whether its POS tag is 'prt' (particle), which is used by spaCy for prepositions that combine with verbs to form phrasal verbs.

4.2.3.3. *Filter 3: First word invalid*

Another relatively simple function checked the POS tag of the first word in a filler. If the word is a preposition, the filler is marked as invalid because noun phrases cannot begin with a preposition.

4.2.3.4. *Filter 4: Nouns controlled by adjectives*

Because this study is focused only on noun-based fillers, adjective phrases, though they may contain nouns, are not considered as valid fillers because they are headed by an adjective rather than a noun. Checking for adjective phrases was necessary because the first filter, which checks for the presence of nouns, would not mark as invalid those fillers that contain a noun that is part of an adjective phrase, which can consist of an adjective modified by adverbs ("considerably better"), comparative phrases ("better **than him**"), or comparative clauses ("better **than he thought**"). In the context of lexical frames, and based on a pilot analysis, this type of structure most often occurs when an adjective serves as the complement of a copular verb. For example, in the sentence "The second group of participants WERE **less proficient than the first group** ON THE speaking test" the adjective "proficient" is the complement of the verb

"WERE" and no noun is required for completeness. There is a noun in the filler ("group"), but it is part of the adjective phrase, so does not meet the validity criteria for this study.

To identify this type of structure, a function was created to first find all nouns and adjectives in a filler. Then a temporary list is made of all descendents of those adjectives, meaning all words directly or indirectly descended from a given word. In this case, the adverb "less" and the prepositional phrase "the first group" are modifying the adjective "proficient", so all of these words are descendents of "proficient". Next, the function checks whether the previously identified nouns are part of this list of descendents (adjective phrase components). If all of the nouns in the filler are part of adjective phrases, this means the filler is invalid.

4.2.3.5. Filter 5: Adverbials

Based on the definition of valid fillers used in this study (see Section 3.3.2.2 for more detail), adverbs can only be present when they modify an adjective (or another adverb) that is part of the noun phrase. The filtering function created to check for these forms marks as invalid any entry containing an adverb that does not meet these criteria. Some less straightforward but valid cases arose when words tagged as verbs function as adjectives, such as in the phrase "the above mentioned methods", where "mentioned" is a verb serving as an adjective premodifier of "methods", and the adverb "above" modifies "mentioned". This required checking the dependency tags (dep = 'amod') as well as POS tags.

4.2.3.6. Filter 6: Adj after noun

Another type of invalid filler structure that needs special attention is complex transitives, which may result in structures with an adjective following a noun, as in "the work drove THE

man crazy IN THE end". Although the filler here only consists of a noun and an adjective, it is invalid because "crazy" is not modifying "the man". In spaCy, this structure is sometimes mistagged, however. For example, while the adjective 'crazy' is correctly tagged as an object predicate when the direct object is a pronoun like 'him' ("the work drove **him** crazy"), it is instead tagged as an adverbial clause when the preceding noun is not a pronoun, suggesting that the word 'crazy' is modifying the verb 'drove'. When the verb 'made' is substituted for "drove" ("the work **made** him crazy"), despite the similar meaning, 'crazy' is tagged as a clausal complement when nouns, complex nouns, or pronouns are in the preceding position.

Based on these problems, a special function was needed to correctly identify this type of structure, a function was created that looks for all nouns in the filler, then checks whether its head is an adjective and whether that adjective has the 'ccomp' dependency tag. Although this was not found to be very common during piloting, it was straightforward to identify and mark such forms as invalid fillers.

4.2.3.7. *Filter 7: adj postmod bad*

While the previous filtering function identified invalid forms when an adjective follows a noun, this function checked for valid structures with an adjective following a noun. For example, 'heavy' multi-word adjective phrases sometimes appear after the nouns, as in the sentence "the man **responsible for the work** was crazy". This type of form is valid if the entirety of the postmodifying adjective phrase falls within the bounds of the lexical frame filler; however, as with some other postmodifiers (see discussion in next section) it is sometimes difficult to automatically identify the boundaries of these phrases. This is particularly the case when the fixed words following the filler represent the start of a prepositional phrase, due to the ambiguity

of prepositional phrase attachment (see Section 3.3.2.5). For example, in "We saw THE **man responsible for the renovations** IN THE office", there are two potential ways of parsing. If "in the office" modifies the verb "saw", the filler is valid because the entire adjectival postmodifier of "man" is contained within the filler. However, if "in the office" modifies the noun "renovations", the filler is an incomplete constituent and therefore invalid. For this reason, human judgment is necessary to determine validity, so rather than marking an instance as valid or invalid, its validity status is marked as 'MANUAL' and checked later using an additional script.

4.2.3.8. Filter 8: Coordinator check

As discussed in Section 3.3.2.2, coordinating conjunctions can appear within noun phrases in multiple ways. This presents some challenges in automatic validity judgment, so, as with some other validity checking functions, a conservative approach was taken in order to avoid falsely tagging a valid filler as invalid (false negatives). Therefore, a lower precision value was deemed acceptable, allowing false positives to be caught with later manual checking of the results (see Section 3.4.6. for more detail on evaluating script accuracy).

Coordinators were checked in specific ways based on the type of words being linked. For example, if adjectives (or other words serving an adjective role) are being joined, the function checks whether both conjuncts, as well as the head noun, are present within the bounds of the filler. If not, the filler is marked invalid. If all are present, there may be other invalidating issues to be caught by other functions, but the coordination is not the issue. In the case of adverbs joined by a coordinator, it must check whether both conjuncts, the word being modified by those adverbials (either an adjective or another adverb), and the head noun are present within the filler. In this case (as well as several others) the order in which the filtering functions are placed is important, as any invalid adverbials will have already been eliminated by a previous check

(3.4.3.5). When checking coordinators that join noun phrases, it is more difficult to check whether a filler is valid, as the boundaries of a noun phrase can be difficult to determine in some cases (e.g. prepositional phrase attachment ambiguity with postmodifiers). This is a good example of why this script was designed not to identify valid fillers as much as to remove invalid fillers, as it is much more difficult to confidently label a filler as valid than to find clear violations of the validity criteria. Therefore, a filler containing a conjunction joining nouns will be marked with a validity value of 'MANUAL' for later analysis.

4.2.3.9. *Filter 9: Verbs in filler*

Because the focus of this study is noun-based lexical frame fillers, there are a limited number of situations where a verb can be part of a valid filler. One case is when a participial verb is playing an adjectival role, as in "the **broken** window" or "the **written** test". These are normally simple to identify using spaCy's POS tags, as many (**broken**) are tagged as adjectives, while others that are tagged as verbs (**written**) have a dependency tag of 'amod', meaning adjectival modifier. Therefore, when checking for the presence of verbs in a filler, the function marks as valid any of those with the 'amod' tag. Also, during this check any verb with the dependency tag 'ROOT' was used as an indicator of an invalid filler, as the root verb of a sentence should not be a component of a noun phrase.

A more complex case of verbs being part of noun modification is when they are part of postmodifying clauses. This can include finite relative clauses, as well as non-finite clauses such as participle clauses (-ing and -ed) and to-infinitive relative clauses, which are given several dependency tags by spaCy. The tag 'relcl' is given to the main verb both in finite relative clauses (*he saw people he **knew** in the store*) and infinitive to-clauses (e.g. *they have a lot **to be** upset about*). The 'acl' or adjective clause tag is used for participle clauses (e.g. *a new approach*

*involving nanoparticles was tested; See the results **illustrated** below*). However, during piloting it was found that it is difficult to identify boundaries of these types of clauses (See 3.3.2.5 for details), so any filler containing a verb with one of these tags was marked with a validity value of 'MANUAL' for later checking.

4.2.3.10. Filter 10: Prepositional phrase postmodifiers

As described in detail in Section 3.3.2.5, prepositional phrase postmodifiers present a particular challenge to automatic, and sometimes manual, checking. Therefore, the function created to identify instances of preposition phrases postmodifying nouns, rather than checking for completeness and therefore validity, marked the validity value as 'MANUAL'. Because of the frequency of this type of noun modification, this meant a considerable amount of manual checking was necessary due to the ongoing challenge of ambiguous prepositional phrase attachment.

4.2.4. Evaluating Script Accuracy

During the early phase of this project, ten research articles from the field of applied linguistics were used as a small pilot corpus to aid in script development. This sample was limited compared to the corpus being used in the study's later stages (270 texts), but its scope allowed for a much more detailed, hands-on analysis. This included manual checking of all of the frame variants (~15,000) generated by the script based on the validity criteria described in the previous section (Figure 3.1). Manual checking resulted in a 'gold standard' spreadsheet of all frame variants with fillers 1-9 words long, each marked as valid or invalid.

Figure 4.1

Sample coding spreadsheet for 2-word fillers of the frame THE + OF THE

	A	B	C	D	E
1	VALIDITY	NGRAM	CONTEXT	FILE	NOTES
2	VALID	the actual execution of the	[] safeguards to the rights of the individual proved in the actual execution of the l	012-prag-n-ql-hp.txt	
3	VALID	the intended target of the	there is no doubt that the intended target of the accusations of quote and quote \	012-prag-n-ql-hp.txt	
4	INVALID	the speaker and of the	however invoking the perspectives of the speaker and of the target it is possible	012-prag-n-ql-hp.txt	
5	INVALID	the war and of the	both newspapers were partisan in their orientation and almost invariably critical c	012-prag-n-ql-hp.txt	
6	VALID	the theme-based approach of the	in view of the theme-based approach of the present study the two newspapers a	012-prag-n-ql-hp.txt	
7	VALID	the professed object of the	the enquiry which we wish again to make is what is there in this state of things o	012-prag-n-ql-hp.txt	
8	VALID	the professed object of the	the rhetorical question in the first sentence serves as a lead-in to the paragraph	012-prag-n-ql-hp.txt	
9	INVALID	the attack is of the	attacks that explicitly identify the target are more face threatening than those phr	012-prag-n-ql-hp.txt	

Using the same ten piloting texts, this gold standard was then used to check the accuracy of the script's output against human judgment, both for individual validity check functions (as described in Section 3.4.3), and the filtering script as a whole using precision and recall as measures (Section 3.4.4.1).

Development of the Python script followed a cyclical approach, with multiple rounds of revision and testing to achieve an acceptably accurate and efficient tool for analysis. Accuracy was tested using an additional script that checked the filtering algorithm against the 'gold standard' spreadsheet developed during piloting, which contains approximately 15,000 hand-checked frame + filler instances. Using this gold standard as a benchmark, the overall output of the filtering algorithm was analyzed for precision and recall.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

Precision measures the proportion of accurately labeled instances over all labeled instances. Frame + filler combinations correctly labeled as valid by the filtering algorithm (as compared to the gold standard) were **true positives**, while invalid instances that were marked

valid were **false positives**. **Recall** provides a measure of how many of all true instances were identified correctly by the algorithm. In this case, **false negatives** refer to the valid instances that the algorithm marked as invalid.

After calculating these measures, false positives and negatives were manually analyzed and the filtering algorithm was updated by adding new filtering functions or fine-tuning existing ones. The process was then repeated using the same pilot corpus and gold standard text until achieving acceptable values. Additional testing with other sample data sets would have improved the script's flexibility and accuracy, but due to the large amount of time necessary for creating additional hand-coded gold standard texts, this was not possible within the time frame of this project. While the overall goal was to achieve high values for both precision and recall, it can often be the case that improving one can negatively impact the other. For example, a stricter filtering approach could reduce the number of false positives, but may inadvertently mark some valid fillers as invalid, resulting in more false negatives. For the purposes of this project, it was important to prioritize recall rather than precision to avoid missing some valid frame + filler combinations, since the overall goal of this project is to give a more complete picture of extended lexical frame fillers. Accepting a lower precision value was also a consequence of limitations of the available parsing tools, as some filler structures cannot be automatically evaluated for validity using computational tools (e.g. nested prepositional phrases, see Section 3.3.2.3), and due to their frequency, precision value is lowered considerably. To address this limitation, the script marked some such instances as 'MANUAL' in the database until later manual checking (see Section 3.4.4.2 for details).

The majority of these challenging cases arise when a filler contains one or more prepositional phrases, making it difficult to determine which head a prepositional phrase is

attached to. For example, in the sentence "I saw the man with a telescope", it is unclear whether the subject of the sentence ("I") is using the telescope to see the man, or if the object ("the man") has the telescope. To give an example from the pilot corpus, the phrase "the degree of definiteness of the subject" could be parsed in either of the following ways:

the degree of [definiteness of the subject]

[the degree of definiteness] of the subject

Which of the above groupings is chosen determines whether the filler "degree of definiteness" is valid for the frame THE + OF THE. If the first parse is used, the filler is invalid because it crosses a constituent boundary, but if the second is chosen, the filler is valid. Judging from context, with knowledge of the discipline (English grammar), the second option was determined to be correct, meaning the frame + filler combination "THE degree of definiteness OF THE" is valid in that it contains a complete constituent. However, parsing this kind of ambiguous prepositional phrase attachment has been an ongoing challenge in natural language processing (e.g. Ratnaparkhi & Kumar, 2021; Zhou et al., 2024). Until further advancements are made in automatic parsing, a human judge of acceptability is needed for these types of structures.

While a low precision value must necessarily be remediated with manual checking, low recall values would be more of a problem, as they indicate that many valid frame + filler instances are being missed, and that an incomplete picture of our feature of interest is being provided. Therefore, the development and testing cycle continued until reaching a recall value of approximately 0.95. Precision, however, was found to be acceptable at a much lower value (~0.50) because the vast majority of false positives were structures that needed to be later manually evaluated. When discounting those instances, the precision is much closer to a conventionally acceptable value above 0.90.

Additionally, the script's output included independent calculations of precision and recall for different filler lengths (1-6 words). This was necessary because the relative ease of analyzing shorter fillers, along with their greater frequency, can obscure problems in longer fillers when all are combined. Given this study's goal of providing insight into the frequency and forms of longer fillers, it was especially important to get a clear picture of these forms. However, analyzing different filler lengths individually introduced other challenges, as it was more difficult to achieve a high recall value for longer sequences for multiple reasons. One of these was the overall smaller number of valid fillers compared to the much larger number of invalid sequences. This meant that even a handful of false negatives often containing uncommon vocabulary or structures could have a considerable impact on recall value. For this reason, it was important to keep in mind the specific false negatives and false positives when calculating precision and recall for longer fillers.

4.2.4.1. Results of Precision and Recall analysis

The first research question presented in Chapter 2 (RQ1) asked what level of precision and recall could be achieved with automatic detection of lexical frames with multi-word noun phrase fillers. This was answered using the pilot analysis of a ten-text subset of the corpus, as analysis of Precision and Recall with the full corpus would require an extensive and impractical amount of manual analysis. There is typically a tradeoff between Precision and Recall values, as increasing Precision to avoid incorrectly identifying a feature may result in also excluding some valid instances. For this study, Recall was prioritized, because it was preferable to allow some invalid entries to pass through the filtering functions to be manually checked later.

Recall measures the percentage of correctly identified valid fillers (True Positives) out of all valid fillers, both identified and not identified (True Positives + False Negatives). At the end

of iterative process described in the previous section, the final Recall value achieved was **0.99**, meaning the script identified 99% of all valid instances in the pilot corpus. Manual analysis showed that those that were not identified (false negatives) often contained unusual or specialized vocabulary or other features that resulted in incorrect POS tags being applied.

Precision, on the other hand, measures the percentage of correctly identified instances (True Positives) out of all identified instances (True Positives + False Positives). The script achieved a precision value of **0.74**, meaning 74% of filler + frame combinations labeled as valid in the pilot corpus were truly valid, while 26% were actually invalid. Analysis of false positives showed that most cases included one or more of three types of structures: possessives (e.g. *the teacher's instruction on the topic*), nested prepositional phrases (*in the office of the chief of staff*), clausal postmodifiers (*the dog that bit the man in the house*), or coordinating conjunctions (*the beginning and the end of the story*). These proved difficult to identify using SpaCy due to ambiguous boundaries (see Section 3.4.7), so a step was added to the script for manual judgment of validity (see Section 3.4.4.3)

4.2.4.2. Individual function accuracy

The second research question (RQ2) asked which syntactic features could be useful for automatically distinguishing valid from invalid fillers using available NLP tools. To answer this question, functions using the SpaCy library were created in the Python filtering script. These functions were based on syntactic features of both false positives (invalid fillers marked as valid by the script) and false negatives (valid fillers falsely marked as invalid) that were identified during piloting by comparing script output with the gold standard text. It was found to be more effective to identify invalid fillers rather than valid, using disqualifying syntactic features derived from the operational definition described in Section 3.3).

Using the gold standard text of manually checked fillers from the pilot corpus (10 texts), these functions were tested and modified in an iterative process until achieving at least 95% accuracy, calculated as the percentage of fillers **correctly** tagged as invalid out of **all** fillers 'caught' by the function. Table 3.1 below details the accuracy of each filtering function used in the final algorithm, with a brief description of the way each of them works (the Python code for each function can be found in **Appendix**). The functions are listed in order of the number of invalid fillers they identified ("correctly tagged" column); however, it is important to note that the order in which the filtering functions were applied was important to the overall function of the script. For example, the "ONE WORD" function must follow the "NO NOUNS" check in order to function correctly, because it assumes that if a noun has been found in the filler, and if that filler is only one word in length, the filler must by default be valid.

Table 4.1.

Accuracy of final filtering functions during pilot analysis

Function	Description	Correctly tagged	Incorrectly tagged	% correct	% of all invalid identified (5286)
BAD VERB	Filler is invalid if it contains a root verb, which indicate a constituent boundary	1632	3	0.998	30.87%
ONE WORD	For 1-word fillers, if it is a noun, it is by default valid	1225	9	0.993	23.17%
LAST FILLER WORD BAD	If the last filler word is a conjunction, or determiner, there is a phrase boundary and filler cannot be valid. Exceptions made for nominalized phrasal verbs (e.g., slowing down), where the last word is a preposition.	1193	0	1.000	22.57%
ACL BAD	If there is a non-finite clause modifying noun (dep tag = 'acl'), check whether the head noun is present within the filler	273	6	0.978	5.16%

NO NOUNS	Invalid if no nouns or noun-functioning words are present among filler words	234	2	0.992	4.43%
PREP HEAD OUTSIDE	If a preposition is present, its head must be within the filler	232	12	0.951	4.39%
DET AFTER NOUN	Check for determiners following nouns, as in "as A result of their actions the third and fourth presidents OF THE". Commas were not always present to separate the NPs.	153	0	1.000	2.89%
BAD ADV	If an adverb is present, it must be modifying an adjective or another adverb that meets the same criteria. Exceptions made for adverbs within dependent clauses.	111	0	1.000	2.1%
REL CLAUSE BAD	If a relative clause is present, the head it modifies must be within the filler boundaries.	94	1	0.989	1.78%
INVALID ADJ AFTER NOUN	If an adjective is present, its head must be within the filler. Created to catch examples like "He found THE solution difficult ON THE exam", where the adjective "difficult" is an object complement rather than a modifier of "solution".	62	2	0.956	1.17%
FIRST WORD BAD	Invalid if the first word is a preposition (adposition), which make the filler a prepositional phrase rather than a noun phrase	47	1	0.979	0.89%
ADJP ONLY	Checks for cases when a noun is present, but it's part of an adjective phrase, so there's no NP within the filler.	22	0	1.000	0.42%
PASSIVE VOICE	Checks if the fixed word preceding the filler is the auxiliary of a passive verb. These would not be caught by "BAD VERB" (see above) because the main (root) verb of the sentence.	8	0	1.000	0.15%

In the full corpus analysis, accuracy was not calculated for each function as that would have required manually checking several million entries. However, the number of entries marked invalid by each function is listed in Table 3.2 below, providing insight into the relative importance of each in automatic identification of invalid fillers.

Table 4.2.

Number of invalid entries identified in the full corpus by each filter function

Function	# of invalid identified	% of all identified
BAD VERB	26,813	39.72%
LAST FILLER WORD BAD	19,911	29.5%
ACL BAD	5,535	8.2%
NO NOUNS	4,399	6.52%
PREP HEAD OUTSIDE	4,535	6.72%
ARTICLE AFTER NOUN	1,500	2.22%
BAD ADV	2,024	3%
REL CLAUSE BAD	1,298	1.92%
FIRST WORD BAD	665	0.99%
INVALID ADJ AFTER NOUN	449	0.67%
ADV CL BAD	126	0.19%
PASSIVE VOICE	203	0.3%
Total	67,503	100%

The most useful function for identifying invalid fillers was 'BAD VERB', as the presence of a root verb is a clear indicator of a constituent boundary. The function with the next highest number of invalid entries caught was 'LAST FILLER WORD BAD', which was also straightforward in identifying boundaries indicating an incomplete noun phrase. The 'NO NOUNS' function also caught a relatively large number of invalid entries, despite the density of

nouns in academic writing (Biber & Gray, 2010). While functions identified relatively smaller proportions of the invalid fillers, their presence reduced the manual checking load considerably while allowing for the more complex filler structures that were a key point of interest of this study.

4.2.4.3. *Manual validity checking*

Due to the limitations of automatic parsing, some invalid entries were not eliminated by the filtering algorithm, and their values in the database under the ‘validity’ column were temporarily marked as 'MANUAL'. The most common of these situations was the case of ambiguous prepositional phrase attachment (Section 3.4.3.9), which has been a particular challenge in automatic parsing (e.g. Ratnaparkhi & Kumar, 2021). Additionally, though less common, it was sometimes difficult to identify the boundaries of postmodifying clauses in order to determine a filler's validity (see Section 3.4.3.8). In cases like these, it was necessary to manually check validity.

One of the goals of this study was to find out to what identification of frame validity could be automated. The total number of invalid fillers identified by the filtering functions in the Python script was 67,503 (detailed in **Error! Reference source not found.** above). However, it was found during piloting that manual checking of fillers containing some grammatical features would be necessary to do ambiguity of phrase boundaries and the limitations of SpaCy's parsing tools. These features included coordinating conjunctions (THE beginning **and** end OF THE story), possessives (IN **students'** production OF THE target feature), prepositions (THE point **of** view OF THE government), and dependent clauses (THE man **who spoke TO THE police**).

The number of invalid fillers caught by manual checking was 7,003 or 9.4% of all invalid fillers. 2,836 valid fillers were identified in this way, or 7.4% of the 38,393 total valid 1–6-word

fillers found for all frames with a normalized frequency of at least 400 per million words.

Error! Reference source not found. below shows a breakdown of the numbers of valid and invalid fillers found by each type of manual validity check.

Table 4.3.

Breakdown of manual validity checking results

Manual check type	# VALID found	# INVALID found	% VALID
Dependent Clause	124	595	20.8%
Possessive	86	339	25.4%
Preposition	1,609	4,801	33.5%
Coordinating Conjunction	1,007	1,268	79.4%
Total	2,826	7,003	9,829 (total checked)

Manual checking took a considerable amount of time and has the limitation of requiring some degree of subjective judgment when evaluating certain nested prepositional phrases. With a very large corpus, especially when also considering fillers longer than 6 words, this may be impractical. However, while these manually checked fillers account for only 7.4% of all valid fillers, they provide a new insight into structural characteristics of extended lexical frames, and phraseological patterning more generally, that may not otherwise be available.

4.3. Minimum Frequency and Overlapping Frames

Initial runs of the script with the full corpus (270 texts) found that it would be necessary to manually check over 50,000 entries containing the features described in the previous section. As one of the goals of this study is finding a useful method for identifying extended lexical frames, this amount of manual checking seemed impractical, and not feasible for the scope of

this study. In addition, the pilot analysis found that 7–9-word fillers collectively represented less than 1% of all valid fillers, with the vast majority of instances initially identified being invalid (usually due to presence of nested prepositional phrases or dependent clauses). For these reasons, the decision was made to pursue structural and linguistic analysis only for fillers of 1-6 words in length, and only frames appearing at least 400 times per million words. This decision greatly reduced the amount of manual checking necessary to complete the analysis to a still considerable 11,486 entries.

In addition, manual analysis showed that there was a significant amount of overlap among the frames identified, which resulted in inflated frequency counts. While most studies have focused on frames with three fixed words, others have also included frames with four, though not specifically for the purpose of reducing overlaps (Cunningham, 2017; Lu et al., 2018). Many studies have addressed overlap by manual inspection (Golparvar & Barabadi, 2020), but automatic approaches like the process described below are less common. The problem of overlap was particularly important to address in a study of lexical frames with multi-word fillers. Because this study focused on noun-phrase fillers, determiners often appeared as the first word in a filler, as in the example IN **the beginning** OF THE. According to the operational definition used in this study this is a valid filler, but the same sequence also could produce THE **beginning** OF THE, or as an example of a 4-word frame, IN THE **beginning** OF THE. To address this problem, the final version of the script also included frames with 4 fixed words of the form A B + C D that appeared over 200 times per million words. Instances of 3-word frames that were part of those frequent 4-word frames were then automatically removed to avoid double-counting. Take for example the phrase ‘in the study of the Cold War’. The script first identified 3-word frames IN THE **study** OF and THE **study** OF THE. In the next step, the 4-

word frame variant IN THE **study** OF THE was extracted; if the frame IN THE + OF THE occurred more than 200 times per million words in the corpus, the 4-word frame instance would be retained while the two instances of IN THE + OF and THE + OF THE were removed to avoid duplication. The choice of frequency threshold was practical rather than theoretical, but the choice of a lower frequency for longer sequences was informed by previous lexical bundles research (see discussion in Bestgen, 2018). This process with the given minimum frequencies resulted in a list of 21 frames for further analysis, including 18 3-word frames and three 4-word frames.

4.4. Phase 1 Summary

The findings presented in this chapter have shown that identifying extended lexical frames using the NLP tools developed in this project is both accurate and practical, though a certain amount of manual validity checking proved to be necessary. Limitations in automatic parsing of sentence structure are an ongoing problem in the field of natural language processing, which meant that certain filler structures had to be checked manually. However, this proved to be a reasonable amount of work for a corpus of this size, and future developments in natural language processing may further reduce the need for manual analysis. While some limitations were identified, a newly proposed phraseological form, the extended lexical frame, was defined, operationalized, and effectively identified in a corpus. Further discussion of the implications and limitations of the work presented in this chapter will be found in Chapter 6.

CHAPTER 5. PHASE 2: METHODS AND RESULTS

This study has two related goals, one concerned with methodology and the other with language description. The methodological steps and results of Phase 1 were described in Chapter 4, including the levels of precision and recall achieved by the Python script developed to identify extended frames. In Phase 2 of the study, the goal was to describe the structural and distributional characteristics of extended lexical frames in order to contribute to a fuller description of the phraseological characteristics of academic journal writing in English. This chapter details the methods of analysis used in Phase 2, as well as the descriptive results.

5.1. Data analysis

After the Python script developed in Phase 1 achieved an acceptable level of accuracy and validity was checked manually in some more ambiguous cases, the results in the form of a SQLite database were used for further analysis. An additional script was created for extracting data from the database, performing calculations, and displaying results in a usable form. These calculations included overall frequencies for extended frames, frequencies of each filler length, variability of filler length across frames, and a frequency analysis of the part-of-speech sequences in valid fillers. Each of these are described in more detail below.

5.2. Phase 2 Methods

5.2.1. Overall frame frequency

The initial step in the analysis was to identify the most frequent extended lexical frames, whose overall frequencies were calculated by summing the frequencies of each individual filler length (1-6 words) for the frame. In order to facilitate comparison with previous studies, these frequency values were normalized per million words. Normalized frequencies for all (or nearly all) frames are likely to be considerably higher than in previous frames studies due to the

inclusion of longer fillers; however, the more interesting question is how relative frequencies of frames might change. For example, there may be some frames with a relatively high proportion of 2+ word fillers, and those may have not appeared among the most frequent frames identified in academic writing in previous studies, similar to the way in Gray and Biber (2013) a frames-first approach enabled identification of some frames previously missed by using continuous bundles.

5.2.2. Frequency and variability across filler lengths

The most central question of this study concerned the degree to which an extended filler approach to lexical frames studies might provide a fuller picture of lexical frame use and contribute to phraseology more generally, specifically how much including longer fillers increases the frequency of frame occurrence. To answer this question, for each frame the number of valid fillers of each length (1-6 words) were counted and divided by the frame's total number of valid fillers to provide a percentage. For example, a given frame may occur 100 times in the corpus, with 70 occurrences containing single word fillers (70%), 20 containing two-word fillers (20%), etc. Descriptive statistics were also calculated, including minimum, maximum, mean, and standard deviation for each filler length frequency, allowing broader insight into variation in filler length frequency across all frames. In addition, as discussed in Section 2.31, entropy is used in this study as a measure of variation in filler length, giving a value that can be interpreted as the degree to which the length of a filler is predictable, with 0 as completely predictable (only 1 filler length present for the frame) and greater values representing increasing unpredictability as the frequency of each filler length become more equally dispersed. The entropy value was calculated for each frame, and across all frames, to give a measure of how dispersed frame fillers are across the different filler lengths (1-6).

5.2.3. Structural analysis (POS sequences)

A new part of the methods introduced this study is the analysis of the structure of multi-word fillers. For example, many of the most common frames in academic writing are function word frames that contain nouns as fillers, such as THE (**noun**) OF THE (Biber, 2009). In this study, multi-word noun phrases may also occupy a variable slot, so the task was to classify them based on structure within the noun phrase (see Table 3.2 for examples below).

Table 3.2

Examples of frame fillers and their structural categories

Frame realization	POS tag sequence
THE + OF THE	
THE <u>relative difficulty</u> OF THE	adj + noun
THE <u>overall factor structure</u> OF THE	adj + noun + noun
THE <u>speed of delivery</u> OF THE	noun + prep + noun
THE <u>mean and standard deviation</u> OF THE	noun + conj + adj + noun
IT IS + TO	
IT IS <u>important</u> TO	adj
IT IS <u>not our intention</u> TO	neg + det + noun
IT IS <u>unfortunately not feasible</u> TO	adv + neg + adj

This classification was done automatically using spaCy's part of speech tagging function. After the words of each filler were tagged for part of speech, frequency of each POS tag sequence was calculated for each frame (e.g. the frame IN THE + OF has 20 *noun*, 5 *adj + noun* fillers), as well as the percentage of all fillers represented by each sequence (e.g. 40% of all fillers for IN THE + OF are a single noun, 10% are *adj + noun*, etc.). Frequencies and percentages were also calculated across all frames for particular filler lengths (e.g. 100 2-word fillers are *adj + noun*, 40% of total).

5.3. Phase 2 Results: Linguistic Description

The following sections will describe findings relating to frame and filler frequency and structure across the whole corpus, as well as going into detail about selected highly frequent frames. First, the most frequent frames and the distribution of filler lengths across the whole corpus will be explored in Section 5.3.1. The following section, 5.3.2, will focus on describing the syntactic structure of multi-word fillers, specifically by looking at POS-tag patterns and their realizations. Finally, Section 5.3.3 will look at the filler frequency and structures associated with the five most frequent frames occurring in the corpus.

5.3.1. Frequent frames and filler length distribution

This analysis required an extensive amount of manual checking of filler validity and is primarily focused on description of multi-word filler structure and frequency distribution. For this reason, the full analysis presented here is limited to 3-word frames with a frequency over 400 per million words and 4-word frames that appeared at least 200 times per million words. This is considerably higher than the minimum frequency used in previous frames studies, which often began analysis with (1-word filler) frames that appeared over 40 times per million words in their corpora (e.g. Gray & Biber, 2015). However, this study found much higher frequencies due to the more flexible variable slot, and the next section will describe how this flexibility impacted frequency in more detail. **Table 5.1** below presents 3- and 4-word frames meeting this frequency minimum. In addition to their raw and normalized frequency, the table (and other following tables) also include range, or the number of texts in which the frame appeared.

Table 5.1.

3- and 4-word frames occurring more than 400 times per million words and 200 times per million words, respectively

Frame	Raw Freq	Norm Freq (pmw)	Range
the + of the	5171	3244	263
in the + of	2075	1302	261
of the + of	1696	1064	245
the + in the	1476	926	248
to the + of	1274	799	246
a + of the	1237	776	245
the + of a	1236	776	229
the + and the	1226	769	233
of + in the	1033	648	225
of the + and	980	615	240
on the + of	884	555	231
and the + of	852	535	230
of the + in	809	508	221
for the + of	743	466	218
the + for the	731	459	204
that the + of	725	455	212
in the + of the	711	446	206
in the + and	688	432	201
of the + of the	670	420	195
and + of the	647	406	206
as a + of	642	403	202
to the + of the	542	340	172
on the + of the	383	240	153
of the + in the	383	240	151

All of the frames above this frequency cutoff are composed entirely of function words, consistent with previous studies of lexical frames in academic writing. Two four-word frames

appeared more than 400 times per million words - IN THE + OF THE and OF THE + OF THE, and four others appeared more than 200 times per million words (TO THE + OF THE, ON THE + OF THE, OF THE + IN THE, AND THE + OF THE). All but one of these were extensions of the most frequent frame, THE + OF THE, whose frequency would be much higher if the analysis had only included 3-word frames. As expected, by allowing multi-word fillers, frame frequencies increased compared with previous studies. **Table 5.2** shows the 10 most frequent frames identified in Gray and Biber (2013) compared with the frequencies found in the present study.

Table 5.2

Comparison of most frequent frames in the present study and Gray & Biber, 2013

Frame	Norm freq (pmw)	
	This study	Gray & Biber, 2013
the + of the	3244	2567
in the + of	1302	1002
of the + of	1064	649
the + in the	926	not reported
to the + of	799	546
a + of the	776	589
the + of a	776	not reported
the + and the	769	not reported
of + in the	648	not reported
of the + and	615	322
and the + of	535	394
on the + of	555	371
for the + of	466	297

For the most part, this study found the same frames identified in Gray and Biber (2013) and with similar relative frequencies; however, three frames are found among the ten most frequent that did not appear in the previous list. These included THE + IN THE, THE + OF A, THE + AND THE, and OF + IN THE. The appearance of these frames in higher frequencies seems to reflect their greater proportion of multi-word fillers, since a traditional frames study allowing only 1-word fillers would miss a majority of the occurrences. For example, the filler length distribution for THE + IN THE shows an especially low percentage of 1-word fillers (26.15%). Similarly, THE + AND THE and OF + IN THE appear only among the top ten most frequent for the present study, and both have relatively low proportions of 1-word fillers (33% and 37%, respectively). On the other hand, THE + OF A does not appear on the Gray & Biber list but has a high proportion of 1-word fillers (77%), and more investigation will be necessary to understand the source of the disparity.

Table 5.3

Frequency distribution of filler lengths for THE + IN THE

Length	Raw Freq	Norm Freq (pmw)	% of total	Range
1	386	242.19	26.15%	166
2	389	244.07	26.36%	173
3	276	173.17	18.70%	141
4	225	141.17	15.24%	127
5	140	87.84	9.49%	95
6	60	37.65	4.07%	53

Table 5.4*Frequency distribution of filler lengths for THE + AND THE*

Length	Raw freq	Norm freq (pmw)	% of total	Range
1	405	254.11	33.03%	154
2	286	179.45	23.33%	137
3	184	115.45	15.01%	114
4	175	109.8	14.27%	102
5	106	66.51	8.65%	79
6	70	43.92	5.71%	54

Table 5.5*Frequency distribution of filler lengths for OF + IN THE*

Length	Raw freq	Norm freq (pmw)	% of total	Range
1	378	237.17	36.59%	153
2	367	230.27	35.53%	166
3	145	90.98	14.04%	96
4	82	51.45	7.94%	69
5	37	23.22	3.58%	30
6	24	15.06	2.32%	19

Table 5.6*Frequency distribution of filler lengths for THE + OF A*

Length	Raw freq	Norm freq (pmw)	% of total	Range
1	956	599.83	77.35%	216
2	203	127.37	16.42%	110
3	53	33.25	4.29%	42
4	17	10.67	1.38%	15
5	1	0.63	0.08%	1
6	6	3.76	0.49%	6

5.3.1.1. Filler Length Distribution: All Frames

Research question 3a asked how much including longer fillers increases the frequency of frame occurrence. To answer this question, the percent of all valid fillers represented by fillers of each length was calculated, with the results shown in **Table 5.7** below.

Table 5.7

Frequency and proportion of each filler length

Filler length	Raw Freq	Norm freq (pmw)	% of valid fillers	Range
1	14776	9271	55%	270
2	7035	4414	27%	270
3	2617	1642	10%	260
4	1199	752	5%	241
5	593	372	2%	206
6	318	200	1%	161

The results showed that 1-word fillers represent just over half of the valid fillers identified in the study (55%), with 2-word fillers appearing around half as frequently (27%) and 3-word fillers also being relatively frequent (10%). 4-word (5%), 5-word (2%) and 6-word (1%) fillers were present, but much less frequent. Additionally, the range values show that all lengths other than 6-word fillers occur in the majority of texts in the corpus. These findings illustrate the value of the extended lexical frame approach, as multi-word fillers represent a large percentage of total instances but are missed by traditional methods of lexical frame identification.

It is also important to note that these proportions vary considerably between specific frames, as we briefly saw in the previous section. As shown in **Table 5.8** below, the most variation in filler length proportion is found for 1-word fillers, which ranges from 26.2% of

instances of the frame THE + IN THE to 82.23% of the frame AS A + OF, and progressively less variation exists as filler lengths increase. The same frame, AS A + OF appears with the highest proportion of 1-word fillers as well as the lowest proportions of 2-,3-,4-, and 5-word fillers. These proportions are reflected in the frame's entropy value, which was the lowest of all frames identified in this study, indicating a high level of filler length predictability (see Section 4.1.3.6 for more detail).

Table 5.8

Descriptive statistics for proportion of each filler length

Length	Min	Max	Mean	StDev
1	26.2%	82.23%	54.6%	0.16
	(THE + IN THE)	(AS A + OF)		
2	14.64%	39.69%	26.74%	0.07
	(AS A + OF)	(A + OF THE)		
3	2.34%	18.70%	10.25%	0.04
	(AS A + OF)	(THE + IN THE)		
4	0.16%	15.24%	4.68%	0.04
	(AS A + OF)	(THE + IN THE)		
5	0.00%	9.49%	2.43%	0.03
	(AS A + OF)	(THE + IN THE)		
6	0.00%	5.71%	1.30%	0.02
	(IN THE + OF THE)	(THE + AND THE)		

Although disciplinary variation is not a specific focus for this study, it is interesting to see that filler length distribution varies across disciplines, though not to a large degree generally (see **Table 5.9**). The largest differences are found between Philosophy and Physics, which have the highest and lowest proportions of 1-word fillers, respectively. Biology, another 'hard science' also has relatively larger proportions of longer fillers. These longer fillers often take the form of nominalizations used to characterize the actions of the item of study, as in the following examples.

*It has been verified that this simplified model provides a correct description OF THE
momentum distribution of the ionized electrons IN THE longitudinal direction (PHYS)*

*THE dynamic behavior of tight junction proteins IN THE --- only one study has directly
assessed THE **dynamic behavior of tight junction proteins** IN THE absence of external
stimuli (BIO)*

While this is a very limited look at specific realizations of multi-word fillers, further details about filler structure and form will be provided in Section 4.2.2. Additional exploration of disciplinary variation in filler length frequencies may provide interesting insights into disciplinary variation in phraseological characteristics, it is beyond the scope of the present study and may provide an interesting direction for future work.

Table 5.9*Filler length frequencies by discipline*

Filler length		AL	BIO	HIST	PHIL	PHYS	POLISCI
1	Norm freq (per 100K)	22689	475	1079	749	758	804
	%	58.17%	53.45%	59.16%	64.62%	45.84%	57.51%
2	norm	298	237	435	273	533	342
	%	25.20%	26.69%	23.84%	23.55%	32.22%	24.45%
3	norm	108	92	170	82	206	130
	%	9.12%	10.33%	9.30%	7.07%	12.48%	9.27%
4	norm	50	47	86	32	85	67
	%	4.26%	5.24%	4.73%	2.77%	5.11%	4.81%
5	norm	25	10	24	10	38	34
	%	2.11%	3.20%	2.04%	1.29%	2.64%	2.52%
6	norm	14	3	11	5	24	19
	%	1.15%	1.09%	0.93%	0.71%	1.70%	1.44%

5.3.1.2. Frequent multi-word fillers

As previous studies have extensively covered single word lexical frame fillers in academic writing (e.g., Gray & Biber, 2013; Casal & Kessler, 2020; Lu et al., 2018, etc.) this discussion will focus on the multi-word fillers identified in the analysis. This section will describe commonly occurring examples of each filler length` and examine possible connections between.

The most frequent 2-word fillers identified are specialized disciplinary vocabulary, often coming from a single text that makes frequent use of the term. For example, the filler *global hegemony* occurred more often than any other two-word filler, and all instances came from a single text from the Applied Linguistics subcorpus, where it most often filled the frame OF THE + OF, as in in the sentence

*It makes sense to study American prestige press representations OF THE **global hegemony** OF English during the time period selected (AL)*

The filler also occurs within THAT THE + OF ("yet this total indicates THAT THE **global hegemony** OF English is very definitely on the media radar screen in the united states"), FOR THE + OF ("the media radar blip FOR THE **global hegemony** OF English"), TO THE + OF ("with respect TO THE **global hegemony** OF English") and IN THE + OF ("relevant to my general interest IN THE **global hegemony** OF English").

The second most frequent 2-word filler, **temperature dependence**, was found in multiple texts, but only within the discipline of Physics, and the third, **coherence length**, appeared only in a single text. This filler occurred most often within the frame THE + OF THE, as in the example "THE **coherence length** OF THE pump laser is much larger than the path length difference"). It also appears in the two 4- word frames, WITH THE + OF THE ("the larger difference between the two interfering beams becomes comparable WITH THE **coherence length** OF THE effective pump laser") and AND THE + OF THE ("checking the relation between the path length difference AND THE coherence length OF THE pump laser").

While slightly less frequent, other 2-word fillers that appear in multiple texts and disciplines were found as well, some of which represent expansions of common single-word fillers. For example, **vast majority** occurred 14 times and across several disciplines:

*THE **vast majority** OF THE members of colonized fervently desire to expel imperium and reestablish a politically independent state (PHIL)*

*what worked FOR THE **vast majority** OF disadvantaged groups might not work for the 'most hard to reach individuals' (POLISCI)*

*we predict THAT THE **vast majority** OF HVCs will elude this technique in its current form (PHYS)*

*are the focus OF THE **vast majority** OF validation studies (AL)*

This can be seen as an extension of the frequent 1-word **majority**, which appeared 64 times the corpus. Similarly, the non-specialized 2-word filler **relative importance** (11 occurrences) relates to the 1-word filler **importance** (62 occurrences)

a concomitant increase IN THE relative importance OF THE pelagic trophic pathway was observed (BIO)

*determine THE **relative importance** OF THE several factors in each model (POLISCI)*

*This will allow delineation OF THE **relative importance** OF THE parameters (PHYS)*

Table 5.10

25 most frequent fillers for each length

1-word		2-word		3-word		4-word		5-word		6-word	
filler	raw freq	filler	freq	filler	freq	filler	freq	filler	freq	filler	freq
case	282	global hegemony	36	right hand side	12	global hegemony of English	4	softer exponential fall off behavior	3	single site and multi site events	3
use	163	temperature dependence	21	presence or absence	9	positive parity intruder band	3	number of semantic relations constructed	2	harassment and arrest of polish officials	2
context	163	coherence length	14	point of view	7	notion of q memory	3	average number of technical items	2	reprogramming of host cell gene expression	2
result	158	vast majority	13	large n limit	6	two loop expectation value	3	corrections to the callan treiman relation	2	distribution of the heaviest fragment size	2
basis	139	total number	13	high energy side	6	second and third terms	3	role of the quantum potential	2	acoustic and optical spin wave modes	2
nature	123	early years	12	assessment of speaking	6	non assessment of speaking	3	coverage of the agrarian question	2	tenth cycle for small grounded electrode	2
absence	122	second half	12	two photon width	5	grammatical forms of interest	2	contingency of the republican experiment	2	direction of the positive z axis	2
function	119	relative importance	11	left hand side	4	number of free parameters	2	main objective of the reorganization	1	political theater of the Weimar period	2
presence	109	sum total	11	exact integral expressions	4	order of the questions	2	mobilization capacity of Montreal citizens	1	5-minute talk in interaction sessions	2
development	102	compositional dependence	11	lower scoring texts	4	relevant photoactive center levels	2	use of a progressive vocabulary	1	second half of the twelfth century	2
number	99	first half	10	house of commons	4	distribution of photoactive centers	2	nature of the classroom activities	1	both native born and naturalized latinos	1
role	94	first stage	10	scalar form factors	4	Canadian private pension system	2	existing or potential aggregate number	1	members of the u.s. house of representatives	1

form	92	other side	9	relatively small proportion	3	strategically sensitive amazon region	2	electorate based on key demographic	1	most passive and least personalized types	1
results	90	utilitarian theory	9	first two courses	3	direct and indirect effects	2	densities of the charged species	1	total global annual averaged dust load	1
end	77	q memory	9	a priori deducibility	3	individuality and family life	2	presence of other Chebyshev particles	1	difference in scattering cross section area	1
creation	76	fuzzy version	9	visually symbolic representation	3	best arguments then extant	2	similar spectral variation of k	1	last quarter of the nineteenth century	1
value	76	realist vocabulary	9	safety and security	3	n type pcco samples	2	Non-European tropical island Edens	1	many university level intensive English programs	1
impact	75	immediate aftermath	8	coordination problem paradigm	3	narrow or holistic view	2	agricultural economy and forested land	1	provision of input and output opportunities	1
part	70	real part	8	positive z axis	3	original phenomenological color solid	2	great french geographer elisee reclus	1	full range of feedback and forms	1
majority	64	overall state	8	power law behavior	3	best state of affairs	2	barren and supposedly uninhabited areas	1	three 50 minute class periods production	1
importance	63	first term	8	infinite linear relations	3	average level of knowledge	2	arab invasions of the seventh	1	selection of a new chief executive	1
effect	60	coherence time	8	number of events	3	story of british imperialism	2	european settler population in algeria	1	existence of more than one party	1
formation	59	psychological context	8	virtue of accuracy	3	real and imaginary parts	2	further colonization and territorial expansion	1	vacuum system of the reaction chamber	1
content	58	electronic properties	7	residual temperature dependence	3	second harmonic generation (shg)	2	high kinetic and thermal stability	1	lp emissions during the reaction process	1
theory	56	large number	7	botswana general certificate	3	division of the brain	2	guest/host molar ratio sq	1	forward elastically diffused proton beams capable	1

Total number is another frequent 2-word filler that represents an expansion of a common single word filler. The base form **number** occurs 99 times and the plural **numbers** (3), and it is also expanded as **total number** (13), **large number** (7), small number (3), **number of states**, and **number of questions**.

No single filler was highly frequent among the most frequent 3-word fillers, and both specialized terms (**large n limit**, **positive z axis**) and more, though fewer, general terms (**point of view**) were found. The filler that appeared most was **right hand side**, though it appeared primarily in Physics texts, where it is used to refer to terms within equations.

*we can replace a b by the first term ON THE **right hand side** OF (103) (PHYS)*

*the higher order term IN THE **right hand side** OF equation (43) is perturbatively evaluated as xxx (PHYS)*

Unexpectedly, **side** appears at a lower frequency (5) than both **right hand side** (12) and **high energy side** (6). Many other low frequency variations occur as well, including **left and right hand sides** (1), **lower energy side** (2), **demand side and supply side** (1), **right side** (2), **left ventral side** (1), **negative side** (1), **opposite side** (1), and many others.

Presence or absence (9) was among the most frequent as well, despite a low overall frequency, and appeared in multiple disciplines.

*the infants reveal a primitive sensitivity TO THE **presence or absence** OF these properties (PHIL)*

*tasks are assigned an involvement load index on the basis OF THE **presence or absence** OF involvement factors (AL)*

*the models were chosen to probe how planarity is affected by THE **presence or absence** OF THE imino proton and the positioning of Zn in the major or minor groove (PHYS)*

Interestingly, the related parallel fillers **absence** (22) and **presence** (109) are very different in their own frequencies. Further expansions were also found, including **absence of legislative powers** (2) and several others that each occurred only once, including **absence of systematic legislative revision**, **absence of comprehensive restructuring**, **absence of intermediate consumers**, and **absence of subjects' positive evaluative responses**

Though not among the ten most frequent, **relatively small proportion** (3) is another interesting case in that it relates to other filler forms of different lengths, including **proportion** (37), **substantial proportion** (4), **large proportion** (3), **high proportion** (2), **relative proportion** (1), **much greater proportion** (1), among others.

Most notable in the 4- 5- and 6-word fillers are the small overall numbers, as well as the dominance of topic or discipline-specific terms that appeared only once. In some cases, such as **softer exponential fall off behavior** and **coverage of the agrarian question**, we see the pre- or post-modification of common single word fillers. For example, **behavior** with no modifications occurs 17 times, as well as many other times in specialized expanded forms such as **bending behavior** (1), **expected behavior** (1), and **dynamic behavior of tight junction proteins** (1). This suggests the need for analysis of more generalized modification patterns in longer frames, such as *adjective + noun* (**expected behavior**) or *adjective + noun + prepositional phrase* (**dynamic behavior of tight junction proteins**). An analysis of these characteristics will be presented in the next section (4.2.2).

One alternative to looking at the most frequent fillers for each length is to first identify frequent single word fillers, then find expansions among the longer fillers, which might be called

a single-word first approach. For example, taking the third most frequent single word filler **context** (163) as a starting point, we can also find **larger context** (3), **specific context** (2), and **political context** (2), as well as more than 30 others that occur only once, including **institutional and social contexts**, **social and political context**, and **historical and contemporary contexts**. None of these expanded forms appeared among the most frequent multi-word fillers, but searching by frequent single word fillers as a starting point appears promising as a way of understanding how noun phrase elaboration is used in the context of lexical frames. On the other hand, the 1-word first approach may have similar limitations to the ‘bundles first approach’ of identifying frames based on frequent continuous bundles, a method whose limitations were pointed in Gray and Biber (2013). One illustration of this potential problem is the filler **strength** (30), which alone did not appear among the most frequent fillers, but combined with its many variants, including **overall strength**, **field strength**, **oscillator strength**, **concentrated strength of labor**, it would be among the most frequent fillers. This suggests that future work might consider filler ‘families’ containing a root word that is expanded upon in longer fillers, and it may be feasible to develop an approach for automatic identification of such families using existing NLP tools.

5.3.2. Structural description of multi-word fillers

As the previous section showed, there is a high degree of variability in multi-word fillers, with few highly frequent fillers and a large presence of specialized vocabulary. For this reason, it is informative to generalize more and look at their syntactic structure, which may show more systematic patterns, and answer RQ4, which asked what structures (e.g. *adj + n*, *n + pp*, etc.) are found in frame fillers of different lengths (and in what proportions). The following sections will explore some frequent POS tag patterns for 2-6-word fillers, as well as examples of how they are realized in the corpus.

5.3.2.1. Two-Word POS patterns

The ten most frequent POS tag patterns for 2-word lexical frame fillers are listed in Table 5.11 below. The most frequent are nouns with adjective (A **significant decrease** IN THE), nouns (THE **rotation period** FOR THE), or participial verbs (THE **underlying culture** OF THE).

Table 5.11. Ten most frequent 2-word filler POS patterns

POS tag pattern	Raw Freq	Range
ADJ NOUN	4695	266
NOUN NOUN	1282	236
VERB NOUN	368	164
NUM NOUN	159	90
DET NOUN	68	53
NOUN ADJ	42	32
PRON NOUN	32	28
NOUN PROP	29	24
NOUN VERB	28	24
NOUN NUM	15	12

Much less common are unexpected structures such as *noun + verb*, which usually represents a one-word non-finite clause postmodifying the noun, as in “given the complexity OF THE **variables involved** AND their interaction” (AL). The pattern *noun + num* appeared as well, taking the form of year + number, as in “this return migration occurred about THE **year 1900** AND THE exodus is said to have been completed within the space of about a month” (HIST), or when referring to a table or equation within the text, as in “using THE **formula (74)** FOR THE product of f's” (PHYS). Another unexpected form was *noun + adj*, which often came from the mis-tagging of terms, as in

links between the use OF THE term "right" AND some deontological property (PHIL)

*t is THE **decay constant** FOR THE transverse relaxation of the mri signal back to equilibrium (PHYS)*

Sue then proceeds to the next sentence and again seems uncertain about THE verb tense OF THE verb 'illustrate' (AL)

The most common POS tag pattern for 2-word fillers was *adj + noun*. It appeared most often in History texts (320 per 100k) and least in Biology texts (148.56 per 100k). Nearly all of the 25 most frequent 2-word fillers mentioned in the previous section were *adj + noun* structures.

Table 5.12.

Frequency of the adj + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
HIST	320	30 (/30)
PHYS	291	59 (/60)
POLISCI	244	60 (/60)
PHIL	222	29 (/30)
AL	201	59 (/60)
BIO	149	29 (/30)

Examples of this form are numerous and highly varied. To mention just a few examples, in History texts it was common to find adjectives modifying time periods (THE **first half** OF THE century; IN THE **recent history** OF THE; IN THE **late decades** OF THE), while in Biology, there were many adjectives describing physical characteristics **finite thickness, lower permeability, structural complexity, dry density**, etc. Although these tendencies were present to some degree, the overall frequency and variety of examples of the *adjective + noun* form prevent any broad description of their behavior.

While only about a quarter as frequent as *adj + noun* sequences, *noun + noun* patterns were also frequent among 2-word fillers. Their relative frequency among the disciplines was

higher in Biology and Applied Linguistics compared to the adj + noun pattern, and their presence was much more common in Physics than all other disciplines.

Table 5.13

Frequency of the noun + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
PHYS	161	58 (/60)
BIO	57	26 (/30)
AL	47	53 (/60)
POLISCI	46	54 (/60)
HIST	37	27 (/30)
PHIL	28	18 (/30)

Examples of the use of noun + noun sequences in filler also varied considerably in type. In Physics texts, these sequences often referred to physical properties or objects (**charge density, dust particles, gas phase, host molecules**), elements of research design or methodology (**sensor response, target input**) or units of measurement (**fermi level, Raman spectrum, f factor**). In contrast, a social science field like Applied Linguistics, less concrete and more research-oriented terms were more common, such as **evaluation component, group statistics, corpus approach, and questionnaire phase**. Like the adj + noun pattern, noun + noun fillers are highly variable, and often field or methodology-specific terms, though there are a few more general academic terms, like **research question**.

The verb + noun pattern was also relatively frequent and consisted of participial -ing and -ed verbs taking an adjectival role premodifying the head noun, as in **rising poverty, varying effectiveness, and reduced certainty**.

Table 5.14

Frequency of the verb + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
PHYS	29	40 (/60)
BIO	7	8 (/30)
AL	18	40 (/60)
POLISCI	18	37(/60)
HIST	26	26 (/30)
PHIL	7	2 (/30)

This type of filler pattern was most common in Physics texts, often with a nominalized verb representing motion or actions of the object of study (**scattering pattern, binding energies**). More general terms also appeared across disciplines, occurring with verbs relating to researcher expectations (**predicted behavior, predicted density**), actions of data analysis (**normalized distribution**), or description of a variable's change over time (**increasing prevalence, rising poverty**).

5.3.2.2. Three-Word POS Patterns

As compared to 2-word POS patterns, we see both a lower overall range and frequency in 3-word patterns, which may be attributed both to the greater variety of possible combinations and the lower overall number of valid 3-word fillers. The most frequent 3-word POS tag patterns can be seen as expansions of the most frequent 2-shorter forms. For example, the most frequent one, *adj + noun + noun*, adds an additional noun to the most frequent 2-word pattern *adj + noun*, and *adj + adj + noun* adds a second adjective. Also, third most frequent, *noun + conj + noun*, expands the basal single word *noun* filler by adding a second unmodified base noun, giving *noun + conjunction + noun*. Adverbs are also added to modify the premodifying adjective relatively frequently (**most unequal distribution, unusually high levels, more detailed portrait**), as are nouns with two nominal premodifiers (**proton tunneling system, voting age population**). One

pattern includes a preposition (with the ‘ADP’ or adposition tag) as well. The three most frequent of these patterns are discussed in more detail below.

Table 5.15

Ten most frequent 3-word filler POS patterns

POS tag pattern	Raw freq	Range
ADJ NOUN NOUN	469	167
ADJ ADJ NOUN	351	152
NOUN CCONJ NOUN	293	134
NOUN ADP NOUN	231	123
ADV ADJ NOUN	221	121
NOUN NOUN NOUN	180	96
NOUN ADJ NOUN	51	44
NOUN ADP PROPN	50	36
NUM NOUN NOUN	49	34
VERB ADJ NOUN	42	35

The most frequent 3-word POS pattern was adj + noun + noun, which appeared far more often in Physics texts than the other disciplines. This was true both for frequency, where it occurred three times as often as the next discipline, and for range, where it appeared in nearly all texts, while in others it appeared in approximately half of fewer of the texts.

Table 5.16

Frequency of the adj + noun + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
PHYS	61	51 (/60)
HIST	19	18 (/30)
POLISCI	18	38 (/60)
AL	14	34 (/60)
BIO	14	16 (/30)
PHIL	8	9 (/30)

One common realization of this form has the adjective serving to judge the degree or significance of a measure, the first noun as the variable of interest, and the second noun as the type of measurement. Examples include **neutral air turbulence**, **suitable sensor properties**, **experimental bowing parameters**, **negative value tails**, **strong NL emission**, and **effective absorption coefficient** in physics as well as **low spoilage rate** and **relative syllable duration** in applied linguistics and **standardized regression coefficients** in political science. *Adj + noun + noun* patterns also often take the form of an adjective + noun combination together premodifying the second noun, as in **old growth forest**, **worst case interpretation**, and **punctuated equilibrium theory**. One filler of that pattern type, **right hand side**, occurred particularly frequently (as discussed in Section 4.2.1.2), though most fillers of this length and pattern are not frequent.

The next most frequent 3-word POS pattern was *adjective + adjective + noun*, which appeared most often in history texts, where it appeared in nearly all texts (27/30). IT was also relatively frequent in physics and political science, but at a lower range.

Table 5.17

Frequency of the adj + adj + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
HIST	30	27 (/30)
PHYS	22	28 (/60)
POLISCI	21	37 (/60)
AL	13	33 (/60)
PHIL	11	17 (/30)
BIO	8	10 (/30)

In this pattern, the first adjective often described size or age/time characteristics of the following *adj* + *noun* combination. Examples from various disciplines include **large regional variation**, **newer suburban regions**, **smaller suburban municipalities**, **low initial plausibility** (POLISCI) **ongoing urban deterioration**, **first few years**, **vast empty desert** (HIST) **central angular positions**, **large systematic uncertainties** (PHYS) **novel grammatical meanings** (AL) **early modern system**, and **prolonged historical decline** (HIST). Only one filler of this form, **infinite linear relations** was among the 25 most frequent 3-word fillers (see Section 4.2.1.2), and it only appear three times.

The third most frequent POS tag pattern for 3-word fillers was *noun* + *conjunction* + *noun*, which occurred in lexical frames 293 times per million words. As seen in the breakdown in **Table 5.18**, it appeared most often in history texts, across 21 of the 30 texts in that subcorpus. It was relatively uncommon in physics texts, unlike the two previously discussed 3-word POS patterns.

Table 5.18

Frequency of the noun + conjunction + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
HIST	29	21 (/30)
PHIL	17	18 (/30)
POLISCI	15	28 (/60)
AL	15	36 (/60)
BIO	8	10 (/30)
PHYS	6	16 (/60)

The *noun + conj + noun* pattern was used most often in history texts, and while this is not a study focused primarily on disciplinary variation, it appears that the pattern is used much more in ‘soft’ than ‘hard’ sciences, as it is rare in both biology and physics, and most common in social sciences like history and philosophy. **Presence or absence** (9) and **safety and security** (3) were the only fillers of this pattern that appeared in the 25 most frequent 3-word fillers, and **security** only appeared in a single text.

Fillers of this type often combine two nominalized actions, such as **arrest and disarmament, organization and implementation** (HIST), **planning and organization** (POLISCI) and **addition or deletion** (BIO). Similarly, nominalized forms of adjectives such as **morality and effectiveness** (HIST) also appear. When the conjunction is “or”, we often see nouns representing opposites, as in **inclusivity or exclusivity** (AL), **success or failure** (POLISCI), **addition or deletion** (BIO), and **generation or dissipation** (PHYS). Sometimes the “or” form shows a second alternative name given for the first noun, as in **gap or pseudogap** (PHYS), **antecedents or determinants** (AL), and **scope or ‘field’** (PHIL).

5.3.2.3. *Four Word POS Patterns*

4-word POS tag patterns have an overall lower frequency and range than 3-word patterns, as expected, and as will be the case with increasingly long patterns. These patterns show increasing levels of complexity, including coordinated adjectives (*adj + conj + adj + noun*) and nouns (*adj + noun + conj + noun*) and prepositional phrases as noun postmodifiers (*noun + adp + adj + noun*), both of which are relatively common among the frequent pattern types. Three of the most frequent patterns will be discussed in detail below.

Table 5.19*Ten most frequent 4-word filler POS patterns*

POS tag pattern	Raw Freq	Range
ADJ CCONJ ADJ NOUN	120	75
NOUN ADP ADJ NOUN	101	74
NOUN ADP DET NOUN	93	71
ADJ NOUN ADP NOUN	64	52
ADJ NOUN NOUN NOUN	62	46
ADJ ADJ NOUN NOUN	47	37
NOUN ADP NOUN NOUN	45	37
ADJ NOUN CCONJ NOUN	38	30
NOUN CCONJ NOUN NOUN	32	30
NOUN CCONJ DET NOUN	18	18

The most frequent 4-word POS tag pattern was *adjective + conjunction + adjective + noun*. This pattern in less than a third of the texts in the corpus, though it appeared in two thirds of the history texts. It had similar frequencies in political science and applied linguistics, though with lower ranges.

Table 5.20*Frequency of the adj + conjunction + adj + noun POS tag pattern across disciplines*

Discipline	Freq per 100k	Range
HIST	38	21 (/30)
POLISCI	35	20 (/60)
AL	30	16 (/60)
PHYS	10	8 (/60)
PHIL	6	6 (/30)
BIO	4	4 (/30)

Overall, there were no frequently recurring 4-word fillers in the corpus; however, a few fillers of this form appeared three times (**second and third terms**) or twice (**direct and indirect effects, real and imaginary parts, human and financial resources, internal and external features**). The coordinated adjectives in this structure often represent contrasting ideas, such as in **real and imaginary, internal and external, contemporary and future significance, global and local claims, static and dynamic properties**, and **narrow or holistic view**. Though it is beyond the scope of this study, it may be interesting to explore in more detail whether patterns exist in the types of words that commonly fill the noun slot in this form, which a future version of the tool developed in this study could facilitate by identifying fillers based on their ‘root’ noun.

The second most frequent 4-word POS tag pattern was *noun + preposition + adjective + noun*, which occurred 101 times per million words and in nearly a third of the texts in the corpus. The pattern was most common in political science texts, and least common in biology.

Table 5.21

Frequency of the noun + preposition + adjective + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
POLISCI	29	22 (/60)
HIST	22	13 (/30)
AL	20	16 (/60)
PHYS	18	13 (/60)
PHIL	7	6 (/30)
BIO	5	4 (/30)

In this form, the first noun often served as a measure of some kind, as in **number of free parameters** (BIO), **level of linguistic awareness** (AL), **lack of sufficient conductivity** (PHYS), and **absence of legislative powers** (POLISCI). It was also common to find nominalized verbs in

the first noun position, **opening of western archives** (HIST), **performance of native speakers** (AL), **diffusion of hybrid corn** (POLISCI). In other cases the first noun was a nominalization of an adjective describing a characteristic of the following noun, as in **legitimacy of many organizations** (POLISCI), **pervasiveness of oral activities** (AL), and **sensitivity of light scattering** (PHYS).

The third most frequent 4-word POS tag pattern was *noun + prep + determiner + noun*, which occurred with a frequency and range nearly as high as the previous pattern. It appeared most often in applied linguistics texts, though it appeared in more political science texts, and least often in biology. Due to its internal determiner, this pattern can be seen as an expansion of single-word filler patterns like THE + OF THE. For example, in the following example

Based on data on plant population dynamics collected since the beginning of the 1970s in the study archipelago [...] (BIO)

two frames can be found; the first is “THE **beginning** OF THE” and the second, which fits this pattern, is “THE **beginning of the 1970s** IN THE”. These overlaps sometimes resulted in ambiguous phrase component boundaries and required careful manual validity checking. This was particularly the case when the closing fixed words of the frame began with a preposition, indicating possible prepositional phrase overlap across the lexical frame boundary.

Consequently, most valid fillers with internal prepositions are found in frames ending in “IN THE” rather than “OF THE”, as “IN” often begins an adverbial prepositional phrase component in a sentence rather than a part of the noun phrase. Fillers with internal prepositional phrases were also commonly found in frames ending in “AND”, though this also brought along some ambiguity when coordinated noun phrases were involved.

Table 5.22

Frequency of the noun + preposition + determiner + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
AL	24	16
POLISCI	22	19
HIST	20	12
PHYS	15	13
PHIL	7	6
BIO	5	5

By far the most common preposition that appeared in this pattern was “of”, and it often served to connect a characteristic in the first noun to the main noun in the second position. Examples include **competitiveness of the race**, **power of the judiciary** (POLISCI), **order of the questions**, **meaning of the word** (AL), and **diffusivities of the ions** (PHYS). Like the previously discussed pattern, it also included many nominalized verbs, with the main verb nearly always acting as the object of the verb, as in **imposition of the duty**, **renovation of the crypt** (HIST), **division of the brain** (PHIL), abolition of the quangos (POLISCI), **determination of the CI** (PHYS).

5.3.2.4. Five Word POS Patterns

None of the five-word POS patterns are highly frequent, nor do they appear in a large number of texts in the corpus, which is not unexpected as fillers get longer and more specialized. The majority of the most frequent 5-word patterns include a prepositional phrase as a nominal postmodifier, some of which include adjectives modifying both of the nouns in the structure. Patterns containing coordinated nouns and adjectives also make up a relatively large portion of the patterns of this length. The three most frequent, all of which include prepositional phrases, are detailed below.

Table 5.23*Ten most frequent 5-word filler POS patterns*

POS tag pattern	Raw freq	Range
NOUN ADP DET ADJ NOUN	42	30
ADJ NOUN ADP DET NOUN	27	26
NOUN ADP DET NOUN NOUN	26	25
ADJ NOUN ADP NOUN NOUN	19	18
ADJ NOUN CCONJ ADJ NOUN	16	14
ADJ NOUN ADP ADJ NOUN	15	12
ADJ ADJ CCONJ ADJ NOUN	13	13
NOUN ADP ADJ NOUN NOUN	13	13
ADV ADJ CCONJ ADJ NOUN	11	9
ADJ ADJ NOUN ADP NOUN	10	8

The most frequent 5-word POS tag pattern was *noun + preposition + determiner + adjective + noun*, a noun with a prepositional phrase as a postmodifier. It occurred about 50% more than the next most frequent pattern but did not have a high overall frequency. The pattern appeared most often in political science and physics texts, and was uncommon in applied linguistics and philosophy.

Table 5.24*Frequency of the noun + preposition + det + adj + noun POS tag pattern across disciplines*

Discipline	Freq per 100k	Range
POLISCI	16	11
PHYS	12	8
HIST	8	5
AL	4	4
PHIL	2	2

Fillers following this pattern often show a nominalized verb in the first noun position, with the second noun serving either as the subject, as in **growth of the environmental movement** (POLISCI) and **decay of the metastable state** (PHYS), or the object of the verb, as in **overthrow of the tsarist autocracy** (HIST) and **introduction of a Latin script** (POLISCI). More abstract nouns representing a characteristic of the second noun were also common in the first noun position, as in **nature of the homo level**, **definition of the pdco ratio** (PHYS) and **violence of the postwar period** (HIST).

The second most frequent 5-word POS tag pattern was *adj + noun + prep + determiner + noun*, which is similar in form to the previous pattern, except that the noun in the first position is premodified by an adjective. The pattern was relatively infrequent in all disciplines, but, similar to the previous pattern, it was most frequent in political science and least frequent in philosophy.

Table 5.25

Frequency of the adj + noun + prep + determiner + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
POLISCI	8	8
PHYS	7	7
BIO	4	3
AL	4	4
HIST	4	4
PHIL	2	2

Fillers of this form most often contain a first noun that represents a characteristic of the second, with the adjective premodifying specifying something about that characteristic. This specification often refers to aspects like size, scope, degree, or importance. Examples related to importance include **main objective of the reorganization** and **broader implications of these**

findings (POLISCI). Others that describe degree of the characteristic in noun 1 include **deepest sensibilities of the Mesoamericans** (HIST), **lower permeability of the biofilm** (BIO), **high uniformity of the SWNTS** (PYS), and **vast majority of the students** (AL).

The third most frequent 5-word POS tag pattern was *noun + prep + determiner + noun + noun*, which is again similar in form to the previous patterns, but in this case the second ‘head’ noun is premodified by another noun. This pattern was generally infrequent but appeared much more often in physics texts than the other disciplines in the corpus.

Table 5.26

Frequency of the noun + prep + det + noun + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
PHYS	14	13
AL	4	4
POLISCI	4	4
HIST	3	3
BIO	1	1

Many examples of this form include an abstract first noun modifying a more concrete second noun + noun sequence. Examples include **nature of the classroom activities**, **role of the corpus approach** (AL) **relevance of this water source** (BIO) and **existence of the anatase-like cores** (PHYS). Another way this pattern is realized is when the first noun is a nominalized action and the second is the object of that action, as in **fulfillment of the commission's mandate** (POLISCI), **defeat of the soviets' allies** (HIST), and **choice of a superscript I** (PHYS).

5.3.2.5. Six Word POS Patterns

As 6-word frame fillers are infrequent in the corpus, 6-word POS tag patterns are as well. This is partly due to the wide variety of structures permitted by this number of words, and no one

pattern was particularly dominant among the 45 patterns that appeared at least twice in the corpus. All of the most frequent 6-word patterns contain a prepositional phrase element, with the noun components often elaborated with adjectives or coordination of multiple nouns. It should also be noted that some 6-word patterns containing subordinate clauses, though they were very infrequent as those types of structures are rarely complete within the bounds of a lexical frame.

Table 5.27

Ten most frequent 6-word filler POS patterns

POS tag pattern	Raw Freq	Range
ADJ NOUN ADP DET ADJ NOUN	11	10
ADJ NOUN ADP DET NOUN NOUN	10	10
NOUN ADP DET ADJ NOUN NOUN	8	8
NOUN ADP NOUN CCONJ NOUN NOUN	6	6
NOUN ADP DET ADJ ADJ NOUN	6	6
NOUN NOUN ADP DET NOUN NOUN	5	4
NOUN CCONJ NOUN ADP ADJ NOUN	5	5
ADJ NOUN CCONJ ADJ NOUN NOUN	5	2
NOUN ADP ADJ NOUN CCONJ NOUN	4	4
ADJ NOUN ADP NOUN CCONJ NOUN	3	3

Overall, 6-word fillers were not highly frequent, and specific patterns among them were therefore even more infrequent. However, the most common POS tag pattern, which appeared 11 times in the corpus, was *the adj + noun + prep + det + adj + noun*. It was more frequent in history than the other disciplines, but because frequencies were so low, comparisons are not extremely meaningful.

Table 5.28

Frequency of the adj + noun + prep + det + adj + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
HIST	4	3
POLISCI	2	2
AL	2	2
PHYS	2	2
BIO	1	1

In all but one instance of this pattern, the preposition in the center was “of”. In many cases, the first noun was a nominalized action with the second noun in the subject role, as in **global rise of the English language** (AL), **gradual opening of the military dictatorship** (POLISCI), and **rapid growth of the new monasticism** (HIST). This pattern is also used to describe time periods, as in **last quarter of the nineteenth century** and **second half of the twelfth century** (HIST), which is a reason for its frequency in history texts, and to highlight properties of the second noun, as in **pedagogical focus of the instructional activity** (AL) and **serious divergences of the Greek version** (HIST).

The second most frequent 6-word POS tag pattern, *adj + noun + prep + det + noun + noun*, is similar to the previous pattern, but rather than an adjective + noun combination in the second noun position, it contains a noun + noun sequence. It occurred at roughly the same frequency as the other three most frequent 6-word POS tag patterns, and like the others, it was both infrequent and occurred in few texts. It did not appear at all in the biology or philosophy subcorpora.

Table 5.29

Frequency of the adj + noun + prep + det + noun + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
PHYS	3	3
HIST	3	3
POLISCI	3	3
AL	1	1

There were very few instances of this form in the corpus, and because it nearly always includes the sequence “of the” in the middle, it may be seen as an extension of shorter frames. For example, in the sentence below, two different frame + filler combinations can be extracted.

some participants complained about THE "undemocratic" nature of the planning process AND THE lack of a written response from the mayor (HIST)

The shorter of the two is THE **"undemocratic" nature OF THE**, but the longer option THE **"undemocratic" nature of the planning process AND THE** is also valid because the filler represents a complete constituent as well.

Similar to the previous pattern, this one most often has a first noun specifying a property or component of the second noun, as in potential endogeneity of the resource variables (POLISCI), relevant content of the web page (AL), and boundary layer of the emitter surface (PHYS).

The pattern *noun + prep + det + adj + noun + noun* occurred at nearly the same very low frequency as the previously discussed 6-word POS tag patterns, and similar to the previous pattern, did not appear in biology or philosophy texts. Unlike the others, the first noun in this sequence is unmodified, while the second is premodified by both an adjective and a second noun.

Table 5.30

Frequency of the the noun + prep + det + adj + noun + noun POS tag pattern across disciplines

Discipline	Freq per 100k	Range
HIST	3	3
POLISCI	2	2
PHYS	2	2
AL	1	1

Similar to other 6-word patterns, this one often contained a nominalized verb in the first position with the second verb in the object role, a structure that may be seen in a way as a nominalized passive voice sentence. For example, the noun phrase **adoption of a human security framework** allows the sentence “a human security framework was adopted” to fill the noun role in the sentence

*Other significant statements include **THE adoption of a human security framework** FOR THE report by the un secretary general Kofi Annan’s 2004 high level panel on threats (POLISCI).*

In other cases, the second noun was the verb’s subject, as in **fall of the liberal Decazes government** (the liberal Decazes government fell) and **decline in the international spice trade** (the international spice trade declined).

Overall, this section has identified some structural tendencies among lexical frame fillers of particular lengths, without consideration of the frames they fill. The next section will take this next step and explore which structural patterns are most common for five of the most frequent frames identified in the corpus, as well as identifying the frames that are most and least variable in terms of filler length.

5.3.3. Structural Description of Frequent Frames

The previous sections described the frequency distribution of different filler lengths across the entire corpus, as well as frequencies of filler POS tag patterns. However, it is also important to consider how filler lengths and POS patterns may be associated to different degrees with specific lexical frames. To provide insight into this aspect of extended lexical frame characteristics, this section will explore how filler lengths and POS patterns vary in frequency among the five most frequent the frames that appeared most often in the corpus, giving examples of fillers matching the three most frequent POS tag patterns for each of the frames. In addition, this section will identify the frames that had the highest and lowest variability in filler length, as measured by entropy.

5.3.3.1. *Frame 1: THE + OF THE*

The most frequent frame found in this corpus, consistent with findings from previous studies of lexical frames in academic writing, was THE + OF THE. It occurred 3,244 times per million words in this corpus and in nearly all texts. It is important to note also that because this study looked at both 3- and 4-word frames, some instances of THE + OF THE were included instead inside the frames IN THE + OF THE (446 pmw), OF THE + OF THE: (420 pmw), TO THE + OF THE (340 pmw), ON THE + OF THE (240 pmw), and AND THE + OF THE (206 pmw). Therefore, the true frequency of this 3-word frame could be interpreted as being even higher. While its overall frequency was high, its frequency varied across disciplines (**Table 5.31**), with physics texts using it the most, and biology texts using it the least.

Table 5.31

Frequency and range of the frame THE + OF THE across disciplinary subcorpora

Discipline	Raw freq	Norm freq (per 100K)	Range
PHYS	1476	391	58
HIST	1037	367	30
POLISCI	1018	241	58
AL	880	200	59
PHIL	559	199	29
BIO	201	130	29

When analyzing the variability of filler lengths with this frame, its entropy value was found to be 1.47, near the average value for the frames found in this study. More than half of fillers were one word long, and over a quarter were two words long. 3+ word fillers together made up a little over 10% of all instances (details in **Table 5.32**).

Table 5.32

Filler length frequencies and proportions for THE + OF THE

Length	Raw freq	Norm Freq (pmw)	% total	Range
1	3163	1985	61.17%	258
2	1374	862	26.57%	247
3	426	267	8.24%	166
4	150	94	2.90%	91
5	33	21	0.64%	30
6	25	16	0.48%	23

Seven POS tag patterns occurred in at least 1% of valid fillers for THE + OF THE. The most common POS tag pattern was a single noun, as the study is only focused on frames with

noun-based fillers. 80% of the instances of this frame include either *noun* or *adj + noun*, which suggests that this frame does not include a lot of structural variety in its fillers; however, as will be seen in the subsequent sections, this value is fairly similar for all of the frequent frames.

Table 5.33

POS tag patterns occurring more in more than 1% of instances of THE + OF THE

Pattern	Raw Freq	% total
NOUN	3134	61.00%
ADJ NOUN	988	19.00%
NOUN NOUN	267	5.00%
ADJ NOUN NOUN	91	2.00%
ADJ ADJ NOUN	82	2.00%
NOUN CCONJ NOUN	70	1.00%
VERB NOUN	69	1.00%

Looking into some of the more frequent multi-word filler patterns for THE + OF THE, we first see the pattern *adjective + noun*. A wide variety of fillers matched this pattern, and while there was no single high frequency two-word sequence that dominated, the head noun **part** with an adjective modifier appeared 30 times, with **dominant part** (2), **first part** (2), **last part** (2), **positive part** (4). There were also many 23 examples expanding on the head noun **nature**, including **personal nature** (2), **precise nature** (2), **public nature** (3), **complex nature** (1), and **controversial nature** (1). These types of examples illustrate the value of the extended lexical frame approach, as this seems to be a salient filler “family” that would not be possible to identify using traditional lexical frame identification approaches.

The majority of *noun + noun* fillers for this frame were specialized and low frequency, but similar groupings could also be found. The most common head noun was **dependence**, which appeared 20 times. The most common premodifying noun was **temperature dependence** (14), and though no others were nearly as frequent, some included **time dependence** (2) and **energy dependence** (2). The head noun **time** also appeared multiple times, with 11 instances including **coherence time** (7), **decoherence time** (1), **relaxation time** (1), **residence time** (1) and **correlation time** (1). The filler **coherence length** appeared 12 times, with no variation in the first noun, indicating that this is more of a fixed unit of specialized vocabulary, though across the all frames the noun **length** did appear with other nominal premodifiers.

Fillers of the pattern *adj + noun + noun* were even more specialized and include few repetitions. The most common head noun was **dependence**, which was also common among *noun + noun* fillers, specifically as **temperature dependence**, which appeared within the *adj + noun + noun* pattern as **residual temperature dependence** (2), **residual temperature dependence** (2), **observed temperature dependence** (2), and **complex temperature dependence** (2). No other head noun recurred more in this pattern and frame, but examples that occur more than once, such as **nonlinear probability distribution**, are likely part of filler families across the whole corpus. This suggests that it would be valuable to augment simple frequency-based frame + filler identification with a new approach to identifying fillers based on noun ‘heads’ and their elaboration. It may be possible to implement this method using existing NLP libraries and will be the subject of future work.

5.3.3.2. Frame 2: *IN THE + OF*

The second most frequent frame found in this corpus was *IN THE + OF*, which occurred 1,302 times per million words in this corpus and in nearly all texts. As in the previous frame, its

frequency would be higher if 4-word word frames had not been included in the study, specifically the frame THE + OF THE (446 pmw). Unlike THE + OF THE, the frequency of IN THE + OF did not vary greatly across disciplinary subcorpora, though it was most frequent in history texts.

Table 5.34

Frequency and range of the frame IN THE + OF across disciplinary subcorpora

Discipline	Raw freq	Norm freq (per 100k)	Range
HIST	430	152	30
PHIL	320	114	30
POLISCI	433	103	59
BIO	151	98	30
PHYS	343	91	56
AL	398	90	56

The entropy value for filler length frequencies of IN THE + OF was found to be 1.09, or below average, indicating a relatively high degree of predictability of filler length. This is because more than 75% of all fillers were one word long, and just under 17% two words long. Combined, all 3+ word fillers constituted just over 7% of all instances (details in **Table 5.35**).

Table 5.35

Filler length frequencies and proportions for IN THE + OF

Length	Raw freq	Norm Freq (pmw)	% total	Range
1	1582	993	76.24%	254
2	345	216	16.63%	173
3	99	62	4.77%	66
4	37	23	1.78%	34
5	9	6	0.43%	9
6	3	2	0.14%	3

Only four POS tag patterns occurred in at least 1% of valid fillers for IN THE + OF (Table 5.36), as compared to seven for the THE + OF THE, which reflects the low entropy value for the frame. Aside from single noun fillers, which were the most common pattern for all frames, the other patterns occurring in more than 1% of frames were *adj + noun* (12%) *noun + noun* (2%) and *noun + conj + noun* (1%).

Table 5.36

POS tag patterns occurring more in more than 1% of instances of IN THE + OF

Pattern	Freq	% total
NOUN	3134	75%
ADJ NOUN	988	12%
NOUN NOUN	267	2%
NOUN CCONJ NOUN	70	1%

There were no highly frequent individual *adj + noun* sequences, and no single head noun that was highly frequent among them, but the most common was **year(s)**, which appeared 8 times with the adjectives **early** (4), **first** (1), **second** (1), **final** (1), and **latter** (1). The noun **column** appeared 4 times, with different numbers in the adjective position, representing references to tables in academic articles. Others included **sense** (5), which was modified by adjectives in phrases like IN THE **literal sense** OF , IN THE **simple sense** OF, and IN THE **classic sense** OF. And **work**, including **late work**, **pioneering work**, and **seminal work**.

Lie algebra (3) was the only recurring noun + noun filler for this frame. However, the most common noun in the *noun + noun* fillers for this frame was **range** (6), which was modified by **energy** (4), **temperature** (1), and **wavelength** (1). **Density** appeared three times, as **number**

density, electron density, and packing density. All others appeared only once or twice, though it is likely that expanding this method of identification across all frames would

Noun + conjunction + noun was the fourth most frequent pattern for this frame, and while it only represents 1% of fillers, it is worth exploring as this pattern was not described for the previous frame. No repeated fillers were found, other than **recruitment and retention**, which appeared twice, and no filler ‘families’ were apparent based on a head noun, possibly in part because in this pattern both nouns have coequal status. The most common noun types occupying these slots are nominalized actions, such as **identification and classification, cultivation and export, and teaching and learning.** In other cases, the nouns represent attributes of a head noun that appears following the lexical frame, as in “IN THE **competitiveness and cost** OF judicial elections”.

5.3.3.3. *Frame 3. OF THE + OF*

The third most frequent frame found in this corpus was OF THE + OF, which occurred 1,064 times per million words and in nearly all texts. Like the previous frames, its frequency would be higher if 4-word word frames had not been included in the study, specifically the frame OF THE + OF THE (420 pmw). It appeared most frequently in political science texts and least often in biology.

Table 5.37

Frequency and range of the frame OF THE + OF across disciplinary subcorpora

Discipline	Raw freq	Norm freq (per 100k)	Range
POLISCI	355	125	56
PHIL	284	100	28
AL	371	98	57
HIST	337	80	29
PHYS	272	72	50
BIO	77	50	25

The entropy value for filler length frequencies of OF THE + OF was 1.55, meaning the predictability of filler length is around average for the frames identified in this study. More than half of all fillers were one word long, and 30% of fillers were 2-word, which was the highest proportion of 3-word fillers found among the most frequent frames. (details in **Table 5.38**). Proportions of 3+ fillers were similar to THE + OF THE and roughly average overall.

Table 5.38

Filler length frequencies and proportions for OF THE + OF

Length	Raw freq	Norm Freq (pmw)	% total	Range
1	949	595	56.0%	225
2	523	328	30.8%	189
3	153	96	9.0%	94
4	47	29	2.8%	40
5	18	11	1.1%	16
6	6	4	0.4%	6

Nine POS tag patterns occurred in at least 1% of valid fillers for OF THE + OF (**Table 5.39**), as compared to seven for the THE + OF THE, which reflects the slightly above average

entropy value for the frame. The same seven patterns from THE + OF THE were present, but with the addition of *number* + *noun* and *adverb* + *adjective* + *noun*, which were less common among the other frames and only appeared in more than 1% of fillers for two frames. Examples of the *number* + *noun* pattern include “each pair shared their language background with one OF THE **six sets** OF candidates” (applied linguistics) and “one or more OF THE **four categories** OF citizens” (political science). Examples of the *adverb* + *adjective* + *noun* pattern include “free OF THE **more pernicious effects** OF colonialism” (history) and “one OF THE most popular telenovelas OF recent years” (political science).

Table 5.39

POS tag patterns occurring more in more than 1% of instances of OF THE + OF

Pattern	Freq	% total
NOUN	907	53%
ADJ NOUN	399	24%
NOUN NOUN	49	3%
VERB NOUN	34	2%
NOUN CCONJ NOUN	29	2%
NUM NOUN	22	2%
ADJ NOUN NOUN	22	1%
ADV ADJ NOUN	21	1%
ADJ ADJ NOUN	19	1%

The most frequent *adj* + *noun* filler for OF THE + OF was **global hegemony**, which appeared 21 times without variation. No other fixed sequences were frequent, but there were variations on several head nouns. For example, **number** was modified by the adjectives **large** (4), **total** (3), **raw** (1), and **uneven** (1). **Form** was also modified by **active** (3), **different**,

earliest (1), grammatical (1), and modular (1). Other head nouns that appeared several times with different premodifying adjectives were **conception (evaluative conception (4), cognitive conception (2), legalistic conception (1)),** and **consequences,** which appeared with several non-recurring adjectives including **potential, key, positive, political, possible, negative,** and **downstream.**

Noun + noun fillers for the frame OF THE + OF only included one recurring sequence, **housing act (2),** and no recurring head nouns. This may be attributed in part to the overall infrequency being only 48, and a search for this pattern across all frame fillers would likely uncover many more. Patterns of this type often include a more concrete or specific first noun premodifying an abstract or non-specific second noun, as in **temperature variation, displacement vectors, literacy experiences,** and **coast ranges.**

Recurring patterns of the *verb + noun* type included the noun **nature,** which was premodified by the participial verbs **changing, contested,** and **nominalized,** and **system,** which was premodified by **proposed** four times and **unchanging** once. Recurring premodifying verbs included **growing,** which appeared as **growing strength, growing prestige, growing group,** and **growing understanding,** and **emerging,** which included **emerging genre** and **emerging system.** Along with **unfolding** (organization) and **changing** (nature), these verbs all carry a meaning of change or increase over time.

5.3.3.4. *Frame 4. THE + IN THE*

The fourth most frequent frame found in this corpus was THE + IN THE, which occurred 926 times per million words in this corpus and in nearly all texts. It overlaps with the 4-word frame OF THE + IN THE (230 pmw), so when comparing to frequencies reported in other frames studies, it should be noted its frequency would be higher if 4-word word frames had not

been included. Its frequency varies considerably by discipline (**Table 5.40**), with the highest in physics (128 per 100k) and the lowest in philosophy (28 per 100k).

Table 5.40

Frequency and range of the frame THE + IN THE across disciplinary subcorpora

Discipline	Raw freq	Norm freq (per 100k)	Range
PHYS	484	128	57
HIST	226	80	29
POLISCI	331	78	57
AL	296	67	55
BIO	59	38	22
PHIL	80	28	28

At 2.39, the entropy value for THE + IN THE was the highest of all the frames identified in this study. This value indicates that the length of a filler is relatively unpredictable, as an entropy value of zero would indicate that filler length is 100% certain, meaning all fillers are of the same length. In this case, there is an especially low proportion of 1-word fillers (26%) when compared to other frames, which averaged above 50%. In fact, 2-word fillers were just as common for this frame, which was an unexpected result, and 3-word fillers were nearly as common. 4- 5- and 6-word fillers all had unusually high frequencies as well.

Table 5.41

Filler length frequencies and proportions for THE + IN THE

Length	Raw freq	Norm Freq (pmw)	% total	Range
1	386	242	26.2%	166
2	389	244	26.4%	173
3	276	173	18.7%	141
4	225	141	15.2%	127
5	140	88	9.5%	95
6	60	38	4.1%	53

This frame included both greater frequency and diversity of POS tag patterns, with 14 occurring in at least 1% of valid fillers (**Table 5.42**), which was considerably more than the other frequent frames described in this section. Patterns that were not present in the other frames included:

noun + prep + noun (5%)

“THE **campaigns against heresy** IN THE later part of the century” (history)

noun + prep + adj + noun (3%)

“manage THE **evolution of antimicrobial resistance** IN THE future” (biology)

noun + prep + det + noun

“accuracy in THE **use of these functions** IN THE pre test” (applied linguistics)

adj + noun + prep + noun

“THE **relative importance of latifundia** IN THE case of the Philippines” (pol sci)

propn + noun

“pressure THE **Sudan government** IN THE lead up to independence” (history)

Propn

“towns across THE **United States** IN THE 1960s” (pol sci)

noun + prep + noun + noun.

“THE **timing of cell division** IN THE future germ line” (biology)

Unexpectedly, however, the pattern *noun + conj + noun* was not present in at least 1% of fillers, unlike many other frames identified.

Table 5.42

POS tag patterns occurring more in more than 1% of instances of THE + IN THE

Pattern	Freq	% total
NOUN	356	24%
ADJ NOUN	214	14%
NOUN NOUN	88	6%
NOUN PREP NOUN	79	5%
ADJ NOUN NOUN	47	3%
NOUN PREP ADJ NOUN	42	3%
NOUN PREP DET NOUN	36	2%
ADJ ADJ NOUN	31	2%
ADJ NOUN PREP NOUN	29	2%
PROPN NOUN	28	2%
NUM NOUN	20	1%
PROPN	19	1%
NOUN PREP NOUN NOUN	18	1%
VERB NOUN	17	1%

Among the fillers for THE + IN THE that had the *adj + noun*, there were no frequently occurring fixed sequences, with only **foreign language** (3) appearing more than twice.

However, the head noun **term(s)** appeared most frequently (10) with a variety of adjectives, including **first** (3), **relevant** (2), **last**, and **subsequent**. Another was **change** (7), which appeared in **large change**, **abrupt change**, **gradual change**, and **substantial change** among

others. If we consider recurring adjectives rather than nouns, however, we find **first** appearing nine times, in forms like **first term**, **first stage**, and **first element**. This approach may be interesting to explore but arguably represents a less coherent grouping than noun-based filler families, and would be beyond the scope of this study, whose focus is on complex noun phrases as fillers.

None of the *noun + noun* fillers in this frame were frequent, though a few specialized terms like **G base**, **bond contraction**, and **population inversion** appeared more than twice. No head noun recurred more than twice, but there were some recurring nouns in the first position, including **energy** (**energy flux**, **energy losses**) and **neutron** (**neutron energy**, **neutron velocity**), cases where the second noun may be seen as describing a characteristic of the first.

When considering *noun + preposition + noun* fillers, it is more informative to look at the first noun as a head rather than the second, which rarely recurred. Most frequent among the first position nouns was **number** (13), which was followed by general terms like **number of questions** (3) and **number of events** (3), as well as more specific, concrete nouns like **number of atoms** (1), **number of threads** (1), and **number of words** (1). Though less frequent, **role** recurred in the first position (**role of language**, **role of metacognition**, **role of party**, all appearing once), as did **use** (**use of charcoal**, **use of ITCS**, and **use of technology**, all once). All of these recurring first nouns are among the most frequent 1-word fillers, which supports the value of developing a method of searching based on a head noun rather than simple frequency, as it may be informative to see how a single frequent filler is elaborated in frames across a corpus.

5.3.3.5. *Frame 5. TO THE + OF*

The fifth most frequent frame found in this corpus was TO THE + OF, which occurred 799 times per million words in this corpus and in the majority of texts. It overlaps with the 4-

word frame TO THE + OF THE (340 pmw), so like most of the others, its frequency would have been higher in a 3-word frame study. TO THE + OF was used most often. in philosophy texts, and least in biology, and frequency varied moderately across disciplines overall.

Table 5.43

Frequency and range of the frame TO THE + OF across disciplinary subcorpora

Discipline	Raw freq	Norm freq (per 100k)	Range
PHIL	248	88	30
HIST	233	82	30
POLISCI	273	65	56
PHYS	220	58	51
AL	233	53	57
BIO	67	43	22

TO THE + OF had a roughly average entropy value (1.34) among these frames. The proportions of 1, 2 and 3-word fillers were similar to many other frames, and 4+ words were very infrequent (details in **Table 5.44**).

Table 5.44

Filler length frequencies and proportions for TO THE + OF

Length	Raw freq	Norm Freq (pmw)	% total	Range
1	845	530	66.3%	227
2	293	184	23.0%	139
3	102	64	8.0%	75
4	21	13	1.7%	21
5	11	7	0.9%	11
6	1	1	0.1%	1

More than 80% of the instances of this frame follow patterns of either *noun* or *adj* + *noun*, and overall the frame does include a lot of structural variety in its fillers; however, it is notable that the frequency of the pattern *noun* + *conj* + *noun* makes up a higher proportion of the fillers for TO THE + OF than any of the other frames examined, though still low overall.

Table 5.45

POS tag patterns occurring more in more than 1% of instances of TO THE + OF

Pattern	Freq	% total
NOUN	822	65%
ADJ NOUN	229	18%
NOUN CCONJ NOUN	34	3%
VERB NOUN	27	2%
NOUN NOUN	25	2%
ADV ADJ NOUN	16	1%
ADJ ADJ NOUN	15	1%

The most frequent *adj* + *noun* sequences in this frame were **global hegemony**, a specialized term from one text that appeared seven times in the corpus, and **overall state**, a more general term. The head noun **number** appeared nine times, with the adjectives **total** (3), **small** (3), **high** (1), **large** (1), and **larger** (1), and **range** appeared six times, as **full range** (3), **diverse range** (1), **entire range** (1), and **broad range** (1). The majority of the other head nouns were general, such as **distribution**, **effects**, **forms**, **value**, and **experience**, while premodifying adjectives included a balance of general (**overall**, **total**, **small**, **full**) and topic or discipline-specific (**epistemological**, **educational**, **psychological**, **radiative**).

The only recurring filler of the *noun + conjunction + noun* form for this frame was **safety and security**, which appeared three times. There were no other recurring nouns in either the first or second position. In many cases the two nouns were nominalized verbs, with the head noun following the frame in the object role, as in

*this feedback was usually related to academic aspects rather than TO THE **organization and delivery** OF their AP [academic presentation] (AL)*

*integral TO THE **demobilisation and reinsertion** OF combatants (POLISCI)*

*these same considerations turn out also to apply TO THE **assertion and justification** OF mathematical propositions (PHIL)*

In other cases, the coordinated nouns are attributes or possessions of the later head noun, and the structure works as an “of” genitive, as in the following examples:

*the experiments seem to reveal a fairly sophisticated sensitivity TO THE **presence and behaviour** OF physical objects (PHIL)*

*URL names and domains can be pointers TO **THE authority and reliability** OF website content (AL)*

There were no verb + noun fillers that repeated more than twice, nor were there any repeating ‘head’ nouns among them. However, it may be worth noting that among the premodifying verbs, many were related to growth or change over time, including **increased** (3), **growing** (2), **emerging** (1), and **unfolding** (1).

5.3.3.6. Lowest entropy frame: AS A + OF

While the highest entropy frame (THE + IN THE, see above) was among the five most frequent frames in this corpus, the least variable (entropy = 0.83) frame AS A + OF ranked 21st in order of frequency, appearing 202 times per million words.

Table 5.46

Frequency and range of the frame TO THE + OF across disciplinary subcorpora

Discipline	Raw freq	Norm freq (per 100k)	Range
POLISCI	147	35	44
PHYS	131	35	44
AL	123	28	43
PHIL	109	39	27
HIST	99	35	27
BIO	33	21	17

The entropy value of 0.83 is especially low because more than 80% of all fillers were one word long, a much higher proportion than any other frames found in this study (**Table 5.47**). It is interesting to note, however, that even for the least variable frame, a traditional 1-word filler approach to lexical frames identification misses more than 15% of instances. For the remaining filler lengths, just under 15% of fillers were two words long, and 3+ word fillers made up a comparatively small proportion, roughly 3% of all instances

Table 5.47

Filler length frequencies and proportions for AS A + OF

Length	Raw freq	Norm Freq (pmw)	% total	Range
1	528	331	82.24%	187
2	94	59	14.64%	69
3	15	9	2.34%	13
4	1	1	0.16%	1
5	0	0	0%	0
6	1	1	0.16%	1

This frame's fillers contain the least structural variety found in this study, with 82% of the instances being a single noun, and only two other patterns present in more than 1% of instances. The pattern *adj + noun* represented 13% of fillers, and *noun + noun* sequences were an additional 1%.

Table 5.48

POS tag patterns occurring more in more than 1% of instances of AS A + OF

Pattern	Freq	% total
NOUN	527	82%
ADJ NOUN	82	13%
NOUN NOUN	7	1%

The most frequent head noun among the *adj + noun* fillers for AS A + OF was **result**, which appeared as **direct result** (2) and **possible result** (2), and represents an expansion of the most frequent 1-word filler **result**, which appeared 91 times for this frame. The head noun **part** appeared three times (**routine part**, **specific part**, **normal part**), nearly as often as the 1-word filler **part**, which only appeared four times. The frequent 1-word filler **function** (79), however, did not appear as part of any *adj + noun* combinations. The third POS tag pattern mentioned in **Table 5.48**, NOUN NOUN, only appeared seven times in this frame with no repeating nouns in either position, making it infeasible to identify any meaningful patterns in their form or usage.

5.4. Phase 2 Summary

The findings presented in this chapter have shown that multi-word fillers are relatively high in frequency, representing nearly half of all lexical frame occurrences in the corpus. Longer fillers tended to be more specialized or topic-specific, though some represented expansions of frequent single word fillers. Including multi-word fillers also shed light on some filler “families”

built around a central noun with various expansions, either pre-modifiers (**case** → **particular case**) or post-modifiers (e.g., **use** → **use of technology**). Some frequent fillers were found to be more associated with one type of modification and learning more about those characteristics may be pedagogically useful. In some cases, while no individual fillers were highly frequent, as a whole the filler family reached high frequency levels. Including these multi-word fillers in the analysis also resulted in identification of some high frequency frames that did not appear among the most frequent frames identified in Gray and Biber (2013). Like the frames-first approach introduced in that study, it is hoped that this extended lexical frame approach might allow researchers to identify frames, or characteristics of frames, that were not visible using previous methods, providing a fuller look into lexical frame behavior and phraseology more generally.

The chapter also identified structural patterns in multi-word fillers and explored the ways these structures are used to package complex information about objects and actions in a noun phrase within the bounds of a lexical frame. This type of analysis may present a potentially fruitful area of overlap between noun phrase complexity and lexical frames studies that merits further exploration.

Overall, the findings of Phase 2 suggest that extended frames merit further exploration in future studies, both in the methodological and language description aspects. This approach to lexical frames provides a new measure of variability not discussed in previous studies: the degree to which particular lexical frames are associated not only with particular fillers, but also with filler structural types. In the past, lexical frame productivity has been measured using type-token ratio (TTR), a measure that is less useful for extended frames, which have a very high proportion of types due to the lower likelihood of multi-word fillers repeating. In this case, comparing the variability in structural types of fillers may be a more appropriate and useful way of measuring

frame variability. The following chapter contains a discussion of how these findings fit in the context of previous work, as well as considerations of the limitations, implications for research and practice, and future directions that may follow from this work.

CHAPTER 6. DISCUSSION AND CONCLUSION

6.1. Summary of Findings

Previous research has demonstrated the value of lexical frames for describing phraseological characteristics of academic writing (Biber, 2009), providing insight into learner writing development (Appel et al., 2024; Garner, 2016; Tan & Römer, 2022; Yan et al., 2024), and as a pedagogical tool for raising learner awareness of formulaic features of expert writing and disciplinary norms (Casal & Kessler, 2010; Cunningham, 2017; Golparvar & Barabadi, 2020; Liu & Chen, 2022; Lu et al., 2018). In academic writing, it is primarily nouns that fill the variable slot in these frames, but the majority studies have only considered single word fillers, despite the frequency of noun phrase elaboration in academic writing (Biber & Gray, 2016). Therefore, this study argued that a fuller picture of the phraseological characteristics of academic writing requires taking into account multi-word noun phrases as lexical frame fillers. However, identifying such forms presents some methodological challenges, and addressing them was the first goal of this study.

The first methodological goal was to establish a clear operational definition of what constitutes a variant of a multi-word lexical frame, as previous frames studies often lacked clear specifications about what variants were to be considered and which were left out (see discussion in Kizil et al., 2024). For example, the method developed in Römer (2010) enabled identification of frame variants with multiple fillers, either adjacent or non-adjacent, such as *it would **be** very interesting, it should also **be** interesting*, and *it seems to me that it would **be** interesting*. However, that approach required starting with previously identified frames with 1-word fillers, as well as a large amount of manual checking that in some cases required subjective judgments of meaningfulness or coherence. Hsu (2020) also identified frames with multi-word fillers, but gave

no specifics as to how frame variants were determined to be of interest or not. Another approach was taken by Geluso (2020), who focused on a specified structure, THE (**Noun1**) OF THE (**Noun2**), while allowing a one-word modifier before each of the nouns in the framework. This study provides well-defined criteria for what constitutes a frame variant of interest, though with a relatively limited scope. It is hoped that the present study, by specifying objective structural criteria to define the extended lexical frames of interest, can contribute to further replicability of frames studies in the future.

After a definition was established based on syntactic characteristics, it was then necessary to develop a tool to identify these extended frames using existing NLP libraries with a minimum of manual checking. The Python scripts developed were found to be both accurate and practical, though some manual validity checking was necessary due to structural ambiguities and parsing limitations. Next, these scripts were applied to the descriptive goals of this study, which sought to answer three questions about the phraseological characteristics of academic journal writing in English, using the Academic Journal Register Corpus (AJRC) (Gray, 2015). The first asked how much including longer fillers increases the frequency of frame occurrences. Results showed that across all of the frames meeting the frequency criteria, nearly half of the variants (frame + filler combinations) identified (45%) included two or more words. As an example of the increased frequency resulting from allowing multi-word fillers, the most frequent frame in the corpus, THE + OF THE, appeared 3,244 times per million words, as compared to 2,567 times in Gray and Biber (2013). This also resulted in several frames appearing with greater relative frequencies than in previous studies. For example, the frame THE + IN THE was the fourth most frequent in this analysis, but did not appear among the 10 most frequent frames in academic prose reported in Gray and Biber (2013). As explained in section 4.2.1, this frame was notable in that only 26%

of variants had 1-word fillers, which explains its lower frequency in previous studies. When examining filler length variation within individual frames and found that even the least variable frame had multi-word fillers in approximately 17% of variants. Taken together, these results demonstrate the value of the extended lexical frame approach introduced in this study for providing a fuller picture of the phraseological characteristics of a corpus.

The second descriptive goal of this study was to investigate the forms and structural characteristics of multi-word fillers found in the most frequent extended lexical frames. The findings showed that longer fillers are often more specialized and lower frequency; however, some represent expansions of frequent single-word fillers, which recalls previous, though less extensive, findings in Eeg-Olofsson & Altenberg (1994). These expansions (e.g. **proportion** → **relative proportion, large proportion, much greater proportion**) suggest the value of a future approach that might more directly identify single-word nominal “roots” and their expansions as filler “families”. This could be especially valuable in cases where no one variant is highly frequent, but the combined variants in the “family” reach a high frequency. Also, as discussed in Chapter 5, some frequent fillers were more likely to have certain modification structures, which may have useful pedagogical implication and merits further exploration as well. This exploration of filler structure builds on the results in Eeg-Olofsson & Altenberg (1994), who found that grammatical functions of filler words varied considerably, though they did not explore that variation in detail. This type of analysis may also provide enable further connections between frames research and studies of noun modification patterns (e.g. Arnaud et al., 2008).

Taken together, these findings suggest the value of further pursuing this intersection of noun phrase complexity and lexical frames studies. While the limitations of this study must be

acknowledged (Section 5.3), its results may still have some useful implications for research and pedagogical practice.

6.2. Implications

6.2.1. Research implications

The results of this project have implications for future research as part of a broader push toward greater replicability in applied linguistics research (e.g. Grieve, 2021). In phraseology studies, and lexical frames studies more specifically, some have pointed to the need for clearer justification for methodological choices, including how and why decisions are made regarding which frames are of interest (Kizil et al., 2024). Previous lexical or phrase frame studies have often determined which frames would be included based on “semantic homogeneity” of the filler words (Fuster-Marquez & Pennock-Speck, 2015), subjective judgments about meaningfulness (Römer, 2010), or avoiding highly idiosyncratic sequences (Fuster-Màrquez, 2014). This study’s efforts to provide an explicit and objective operational definition for extended lexical frames contribute to replicability while also facilitating automation using NLP computational tools.

Another area in which replicability of lexical frames or bundles research may be improved is the process in which overlapping sequences are treated. This study used an automatic approach, combining overlapping frames and deleting repetitions when the combinations reach a target frequency. While this may not be the best option for all studies, as sequence length has a large effect on results in phraseology studies, it offers useful suggestions for future work to avoid the need for manual checking, improving both efficiency and ease of replication. Also, as lexical frames with multi-word fillers have not been the topic of previous frames studies, this study argues for entropy as a useful measure of overall filler length variation, extending its previous use for measuring variation in single word fillers (Eeg-Olofsson &

Altenberg, 1994). As was seen in this study, the measure facilitated identification of frames that may be of interest due to particularly high or low variation.

In addition to the methodological implications of this study, the finding that multi-word fillers are very frequent, making up nearly half of all frame instances in this analysis, gives further support to previous findings about the high frequency of formulaic language (Nelson, 2018). In addition, the combination of flexibility and formulaicity that define extended lexical frames reflect Sinclair's discussions about the interaction between the Idiom Principle and Open Choice Principle (Sinclair, 1991)

6.2.2. Pedagogical implications

While this study's goals were not pedagogical in nature, some aspects of the findings can inform language pedagogy. One way is through the study of lexical frames use as a marker of writer development. Previous studies have found that developing writers show greater frequency and variability in lexical frame use as their proficiency increases (e.g. Appel et al., 2024; Garner, 2016; Tan & Römer, 2010, 2022). This paper argues that extended lexical frames offer insight into the intersection between phraseological patterning and noun phrase complexity in a way previous single-word filler lexical frames studies could not. This suggests the potential of linking studies of learner frame use with others that examine increasing noun phrase elaboration in learners (e.g. Staples et al., 2016). Also, many previous studies have developed pedagogically oriented lists of phrase frames (Barabadi et al., 2020; Casal & Yoon, 2023; Cunningham, 2015; Lu et al., 2018). While the purpose of the present study was not to identify pedagogically valuable frames, the extended lexical frames approach might provide more richness to future pedagogical materials using frames by also highlighting the ways frame fillers are elaborated in different contexts.

6.3. Limitations

While this study offers potentially useful findings in an under-explored area, limitations of the methods used must be considered. One important limitation in the automatic filler validity checking script is that it was developed and tested using only a small test corpus of 10 research articles from the field of applied linguistics. Due to potential linguistic variation across disciplines, accuracy would likely be improved by using a larger and more varied dataset, which was not possible for this study due to the many hours of manual checking required to create a gold standard coded text for even a small corpus. However, future studies may explore the use of AI tools to streamline this process and reduce the human workload.

A related practical limitation is the fact that several thousand frame variants had to be checked manually due to structural ambiguity and limitations of automatic sentence parsing tools (see Ratnaparkhi & Kumar, 2021; Zhou et al., 2024). This manual checking procedure requires a certain degree of subjective judgment, which was mitigated as best as possible through establishment of an explicit operational definition for acceptable frame variants based on syntactic structure. While other researchers may take a different approach to the criteria they include in their own definitions, it is hoped that by providing a clear definition this study will contribute to the replicability of future lexical frames studies.

The limitations of automatic sentence parsing also caused some words to be mis-tagged, resulting either in false positives or false negatives. To address this, the filtering scripts were designed to favor false positives, which could then be manually removed, rather than false negatives, which would cause some true frame realizations to be overlooked. However, in the case of some texts with highly specialized vocabulary, it may be valuable to make modifications to the SpaCy processing pipeline in order to increase accuracy. One type of special case is the use of multi-word proper nouns, which occur with much more frequency in history texts, for

example. By default, SpaCy gives each word in the proper noun the PROPN part of speech tag, meaning ‘New York City’ or ‘Jerry Lee Lewis’ would be counted as three tokens. However, with a study focused on frame filler length, such terms may be more properly considered as a single unit, and future development of this script would adjust accordingly. Another special case is when commas appear inside fillers, such as complex coordinated noun phrases such as “*comfort, happiness, and accommodation*” and coordinate adjectives such as “*bulky, heavy box*”. This study followed the conventions of previous research by not crossing comma boundaries, and initial analysis found this type of structure to be rare in relation to the false positives that resulted from allowing them; however, it may be interesting to explore this type of structure in more detail in the future and find ways of efficiently identifying them.

The method developed in this study was designed for analysis of written text, and more specifically for the kind of writing found in academic journal articles. Therefore, any future studies interested in extended lexical frames in spoken texts or other registers with speech-like characteristics (e.g. tweets (Baker & McEnery, 2015)) would require significant adaptations. First of all, previous studies comparing frames in speech and writing found that spoken conversation is characterized more by continuous sequences rather than frames with variable slots, and the frames that are found typically include function word fillers rather than content words (Biber, 2009). Additionally, complexity in spoken English is more clausal than phrasal (Biber et al., 2016), so there is less reason to focus on the noun phrase elaboration which was central to this project. It may be the case that extended lexical frames as defined in this study are not a relevant phraseological form for spoken language, so future studies may choose to instead explore semi-fixed patterns using other approaches.

Another limitation of the approach developed in the study relates to corpus size. While the corpus used in this study was large and representative of the register of academic journal writing, many corpora developed in recent years are much larger and would likely present processing challenges for the scripts used in this study. Identifying extended lexical frames in a corpus of this size required significant memory and processing time due to the number of possible n-grams when allowing for up to nine filler words. For example, a 15-word sentence can be broken into 12 4-grams, 11 5-grams, 10 6-grams, etc., resulting in 57 total ngrams, each of which required further processing. Across an entire corpus, this resulted in very large file sizes and processing challenges, necessitating use of large SQL databases to manage such a large dataset. With a much larger corpus, processing times may become impractical, as would the necessity of manual checking of certain structures. Some of these issues may be lessened by further development of technology, but for the present time, this method may be best suited to small or medium-sized corpora.

A common methodological challenge in frequency-based phraseological studies is the choice of minimum frequency when identifying items of interest. Because previous frames studies have not considered variants with multi-word fillers, the frames found in this study would be expected to be more frequent than in previous studies; however, to what degree was unknown before carrying out the study, and there was no clear precedent in setting a frequency cutoff. Therefore, partly for practical reasons, this study began by identifying frames with three fixed words that occurred at least 400 times per million words, a number that provided a number of frames that provided some insight into broad patterns of variation while remaining manageable for more detailed analysis within the scope of this study. It is possible that some lower frequency frames may have been of interest; for example, by having a higher proportion of longer fillers

than more frequent frames. However, the frames reported represented a group similar to previous studies of frames in academic writing (Gray & Biber, 2013), allowing for some useful comparisons. The effect of choice of frame length has also been a topic of previous studies (Liu & Chen, 2022; Lij et al., 2023), as no established cutoff frequencies exist related to length. This was, of course, complicated by the fact that the frames identified in this study varied in length of filler as well. The decision was made to include frames with four fixed words appearing at least 200 times per million words in order to account for the many overlaps observed in the preliminary analyses, particularly when a determiner appears within the filler (e.g. **AT the beginning OF THE**, **AT THE beginning OF**, **AT THE beginning OF THE**). Future studies may provide a more fully developed approach to making these types of decisions to best balance frequency and variation while remaining sensitive to the wider co-textual environment (Kizil et al., 2024; Marco, 2000)

6.4. Future directions

Though this study provided some useful insight into new methodological possibilities for research in phraseology, there remains much more to be done to develop both theoretical and practical sides of the ideas it has suggested.

6.4.1. NLP

First of all, the implementation of this study's operational definition of extended lexical frames relied on the SpaCy NLP library and the large English language model packaged with that library. Since the project began, large language models (LLMs) such as ChatGPT have developed at a rapid pace, increasing both their scope and accuracy. At the time of this writing ChatGPT has gained the ability to produce sentence trees and constituency parses, though their accuracy is not perfect based on some informal testing. This offers possibilities for improving the parsing of ambiguous prepositional phrase attachment. For example, ChatGPT was able to

identify and describe the ambiguity in the sentence ‘the man saw a dog with a telescope’, give tree diagrams for both interpretations, and choose the more likely (i.e. common in its training data) interpretation. However, with the example phrase “the point of view of the community”, it first failed to see “point of view” as a fixed unit, instead describing “of the community” as modifying “view”. It was able to catch this problem when it was pointed out, which suggests that the use of specialized prompts or training may make such tools a useful tool for parsing the more ambiguous structures that sometimes appeared in longer lexical frame fillers, thus reducing the amount of manual checking that was necessary for this study.

During piloting, a constituency parsing tool called Constituent treelib (Halvani, 2023) was tested to simplify the identification of noun phrase boundaries in fillers and reduce the number of filtering scripts. However, similar to other NLP tools, had difficulties with prepositional phrase attachment, and ran much slower than the custom-written filtering rules, resulting in excessive processing time that was not usable for a full corpus. No alternatives that could achieve the same function were found at the time of this study, but future research may take advantage of new parsing tools or advancing computational efficiency to streamline the analysis process. Moving forward, it may also be possible to use the hand-checked dataset from this study to train a machine learning classification model to identify acceptable or unacceptable frame variants, though this possibility was not explored in any depth for this study.

6.4.2. Language description and pedagogy

As discussed in Chapter 5, it was observed that many multi-word fillers represent expansions of frequent single-word filler nouns. In some cases, the single-word variant is not very frequent, but taken together, all variants reach a high frequency, similar in a way to how the bundles- vs frames-first approaches were contrasted in Gray and Biber (2013). This finding

illustrates how the extended frames approach may help future studies identify more patterns that have been invisible when using traditional approaches. Also, this was noticed through manual analysis, but future studies may develop tools to directly identify lexical frame filler “families” of a central noun and its elaborated variants. Similar to the argument made by Casal and Yoon (2023) regarding frames with single-word fillers and their lead variants (most common fillers), identifying these families may prove useful for pedagogical purposes; the central noun may be taught as the prototypical form along with the functions it may serve, and the pre- or post-modified variants could be introduced gradually to illustrate how the functions can become more flexible or complex.

While disciplinary variation was not the focus of this study, some variation was found across the disciplinary subcorpora. For example, the 3-word filler pattern *noun* + *conjunction* + *noun* was much more frequent in history texts than in physics. Future studies may take a closer look at how noun phrase elaboration patterns in lexical frame fillers vary across disciplines, registers, text types, or other parameters, building on findings on the relationship of discipline with frame variability (J Ren, 2021) and phrasal complexity features (Gray, 2015).

Future work may also focus on how extended lexical frames can inform studies of writing development, for example, by combining aspects of studies of frames across learner proficiency levels (Garner, 2016; Tan & Römer, 2022) and studies of noun phrase complexity as a marker of learner development (Parkinson & Musgraves, 2014). In addition to describing features of learner developmental stages, pedagogical value may come from studies exploring extended lexical frames from a discourse function perspective, as previous studies have done with traditional single word filler lexical frames (Cao & Wu, 2022; J Ren, 2021; Lu et al., 2021; Zhou & Li, 2014).

6.5. Concluding remarks

In summary, this study has attempted to explore a new path in phraseology research, taking a novel approach to lexical frames research with a different balance of fixedness and variation. The project grew from a curiosity about both noun phrase complexity and formulaic language, finding an intersection with potentially useful implications for understanding, describing, and teaching the characteristics of academic. One goal of the study was to contribute to transparency and reproducibility in phraseology research through clearly defined operational definitions, and in that spirit the code used in this analysis will be available on GitHub (add link).

While the scope of the study was limited and to some extent exploratory, I hope some of the ideas introduced here will continued to be refined in my own future work or that of others whose curiosity may be sparked by these preliminary findings. From a language description perspective, there are several potentially fruitful avenues that could be pursued. For example, this study looked briefly at disciplinary variation in extended frames with promising results, and I hope to pursue this path further in future work. The extended frame approach could offer valuable pedagogical insights as well, and I look forward to exploring how the frequency and structure of multi-word fillers vary among learners as well. From a methodological perspective, there may be value in explore integrating LLMs to improve parsing ambiguity issues and reduce the need for manual checking, making this type of analysis possible the larger corpora that are becoming more and more common in recent studies.

The focus of this dissertation was on recurrent discontinuous sequences of words, but it must be remembered that “users of a language do not use p-frames, they use phrases” (Kizil et al., 2024, p. 15). With that in mind, the methodological goals and technical aspects explored here should not be the end goal of this work, but a means toward the end of greater understanding of language in context.

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